# DISCO-TAB — Comprehensive Benchmark

all generators × 24 datasets × all metrics; cells are **mean ± 95% confidence interval** over random seeds (Student-*t*, up to 3 seeds); **bold** = best per row (by mean, where a direction is defined); – = missing run/metric; ± absent where only one seed is available.

## Utility (downstream TSTR)

Table 1: TSTR F1-macro (↑) — mean ± 95% CI over seeds

| Dataset | DISCO-TAB | CTGAN | TVAE | GaussCop | TabDDPM | ADS-GAN | PATE-GAN | RTVAE | NFlow | ARF | BayesNet | Marginals | GReaT |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| heart_failure | **0.783 ± 0.184** | 0.483 ± 0.237 | 0.135 ± 0.582 | 0.591 ± 0.252 | 0.406 | 0.635 ± 0.151 | 0.396 ± 0.174 | 0.600 ± 0.270 | 0.559 ± 0.371 | 0.649 ± 0.296 | 0.735 ± 0.205 | 0.394 ± 0.217 | – |
| breast_cancer | 0.930 ± 0.035 | 0.295 ± 0.304 | 0.934 ± 0.084 | 0.912 ± 0.041 | 0.924 | 0.934 ± 0.044 | 0.497 ± 0.461 | 0.937 ± 0.047 | 0.855 ± 0.082 | 0.940 ± 0.034 | **0.947 ± 0.046** | 0.513 ± 0.508 | – |
| liver_disorders | 0.509 ± 0.208 | 0.473 ± 0.224 | 0.527 ± 0.171 | 0.553 ± 0.098 | 0.592 | 0.542 ± 0.174 | 0.488 ± 0.263 | 0.490 ± 0.333 | 0.500 ± 0.106 | **0.593 ± 0.140** | 0.561 ± 0.271 | 0.380 ± 0.208 | – |
| parkinsons | 0.646 ± 0.517 | 0.480 ± 0.344 | 0.000 ± 0.000 | 0.630 ± 0.104 | **0.928** | 0.737 ± 0.215 | 0.424 ± 0.012 | 0.704 ± 0.131 | 0.695 ± 0.096 | 0.688 ± 0.183 | 0.759 ± 0.296 | 0.446 ± 0.139 | – |
| obesity | **0.886 ± 0.011** | 0.138 ± 0.161 | 0.490 ± 0.086 | 0.243 ± 0.045 | 0.795 | 0.655 ± 0.097 | 0.115 ± 0.033 | 0.607 ± 0.092 | 0.171 ± 0.091 | 0.757 ± 0.046 | 0.821 ± 0.023 | 0.102 ± 0.026 | – |
| diabetes | 0.843 ± 0.186 | 0.836 ± 0.042 | 0.875 ± 0.003 | 0.485 ± 0.014 | **0.892 ± 0.006** | 0.797 ± 0.086 | 0.478 ± 0.000 | 0.727 ± 0.121 | 0.658 ± 0.070 | 0.853 ± 0.012 | 0.704 ± 0.071 | 0.377 ± 0.082 | – |
| german_credit | 0.516 ± 0.572 | 0.448 ± 0.046 | 0.137 ± 0.591 | 0.496 ± 0.118 | 0.454 ± 0.107 | 0.594 ± 0.194 | 0.468 ± 0.203 | 0.501 ± 0.164 | 0.449 ± 0.145 | 0.584 ± 0.121 | **0.638 ± 0.179** | 0.459 ± 0.077 | – |
| bank_marketing | **0.673 ± 0.205** | 0.604 ± 0.039 | 0.673 ± 0.043 | 0.469 ± 0.001 | 0.637 ± 0.018 | 0.608 ± 0.082 | 0.469 ± 0.000 | 0.498 ± 0.058 | 0.471 ± 0.005 | 0.558 ± 0.054 | 0.551 ± 0.100 | 0.236 ± 0.317 | – |
| census_income | 0.765 ± 0.039 | 0.727 ± 0.031 | 0.746 ± 0.014 | 0.459 ± 0.038 | 0.750 ± 0.020 | 0.669 ± 0.126 | 0.433 ± 0.002 | 0.549 ± 0.126 | 0.615 ± 0.145 | 0.744 ± 0.011 | **0.767 ± 0.005** | 0.524 ± 0.264 | – |
| surv_melanoma | 0.536 ± 0.169 | 0.406 ± 0.142 | 0.000 ± 0.000 | **0.606 ± 0.185** | 0.510 ± 0.063 | 0.557 ± 0.099 | 0.442 ± 0.070 | 0.529 ± 0.128 | 0.508 ± 0.201 | 0.573 ± 0.354 | 0.570 ± 0.345 | 0.445 ± 0.588 | – |
| surv_pbc | 0.522 ± 0.515 | 0.490 ± 0.120 | 0.376 ± 0.000 | 0.534 ± 0.108 | 0.564 ± 0.019 | 0.590 ± 0.152 | 0.492 ± 0.182 | 0.646 ± 0.177 | 0.566 ± 0.220 | 0.634 ± 0.162 | **0.679 ± 0.101** | 0.428 ± 0.123 | – |
| surv_pbc3 | 0.468 ± 0.398 | 0.475 ± 0.105 | 0.000 ± 0.000 | 0.530 ± 0.014 | 0.453 ± 0.000 | 0.632 ± 0.095 | 0.453 ± 0.000 | 0.469 ± 0.098 | 0.477 ± 0.101 | 0.553 ± 0.215 | **0.707 ± 0.091** | 0.497 ± 0.131 | 0.453 |
| surv_retinopathy | 0.518 ± 0.146 | 0.509 ± 0.257 | 0.124 ± 0.535 | 0.484 ± 0.040 | 0.452 ± 0.101 | 0.539 ± 0.211 | 0.480 ± 0.201 | 0.478 ± 0.147 | 0.513 ± 0.146 | **0.586 ± 0.116** | 0.493 ± 0.159 | 0.511 ± 0.042 | – |
| surv_whas500 | 0.739 ± 0.032 | 0.428 ± 0.040 | 0.476 ± 0.227 | 0.670 ± 0.213 | **0.763 ± 0.059** | 0.699 ± 0.159 | 0.395 ± 0.161 | 0.619 ± 0.114 | 0.709 ± 0.046 | 0.671 ± 0.081 | 0.673 ± 0.189 | 0.324 ± 0.096 | – |
| surv_gbsg2 | 0.580 ± 0.067 | 0.438 ± 0.073 | 0.505 ± 0.182 | 0.526 ± 0.181 | 0.547 ± 0.135 | 0.538 ± 0.086 | 0.474 ± 0.204 | 0.556 ± 0.082 | **0.590 ± 0.196** | 0.572 ± 0.067 | 0.547 ± 0.061 | 0.475 ± 0.310 | 0.460 |
| surv_colon | **0.587 ± 0.120** | 0.506 ± 0.146 | 0.534 ± 0.031 | 0.541 ± 0.033 | 0.577 ± 0.063 | 0.497 ± 0.250 | 0.426 ± 0.203 | 0.564 ± 0.075 | 0.526 ± 0.083 | 0.581 ± 0.037 | 0.539 ± 0.147 | 0.450 ± 0.097 | – |
| surv_grace | 0.609 ± 0.052 | 0.456 ± 0.103 | 0.437 ± 0.083 | **0.610 ± 0.067** | 0.576 ± 0.068 | 0.591 ± 0.100 | 0.456 ± 0.074 | 0.508 ± 0.058 | 0.543 ± 0.224 | 0.593 ± 0.072 | 0.521 ± 0.254 | 0.463 ± 0.033 | – |
| surv_trace | 0.676 ± 0.019 | 0.513 ± 0.077 | 0.666 ± 0.081 | 0.624 ± 0.028 | 0.660 ± 0.121 | 0.646 ± 0.163 | 0.445 ± 0.065 | 0.516 ± 0.148 | 0.609 ± 0.192 | **0.704 ± 0.042** | 0.684 ± 0.036 | 0.516 ± 0.045 | – |
| surv_framingham | **0.581 ± 0.088** | 0.577 ± 0.090 | 0.455 ± 0.046 | 0.488 ± 0.036 | 0.556 ± 0.031 | 0.546 ± 0.088 | 0.474 ± 0.112 | 0.486 ± 0.104 | 0.453 ± 0.122 | 0.532 ± 0.033 | 0.475 ± 0.112 | 0.448 ± 0.110 | – |
| tab_blood_transfusion | 0.351 ± 0.755 | 0.481 ± 0.151 | 0.000 ± 0.000 | 0.579 ± 0.146 | 0.541 ± 0.065 | 0.498 ± 0.141 | 0.510 ± 0.243 | 0.517 ± 0.078 | 0.456 ± 0.215 | **0.580 ± 0.132** | 0.528 ± 0.157 | 0.502 ± 0.198 | – |
| tab_credit_g | 0.643 ± 0.103 | 0.421 ± 0.034 | 0.412 ± 0.000 | 0.476 ± 0.143 | 0.491 ± 0.014 | 0.577 ± 0.077 | 0.435 ± 0.099 | 0.465 ± 0.128 | 0.448 ± 0.085 | 0.526 ± 0.095 | **0.653 ± 0.076** | 0.434 ± 0.200 | – |
| tab_qsar_biodeg | 0.796 ± 0.129 | 0.419 ± 0.028 | 0.662 ± 0.126 | 0.712 ± 0.106 | 0.391 ± 0.025 | 0.729 ± 0.081 | 0.428 ± 0.232 | 0.770 ± 0.071 | 0.612 ± 0.298 | 0.780 ± 0.119 | **0.850 ± 0.139** | 0.393 ± 0.338 | – |
| tab_maternal_risk | **0.624 ± 0.257** | 0.305 ± 0.042 | 0.000 ± 0.000 | 0.375 ± 0.160 | 0.296 ± 0.087 | 0.463 ± 0.240 | 0.235 ± 0.187 | 0.527 ± 0.099 | 0.395 ± 0.082 | 0.573 ± 0.106 | 0.214 ± 0.160 | 0.234 ± 0.260 | – |
| tab_website_phishing | **0.614 ± 0.075** | 0.329 ± 0.130 | 0.521 ± 0.049 | 0.359 ± 0.081 | 0.604 ± 0.064 | 0.555 ± 0.090 | 0.263 ± 0.217 | 0.492 ± 0.047 | 0.454 ± 0.013 | 0.562 ± 0.101 | 0.551 ± 0.100 | 0.286 ± 0.087 | – |

Table 2: TSTR F1-weighted (↑) — mean ± 95% CI over seeds

| Dataset | DISCO-TAB | CTGAN | TVAE | GaussCop | TabDDPM | ADS-GAN | PATE-GAN | RTVAE | NFlow | ARF | BayesNet | Marginals | GReaT |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| heart_failure | **0.799 ± 0.195** | 0.559 ± 0.243 | 0.185 ± 0.796 | 0.679 ± 0.178 | 0.555 | 0.710 ± 0.122 | 0.500 ± 0.317 | 0.676 ± 0.173 | 0.655 ± 0.256 | 0.722 ± 0.219 | 0.768 ± 0.183 | 0.416 ± 0.438 | – |
| breast_cancer | 0.935 ± 0.033 | 0.284 ± 0.354 | 0.939 ± 0.079 | 0.920 ± 0.038 | 0.929 | 0.939 ± 0.042 | 0.489 ± 0.601 | 0.942 ± 0.045 | 0.869 ± 0.068 | 0.944 ± 0.033 | **0.950 ± 0.044** | 0.558 ± 0.428 | – |
| liver_disorders | 0.511 ± 0.205 | 0.472 ± 0.228 | 0.529 ± 0.169 | 0.553 ± 0.099 | 0.592 | 0.543 ± 0.175 | 0.488 ± 0.265 | 0.490 ± 0.333 | 0.501 ± 0.105 | **0.593 ± 0.140** | 0.562 ± 0.271 | 0.384 ± 0.206 | – |
| parkinsons | 0.732 ± 0.378 | 0.632 ± 0.164 | 0.000 ± 0.000 | 0.729 ± 0.116 | **0.947** | 0.801 ± 0.156 | 0.630 ± 0.018 | 0.784 ± 0.111 | 0.768 ± 0.100 | 0.756 ± 0.184 | 0.829 ± 0.201 | 0.584 ± 0.126 | – |
| obesity | **0.888 ± 0.010** | 0.138 ± 0.168 | 0.493 ± 0.088 | 0.250 ± 0.046 | 0.796 | 0.655 ± 0.108 | 0.120 ± 0.036 | 0.610 ± 0.098 | 0.176 ± 0.096 | 0.759 ± 0.050 | 0.827 ± 0.022 | 0.101 ± 0.023 | – |
| diabetes | 0.956 ± 0.045 | 0.952 ± 0.017 | 0.962 ± 0.001 | 0.876 ± 0.003 | **0.968 ± 0.002** | 0.945 ± 0.021 | 0.874 ± 0.001 | 0.927 ± 0.029 | 0.901 ± 0.039 | 0.959 ± 0.003 | 0.923 ± 0.016 | 0.584 ± 0.144 | – |
| german_credit | 0.502 ± 0.722 | 0.585 ± 0.036 | 0.192 ± 0.827 | 0.624 ± 0.074 | 0.604 ± 0.067 | 0.680 ± 0.112 | 0.601 ± 0.099 | 0.631 ± 0.105 | 0.587 ± 0.049 | 0.687 ± 0.082 | **0.715 ± 0.119** | 0.516 ± 0.219 | – |
| bank_marketing | 0.865 ± 0.036 | 0.846 ± 0.049 | 0.854 ± 0.030 | 0.828 ± 0.000 | **0.872 ± 0.004** | 0.862 ± 0.019 | 0.828 ± 0.000 | 0.835 ± 0.015 | 0.829 ± 0.001 | 0.852 ± 0.014 | 0.848 ± 0.024 | 0.273 ± 0.618 | – |
| census_income | 0.830 ± 0.014 | 0.812 ± 0.018 | 0.818 ± 0.016 | 0.671 ± 0.019 | 0.826 ± 0.013 | 0.782 ± 0.067 | 0.656 ± 0.002 | 0.718 ± 0.069 | 0.748 ± 0.076 | 0.824 ± 0.006 | **0.838 ± 0.009** | 0.672 ± 0.164 | – |
| surv_melanoma | 0.561 ± 0.205 | 0.573 ± 0.124 | 0.000 ± 0.000 | **0.706 ± 0.136** | 0.660 ± 0.041 | 0.676 ± 0.045 | 0.588 ± 0.059 | 0.645 ± 0.130 | 0.614 ± 0.026 | 0.684 ± 0.262 | 0.672 ± 0.247 | 0.454 ± 0.688 | – |
| surv_pbc | 0.507 ± 0.610 | 0.502 ± 0.119 | 0.454 ± 0.000 | 0.544 ± 0.102 | 0.584 ± 0.026 | 0.592 ± 0.167 | 0.528 ± 0.205 | 0.662 ± 0.191 | 0.586 ± 0.156 | 0.648 ± 0.161 | **0.692 ± 0.107** | 0.444 ± 0.124 | – |
| surv_pbc3 | 0.541 ± 0.652 | 0.756 ± 0.039 | 0.000 ± 0.000 | 0.780 ± 0.013 | 0.751 ± 0.000 | 0.786 ± 0.079 | 0.751 ± 0.000 | 0.749 ± 0.037 | 0.748 ± 0.012 | 0.791 ± 0.087 | **0.845 ± 0.042** | 0.675 ± 0.082 | 0.751 |
| surv_retinopathy | 0.542 ± 0.133 | 0.526 ± 0.279 | 0.148 ± 0.636 | 0.515 ± 0.040 | 0.480 ± 0.099 | 0.565 ± 0.183 | 0.481 ± 0.265 | 0.507 ± 0.136 | 0.531 ± 0.148 | **0.601 ± 0.135** | 0.513 ± 0.132 | 0.517 ± 0.032 | – |
| surv_whas500 | 0.752 ± 0.024 | 0.493 ± 0.047 | 0.538 ± 0.159 | 0.694 ± 0.198 | **0.779 ± 0.055** | 0.719 ± 0.127 | 0.428 ± 0.101 | 0.642 ± 0.126 | 0.721 ± 0.042 | 0.697 ± 0.071 | 0.696 ± 0.178 | 0.339 ± 0.098 | – |
| surv_gbsg2 | 0.587 ± 0.063 | 0.461 ± 0.045 | 0.530 ± 0.145 | 0.539 ± 0.168 | 0.558 ± 0.123 | 0.542 ± 0.110 | 0.486 ± 0.213 | 0.564 ± 0.073 | **0.596 ± 0.186** | 0.582 ± 0.065 | 0.564 ± 0.061 | 0.474 ± 0.344 | 0.477 |
| surv_colon | **0.586 ± 0.122** | 0.506 ± 0.148 | 0.534 ± 0.033 | 0.541 ± 0.032 | 0.577 ± 0.062 | 0.495 ± 0.255 | 0.429 ± 0.200 | 0.565 ± 0.074 | 0.526 ± 0.084 | 0.582 ± 0.037 | 0.540 ± 0.149 | 0.451 ± 0.092 | – |
| surv_grace | 0.639 ± 0.066 | 0.551 ± 0.075 | 0.566 ± 0.057 | **0.668 ± 0.066** | 0.651 ± 0.047 | 0.656 ± 0.050 | 0.526 ± 0.150 | 0.591 ± 0.033 | 0.634 ± 0.153 | 0.652 ± 0.061 | 0.617 ± 0.157 | 0.503 ± 0.096 | – |
| surv_trace | 0.676 ± 0.018 | 0.513 ± 0.075 | 0.665 ± 0.082 | 0.624 ± 0.029 | 0.660 ± 0.120 | 0.645 ± 0.166 | 0.444 ± 0.069 | 0.516 ± 0.147 | 0.609 ± 0.191 | **0.704 ± 0.042** | 0.684 ± 0.036 | 0.516 ± 0.047 | – |
| surv_framingham | 0.622 ± 0.118 | **0.641 ± 0.084** | 0.586 ± 0.025 | 0.606 ± 0.025 | 0.639 ± 0.022 | 0.635 ± 0.054 | 0.590 ± 0.057 | 0.580 ± 0.101 | 0.580 ± 0.057 | 0.628 ± 0.030 | 0.595 ± 0.064 | 0.490 ± 0.119 | – |
| tab_blood_transfusion | 0.417 ± 0.898 | 0.520 ± 0.232 | 0.000 ± 0.000 | 0.676 ± 0.122 | 0.647 ± 0.058 | 0.607 ± 0.048 | 0.636 ± 0.128 | 0.615 ± 0.105 | 0.590 ± 0.192 | **0.678 ± 0.075** | 0.652 ± 0.094 | 0.551 ± 0.266 | – |
| tab_credit_g | 0.675 ± 0.121 | 0.577 ± 0.018 | 0.576 ± 0.000 | 0.618 ± 0.097 | 0.617 ± 0.004 | 0.657 ± 0.038 | 0.582 ± 0.025 | 0.606 ± 0.075 | 0.588 ± 0.063 | 0.645 ± 0.065 | **0.725 ± 0.053** | 0.501 ± 0.385 | – |
| tab_qsar_biodeg | 0.819 ± 0.106 | 0.541 ± 0.016 | 0.709 ± 0.105 | 0.755 ± 0.090 | 0.519 ± 0.034 | 0.767 ± 0.066 | 0.465 ± 0.314 | 0.799 ± 0.059 | 0.674 ± 0.210 | 0.809 ± 0.101 | **0.870 ± 0.119** | 0.361 ± 0.442 | – |
| tab_maternal_risk | **0.660 ± 0.234** | 0.340 ± 0.080 | 0.000 ± 0.000 | 0.447 ± 0.165 | 0.386 ± 0.041 | 0.510 ± 0.333 | 0.306 ± 0.153 | 0.592 ± 0.068 | 0.485 ± 0.087 | 0.637 ± 0.113 | 0.191 ± 0.165 | 0.251 ± 0.394 | – |
| tab_website_phishing | **0.746 ± 0.068** | 0.419 ± 0.082 | 0.717 ± 0.068 | 0.497 ± 0.115 | 0.737 ± 0.041 | 0.694 ± 0.052 | 0.346 ± 0.299 | 0.662 ± 0.035 | 0.614 ± 0.058 | 0.734 ± 0.045 | 0.727 ± 0.070 | 0.368 ± 0.100 | – |

Table 3: TSTR accuracy (↑) — mean ± 95% CI over seeds

| Dataset | DISCO-TAB | CTGAN | TVAE | GaussCop | TabDDPM | ADS-GAN | PATE-GAN | RTVAE | NFlow | ARF | BayesNet | Marginals | GReaT |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| heart_failure | **0.794 ± 0.204** | 0.572 ± 0.305 | 0.228 ± 0.980 | 0.733 ± 0.110 | 0.683 | 0.750 ± 0.110 | 0.572 ± 0.514 | 0.711 ± 0.086 | 0.717 ± 0.143 | 0.761 ± 0.157 | 0.767 ± 0.190 | 0.472 ± 0.335 | – |
| breast_cancer | 0.936 ± 0.033 | 0.316 ± 0.278 | 0.939 ± 0.079 | 0.921 ± 0.038 | 0.930 | 0.939 ± 0.044 | 0.561 ± 0.437 | 0.942 ± 0.045 | 0.874 ± 0.055 | 0.944 ± 0.033 | **0.950 ± 0.045** | 0.579 ± 0.363 | – |
| liver_disorders | 0.546 ± 0.150 | 0.502 ± 0.181 | 0.551 ± 0.144 | 0.556 ± 0.091 | **0.594** | 0.551 ± 0.191 | 0.493 ± 0.252 | 0.493 ± 0.330 | 0.507 ± 0.095 | **0.594 ± 0.144** | 0.565 ± 0.272 | 0.473 ± 0.162 | – |
| parkinsons | 0.735 ± 0.368 | 0.675 ± 0.074 | 0.000 ± 0.000 | 0.744 ± 0.169 | **0.949** | 0.803 ± 0.147 | 0.735 ± 0.037 | 0.795 ± 0.127 | 0.769 ± 0.127 | 0.752 ± 0.224 | 0.846 ± 0.169 | 0.607 ± 0.224 | – |
| obesity | **0.887 ± 0.012** | 0.154 ± 0.127 | 0.499 ± 0.077 | 0.297 ± 0.056 | 0.801 | 0.655 ± 0.120 | 0.149 ± 0.042 | 0.627 ± 0.083 | 0.270 ± 0.101 | 0.766 ± 0.048 | 0.828 ± 0.019 | 0.115 ± 0.024 | – |
| diabetes | 0.961 ± 0.031 | 0.954 ± 0.022 | 0.963 ± 0.000 | 0.915 ± 0.001 | **0.970 ± 0.002** | 0.953 ± 0.015 | 0.915 ± 0.001 | 0.941 ± 0.020 | 0.909 ± 0.095 | 0.963 ± 0.002 | 0.938 ± 0.010 | 0.479 ± 0.151 | – |
| german_credit | 0.542 ± 0.498 | 0.663 ± 0.147 | 0.233 ± 1.004 | 0.700 ± 0.033 | 0.710 ± 0.022 | 0.712 ± 0.040 | 0.680 ± 0.045 | 0.715 ± 0.054 | 0.668 ± 0.094 | **0.743 ± 0.050** | 0.742 ± 0.068 | 0.545 ± 0.255 | – |
| bank_marketing | 0.867 ± 0.064 | 0.858 ± 0.094 | 0.845 ± 0.072 | 0.883 ± 0.000 | **0.896 ± 0.002** | 0.889 ± 0.003 | 0.883 ± 0.000 | 0.884 ± 0.004 | 0.883 ± 0.000 | 0.891 ± 0.004 | 0.886 ± 0.003 | 0.266 ± 0.436 | – |
| census_income | 0.832 ± 0.004 | 0.823 ± 0.016 | 0.821 ± 0.022 | 0.766 ± 0.005 | 0.835 ± 0.012 | 0.812 ± 0.032 | 0.758 ± 0.007 | 0.782 ± 0.034 | 0.784 ± 0.036 | 0.836 ± 0.004 | **0.846 ± 0.015** | 0.698 ± 0.183 | – |
| surv_melanoma | 0.545 ± 0.185 | 0.626 ± 0.152 | 0.000 ± 0.000 | **0.724 ± 0.126** | **0.724 ± 0.070** | 0.707 ± 0.000 | 0.626 ± 0.185 | 0.667 ± 0.195 | 0.634 ± 0.160 | 0.707 ± 0.242 | 0.683 ± 0.218 | 0.463 ± 0.655 | – |

| Dataset | DISCO-TAB | CTGAN | TVAE | GaussCop | TabDDPM | ADS-GAN | PATE-GAN | RTVAE | NFlow | ARF | BayesNet | Marginals | GReaT |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| surv_pbc | 0.566 ± 0.373 | 0.497 ± 0.120 | 0.603 ± 0.000 | 0.540 ± 0.104 | 0.587 ± 0.039 | 0.593 ± 0.159 | 0.561 ± 0.277 | 0.667 ± 0.209 | 0.603 ± 0.079 | 0.651 ± 0.172 | **0.693 ± 0.120** | 0.439 ± 0.127 | – |
| surv_pbc3 | 0.543 ± 0.663 | 0.824 ± 0.020 | 0.000 ± 0.000 | 0.838 ± 0.020 | 0.829 ± 0.000 | 0.781 ± 0.102 | 0.829 ± 0.000 | 0.810 ± 0.020 | 0.805 ± 0.102 | 0.848 ± 0.041 | **0.857 ± 0.035** | 0.652 ± 0.164 | 0.829 |
| surv_retinopathy | 0.554 ± 0.121 | 0.532 ± 0.301 | 0.198 ± 0.853 | 0.541 ± 0.089 | 0.491 ± 0.103 | 0.581 ± 0.146 | 0.514 ± 0.220 | 0.527 ± 0.146 | 0.532 ± 0.155 | **0.608 ± 0.154** | 0.518 ± 0.108 | 0.514 ± 0.034 | – |
| surv_whas500 | 0.753 ± 0.027 | 0.573 ± 0.108 | 0.627 ± 0.015 | 0.706 ± 0.193 | **0.785 ± 0.053** | 0.728 ± 0.086 | 0.452 ± 0.080 | 0.649 ± 0.147 | 0.720 ± 0.046 | 0.713 ± 0.067 | 0.706 ± 0.178 | 0.333 ± 0.093 | – |
| surv_gbsg2 | 0.589 ± 0.068 | 0.505 ± 0.068 | 0.589 ± 0.063 | 0.548 ± 0.146 | 0.563 ± 0.105 | 0.560 ± 0.153 | 0.493 ± 0.230 | 0.565 ± 0.065 | **0.597 ± 0.180** | 0.587 ± 0.065 | 0.587 ± 0.082 | 0.488 ± 0.280 | 0.493 |
| surv_colon | **0.593 ± 0.098** | 0.508 ± 0.149 | 0.542 ± 0.014 | 0.542 ± 0.028 | 0.578 ± 0.065 | 0.550 ± 0.089 | 0.492 ± 0.193 | 0.569 ± 0.071 | 0.535 ± 0.066 | 0.582 ± 0.037 | 0.542 ± 0.147 | 0.475 ± 0.056 | – |
| surv_grace | 0.630 ± 0.069 | 0.592 ± 0.050 | 0.680 ± 0.022 | 0.680 ± 0.078 | 0.685 ± 0.033 | 0.678 ± 0.029 | 0.545 ± 0.228 | 0.625 ± 0.062 | **0.692 ± 0.072** | 0.663 ± 0.059 | 0.682 ± 0.014 | 0.495 ± 0.097 | – |
| surv_trace | 0.679 ± 0.017 | 0.516 ± 0.070 | 0.670 ± 0.069 | 0.625 ± 0.029 | 0.660 ± 0.120 | 0.660 ± 0.119 | 0.501 ± 0.159 | 0.518 ± 0.148 | 0.611 ± 0.187 | **0.704 ± 0.042** | 0.684 ± 0.036 | 0.519 ± 0.040 | – |
| surv_framingham | 0.616 ± 0.140 | 0.649 ± 0.099 | 0.683 ± 0.008 | **0.684 ± 0.015** | 0.667 ± 0.024 | 0.674 ± 0.046 | 0.660 ± 0.046 | 0.613 ± 0.132 | 0.668 ± 0.046 | 0.674 ± 0.036 | 0.676 ± 0.023 | 0.472 ± 0.125 | – |
| tab_blood_transfusion | 0.427 ± 0.927 | 0.511 ± 0.195 | 0.000 ± 0.000 | 0.695 ± 0.128 | 0.667 ± 0.067 | 0.645 ± 0.163 | 0.682 ± 0.061 | 0.623 ± 0.155 | 0.629 ± 0.228 | **0.701 ± 0.061** | 0.698 ± 0.075 | 0.542 ± 0.304 | – |
| tab_credit_g | 0.663 ± 0.127 | 0.685 ± 0.012 | 0.700 ± 0.000 | 0.717 ± 0.052 | 0.687 ± 0.026 | 0.683 ± 0.133 | 0.678 ± 0.093 | 0.700 ± 0.012 | 0.668 ± 0.050 | 0.712 ± 0.040 | **0.747 ± 0.047** | 0.540 ± 0.388 | – |
| tab_qsar_biodeg | 0.820 ± 0.096 | 0.660 ± 0.018 | 0.724 ± 0.094 | 0.771 ± 0.084 | 0.643 ± 0.068 | 0.779 ± 0.058 | 0.485 ± 0.296 | 0.806 ± 0.059 | 0.709 ± 0.106 | 0.815 ± 0.093 | **0.874 ± 0.111** | 0.423 ± 0.276 | – |
| tab_maternal_risk | 0.656 ± 0.262 | 0.341 ± 0.109 | 0.000 ± 0.000 | 0.487 ± 0.192 | 0.436 ± 0.140 | 0.524 ± 0.417 | 0.381 ± 0.316 | 0.626 ± 0.082 | 0.557 ± 0.057 | **0.678 ± 0.167** | 0.278 ± 0.103 | 0.289 ± 0.356 | – |
| tab_website_phishing | **0.761 ± 0.049** | 0.441 ± 0.062 | 0.749 ± 0.071 | 0.510 ± 0.129 | 0.749 ± 0.055 | 0.708 ± 0.079 | 0.434 ± 0.223 | 0.683 ± 0.030 | 0.637 ± 0.081 | 0.749 ± 0.036 | 0.747 ± 0.065 | 0.336 ± 0.109 | – |

## Statistical fidelity

Table 4: Jensen–Shannon div. (↓) — mean ± 95% CI over seeds

| Dataset | DISCO-TAB | CTGAN | TVAE | GaussCop | TabDDPM | ADS-GAN | PATE-GAN | RTVAE | NFlow | ARF | BayesNet | Marginals | GReaT |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| heart_failure | 0.054 ± 0.054 | 0.094 ± 0.020 | 0.200 ± 0.033 | 0.042 ± 0.002 | 0.194 | 0.070 ± 0.006 | 0.192 ± 0.132 | 0.265 ± 0.048 | 0.040 ± 0.004 | 0.041 ± 0.007 | **0.034 ± 0.007** | 0.189 ± 0.014 | – |
| breast_cancer | 0.062 ± 0.010 | 0.143 ± 0.007 | 0.111 ± 0.022 | 0.027 ± 0.011 | 0.294 | 0.072 ± 0.019 | 0.568 ± 0.357 | 0.466 ± 0.024 | 0.024 ± 0.014 | 0.016 ± 0.005 | **0.013 ± 0.002** | 0.293 ± 0.032 | – |
| liver_disorders | 0.068 ± 0.052 | 0.184 ± 0.038 | 0.129 ± 0.058 | 0.028 ± 0.005 | 0.028 | 0.113 ± 0.031 | 0.118 ± 0.095 | 0.450 ± 0.074 | 0.040 ± 0.004 | **0.026 ± 0.002** | 0.029 ± 0.001 | 0.316 ± 0.019 | – |
| parkinsons | 0.082 ± 0.044 | 0.175 ± 0.029 | 0.136 ± 0.013 | 0.056 ± 0.013 | 0.227 | 0.120 ± 0.015 | 0.143 ± 0.175 | 0.578 ± 0.075 | 0.049 ± 0.004 | 0.053 ± 0.016 | **0.039 ± 0.013** | 0.269 ± 0.016 | – |
| obesity | 0.022 ± 0.002 | 0.044 ± 0.017 | 0.090 ± 0.015 | 0.057 ± 0.011 | 0.044 | 0.047 ± 0.075 | 0.126 ± 0.168 | 0.158 ± 0.006 | 0.033 ± 0.011 | 0.030 ± 0.004 | **0.010 ± 0.001** | 0.200 ± 0.006 | – |
| diabetes | 0.024 ± 0.016 | 0.030 ± 0.003 | 0.035 ± 0.014 | 0.067 ± 0.007 | **0.002 ± 0.001** | 0.062 ± 0.041 | 0.174 ± 0.103 | 0.094 ± 0.055 | 0.032 ± 0.020 | 0.039 ± 0.005 | 0.007 ± 0.001 | 0.225 ± 0.020 | – |
| german_credit | 0.207 ± 0.571 | 0.020 ± 0.008 | 0.171 ± 0.014 | 0.008 ± 0.003 | 0.007 ± 0.001 | 0.019 ± 0.001 | 0.046 ± 0.025 | 0.067 ± 0.009 | 0.023 ± 0.002 | 0.007 ± 0.004 | **0.005 ± 0.004** | 0.140 ± 0.009 | – |
| bank_marketing | 0.016 ± 0.004 | 0.009 ± 0.005 | 0.040 ± 0.019 | 0.015 ± 0.006 | **0.004 ± 0.003** | 0.040 ± 0.016 | 0.088 ± 0.110 | 0.083 ± 0.034 | 0.040 ± 0.017 | 0.014 ± 0.002 | 0.014 ± 0.001 | 0.275 ± 0.032 | – |
| census_income | 0.050 ± 0.039 | 0.019 ± 0.005 | 0.055 ± 0.022 | 0.049 ± 0.060 | **0.007 ± 0.003** | 0.082 ± 0.067 | 0.182 ± 0.344 | 0.145 ± 0.108 | 0.085 ± 0.029 | 0.037 ± 0.007 | 0.052 ± 0.020 | 0.347 ± 0.009 | – |
| surv_melanoma | 0.064 ± 0.119 | 0.118 ± 0.005 | 0.147 ± 0.076 | 0.038 ± 0.013 | 0.044 ± 0.022 | 0.077 ± 0.079 | 0.187 ± 0.239 | 0.331 ± 0.072 | 0.033 ± 0.005 | 0.027 ± 0.016 | **0.023 ± 0.005** | 0.127 ± 0.033 | – |
| surv_pbc | 0.054 ± 0.086 | 0.022 ± 0.018 | 0.072 ± 0.011 | 0.005 ± 0.007 | 0.006 ± 0.002 | 0.021 ± 0.011 | 0.053 ± 0.012 | 0.093 ± 0.163 | 0.014 ± 0.009 | 0.005 ± 0.001 | **0.004 ± 0.003** | 0.070 ± 0.011 | – |
| surv_pbc3 | 0.045 ± 0.023 | 0.070 ± 0.005 | 0.141 ± 0.039 | 0.022 ± 0.010 | 0.155 ± 0.061 | 0.051 ± 0.009 | 0.182 ± 0.169 | 0.303 ± 0.043 | 0.031 ± 0.002 | 0.023 ± 0.008 | **0.016 ± 0.002** | 0.179 ± 0.015 | 0.041 |
| surv_retinopathy | 0.126 ± 0.376 | 0.019 ± 0.023 | 0.168 ± 0.018 | 0.009 ± 0.004 | 0.006 ± 0.005 | 0.025 ± 0.022 | 0.043 ± 0.044 | 0.092 ± 0.007 | 0.020 ± 0.010 | 0.005 ± 0.004 | **0.004 ± 0.003** | 0.032 ± 0.006 | – |
| surv_whas500 | 0.034 ± 0.030 | 0.075 ± 0.051 | 0.131 ± 0.032 | 0.011 ± 0.001 | 0.063 ± 0.029 | 0.030 ± 0.013 | 0.129 ± 0.116 | 0.244 ± 0.020 | 0.015 ± 0.005 | 0.010 ± 0.001 | **0.008 ± 0.003** | 0.129 ± 0.010 | – |
| surv_gbsg2 | 0.047 ± 0.074 | 0.062 ± 0.044 | 0.173 ± 0.026 | 0.015 ± 0.006 | 0.023 ± 0.008 | 0.081 ± 0.110 | 0.131 ± 0.218 | 0.163 ± 0.058 | 0.016 ± 0.003 | **0.011 ± 0.003** | 0.033 ± 0.002 | 0.258 ± 0.017 | 0.046 |
| surv_colon | 0.012 ± 0.011 | 0.020 ± 0.013 | 0.104 ± 0.051 | 0.003 ± 0.001 | 0.005 ± 0.002 | 0.051 ± 0.064 | 0.104 ± 0.082 | 0.087 ± 0.019 | 0.008 ± 0.004 | **0.003 ± 0.000** | 0.005 ± 0.001 | 0.132 ± 0.017 | – |
| surv_grace | 0.023 ± 0.008 | 0.085 ± 0.052 | 0.120 ± 0.043 | 0.008 ± 0.001 | 0.013 ± 0.005 | 0.069 ± 0.042 | 0.225 ± 0.495 | 0.269 ± 0.055 | 0.015 ± 0.012 | **0.005 ± 0.001** | 0.007 ± 0.001 | 0.149 ± 0.019 | – |
| surv_trace | 0.010 ± 0.012 | 0.055 ± 0.032 | 0.111 ± 0.078 | 0.009 ± 0.001 | **0.003 ± 0.002** | 0.058 ± 0.041 | 0.100 ± 0.047 | 0.180 ± 0.130 | 0.020 ± 0.009 | 0.005 ± 0.000 | 0.006 ± 0.001 | 0.134 ± 0.003 | – |
| surv_framingham | 0.025 ± 0.005 | 0.059 ± 0.052 | 0.095 ± 0.012 | 0.010 ± 0.003 | **0.004 ± 0.002** | 0.060 ± 0.020 | 0.297 ± 0.057 | 0.226 ± 0.165 | 0.035 ± 0.011 | 0.009 ± 0.002 | 0.007 ± 0.002 | 0.205 ± 0.008 | – |
| tab_blood_transfusion | 0.123 ± 0.173 | 0.135 ± 0.061 | 0.193 ± 0.072 | 0.037 ± 0.018 | 0.029 ± 0.004 | 0.173 ± 0.049 | 0.228 ± 0.092 | 0.425 ± 0.105 | 0.043 ± 0.020 | 0.025 ± 0.014 | **0.024 ± 0.002** | 0.333 ± 0.025 | – |
| tab_credit_g | 0.185 ± 0.003 | 0.020 ± 0.013 | 0.166 ± 0.034 | 0.007 ± 0.001 | 0.008 ± 0.002 | 0.019 ± 0.016 | 0.067 ± 0.091 | 0.063 ± 0.006 | 0.021 ± 0.009 | 0.007 ± 0.001 | **0.004 ± 0.000** | 0.143 ± 0.004 | – |
| tab_qsar_biodeg | 0.071 ± 0.009 | 0.065 ± 0.014 | 0.090 ± 0.010 | 0.094 ± 0.027 | 0.509 ± 0.001 | 0.039 ± 0.016 | 0.272 ± 0.226 | 0.144 ± 0.022 | 0.029 ± 0.005 | 0.016 ± 0.003 | **0.008 ± 0.001** | 0.391 ± 0.011 | – |
| tab_maternal_risk | **0.043 ± 0.035** | 0.289 ± 0.069 | 0.264 ± 0.025 | 0.249 ± 0.020 | 0.275 ± 0.119 | 0.235 ± 0.132 | 0.362 ± 0.175 | 0.476 ± 0.047 | 0.140 ± 0.020 | 0.125 ± 0.022 | 0.125 ± 0.015 | 0.360 ± 0.014 | – |
| tab_website_phishing | 0.100 ± 0.394 | 0.224 ± 0.001 | 0.259 ± 0.003 | 0.223 ± 0.000 | 0.001 ± 0.001 | 0.005 ± 0.006 | 0.060 ± 0.052 | 0.003 ± 0.005 | 0.008 ± 0.005 | **0.000 ± 0.000** | 0.001 ± 0.001 | 0.028 ± 0.004 | – |

Table 5: KS statistic (↓) — mean ± 95% CI over seeds

| Dataset | DISCO-TAB | CTGAN | TVAE | GaussCop | TabDDPM | ADS-GAN | PATE-GAN | RTVAE | NFlow | ARF | BayesNet | Marginals | GReaT |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| heart_failure | 0.190 ± 0.134 | 0.201 ± 0.045 | 0.335 ± 0.035 | **0.084 ± 0.022** | 0.262 | 0.162 ± 0.021 | 0.356 ± 0.130 | 0.246 ± 0.036 | 0.101 ± 0.045 | 0.092 ± 0.008 | 0.089 ± 0.061 | 0.313 ± 0.032 | – |
| breast_cancer | 0.206 ± 0.055 | 0.329 ± 0.017 | 0.252 ± 0.085 | 0.103 ± 0.016 | 0.348 | 0.191 ± 0.029 | 0.736 ± 0.296 | 0.432 ± 0.062 | 0.088 ± 0.040 | 0.073 ± 0.035 | **0.059 ± 0.016** | 0.489 ± 0.028 | – |
| liver_disorders | 0.229 ± 0.141 | 0.415 ± 0.127 | 0.265 ± 0.099 | 0.074 ± 0.014 | 0.073 | 0.241 ± 0.021 | 0.278 ± 0.215 | 0.358 ± 0.098 | 0.123 ± 0.082 | **0.071 ± 0.015** | 0.080 ± 0.019 | 0.479 ± 0.063 | – |
| parkinsons | 0.184 ± 0.046 | 0.382 ± 0.051 | 0.268 ± 0.028 | 0.098 ± 0.016 | 0.293 | 0.253 ± 0.069 | 0.301 ± 0.247 | 0.454 ± 0.050 | 0.112 ± 0.017 | 0.106 ± 0.059 | **0.074 ± 0.014** | 0.454 ± 0.037 | – |
| obesity | 0.258 ± 0.011 | 0.239 ± 0.111 | 0.235 ± 0.033 | 0.216 ± 0.024 | 0.156 | 0.217 ± 0.063 | 0.354 ± 0.349 | 0.289 ± 0.023 | 0.161 ± 0.023 | 0.123 ± 0.022 | **0.113 ± 0.003** | 0.305 ± 0.017 | – |
| diabetes | 0.150 ± 0.075 | 0.121 ± 0.025 | 0.097 ± 0.020 | 0.120 ± 0.042 | **0.014 ± 0.003** | 0.129 ± 0.030 | 0.262 ± 0.082 | 0.158 ± 0.058 | 0.130 ± 0.050 | 0.067 ± 0.010 | 0.046 ± 0.001 | 0.400 ± 0.028 | – |
| german_credit | 0.148 ± 0.144 | 0.125 ± 0.034 | 0.381 ± 0.075 | 0.046 ± 0.022 | **0.039 ± 0.002** | 0.099 ± 0.049 | 0.249 ± 0.094 | 0.143 ± 0.012 | 0.095 ± 0.041 | 0.043 ± 0.006 | 0.040 ± 0.003 | 0.359 ± 0.017 | – |
| bank_marketing | 0.104 ± 0.024 | 0.126 ± 0.022 | 0.171 ± 0.025 | 0.212 ± 0.048 | **0.020 ± 0.007** | 0.169 ± 0.144 | 0.414 ± 0.263 | 0.285 ± 0.027 | 0.140 ± 0.018 | 0.061 ± 0.007 | 0.110 ± 0.011 | 0.643 ± 0.015 | – |
| census_income | 0.445 ± 0.094 | 0.189 ± 0.135 | 0.227 ± 0.051 | 0.282 ± 0.273 | **0.023 ± 0.004** | 0.272 ± 0.021 | 0.492 ± 0.639 | 0.211 ± 0.127 | 0.224 ± 0.167 | 0.126 ± 0.009 | 0.197 ± 0.007 | 0.607 ± 0.021 | – |
| surv_melanoma | 0.182 ± 0.297 | 0.257 ± 0.006 | 0.309 ± 0.128 | 0.072 ± 0.017 | 0.079 ± 0.011 | 0.156 ± 0.117 | 0.317 ± 0.285 | 0.239 ± 0.080 | 0.098 ± 0.016 | **0.068 ± 0.015** | 0.071 ± 0.033 | 0.236 ± 0.016 | – |
| surv_pbc | 0.116 ± 0.054 | 0.084 ± 0.025 | 0.193 ± 0.048 | 0.042 ± 0.045 | **0.025 ± 0.014** | 0.119 ± 0.058 | 0.189 ± 0.073 | 0.147 ± 0.295 | 0.103 ± 0.044 | 0.032 ± 0.038 | 0.030 ± 0.019 | 0.174 ± 0.008 | – |
| surv_pbc3 | 0.143 ± 0.067 | 0.216 ± 0.066 | 0.243 ± 0.048 | 0.073 ± 0.015 | 0.205 ± 0.053 | 0.178 ± 0.041 | 0.347 ± 0.210 | 0.302 ± 0.049 | 0.103 ± 0.027 | 0.080 ± 0.013 | **0.057 ± 0.018** | 0.360 ± 0.006 | 0.112 |
| surv_retinopathy | 0.098 ± 0.077 | 0.101 ± 0.087 | 0.370 ± 0.069 | 0.058 ± 0.027 | 0.033 ± 0.016 | 0.107 ± 0.050 | 0.164 ± 0.162 | 0.138 ± 0.025 | 0.105 ± 0.053 | 0.039 ± 0.007 | **0.032 ± 0.015** | 0.174 ± 0.014 | – |
| surv_whas500 | 0.115 ± 0.088 | 0.182 ± 0.072 | 0.244 ± 0.024 | 0.040 ± 0.005 | 0.091 ± 0.034 | 0.092 ± 0.041 | 0.230 ± 0.154 | 0.185 ± 0.060 | 0.065 ± 0.016 | 0.040 ± 0.012 | **0.037 ± 0.004** | 0.268 ± 0.017 | – |
| surv_gbsg2 | 0.158 ± 0.207 | 0.290 ± 0.109 | 0.400 ± 0.060 | 0.132 ± 0.018 | 0.083 ± 0.029 | 0.286 ± 0.231 | 0.397 ± 0.292 | 0.321 ± 0.052 | 0.076 ± 0.012 | 0.095 ± 0.022 | **0.054 ± 0.019** | 0.570 ± 0.040 | 0.214 |
| surv_colon | 0.068 ± 0.031 | 0.100 ± 0.021 | 0.226 ± 0.063 | 0.035 ± 0.014 | **0.023 ± 0.005** | 0.161 ± 0.150 | 0.254 ± 0.205 | 0.102 ± 0.028 | 0.067 ± 0.027 | 0.032 ± 0.004 | 0.045 ± 0.003 | 0.320 ± 0.027 | – |
| surv_grace | 0.087 ± 0.040 | 0.190 ± 0.049 | 0.302 ± 0.057 | 0.040 ± 0.013 | 0.044 ± 0.015 | 0.185 ± 0.116 | 0.363 ± 0.481 | 0.160 ± 0.028 | 0.091 ± 0.063 | **0.030 ± 0.013** | 0.050 ± 0.009 | 0.241 ± 0.024 | – |
| surv_trace | 0.061 ± 0.066 | 0.142 ± 0.066 | 0.227 ± 0.111 | 0.024 ± 0.012 | **0.017 ± 0.003** | 0.110 ± 0.041 | 0.230 ± 0.058 | 0.190 ± 0.231 | 0.086 ± 0.036 | 0.025 ± 0.006 | 0.024 ± 0.003 | 0.285 ± 0.013 | – |
| surv_framingham | 0.079 ± 0.039 | 0.174 ± 0.090 | 0.171 ± 0.035 | 0.035 ± 0.013 | **0.017 ± 0.004** | 0.132 ± 0.066 | 0.436 ± 0.021 | 0.297 ± 0.357 | 0.122 ± 0.056 | 0.042 ± 0.009 | 0.032 ± 0.001 | 0.316 ± 0.018 | – |
| tab_blood_transfusion | 0.264 ± 0.336 | 0.327 ± 0.185 | 0.396 ± 0.072 | 0.131 ± 0.038 | 0.085 ± 0.021 | 0.336 ± 0.008 | 0.458 ± 0.259 | 0.280 ± 0.087 | 0.133 ± 0.064 | **0.067 ± 0.038** | 0.099 ± 0.009 | 0.537 ± 0.029 | – |
| tab_credit_g | 0.110 ± 0.034 | 0.119 ± 0.027 | 0.383 ± 0.102 | **0.040 ± 0.013** | 0.054 ± 0.019 | 0.113 ± 0.075 | 0.273 ± 0.324 | 0.137 ± 0.012 | 0.091 ± 0.024 | 0.056 ± 0.017 | 0.042 ± 0.013 | 0.365 ± 0.012 | – |
| tab_qsar_biodeg | 0.192 ± 0.065 | 0.209 ± 0.022 | 0.204 ± 0.021 | 0.190 ± 0.031 | 0.425 ± 0.035 | 0.131 ± 0.030 | 0.412 ± 0.171 | 0.206 ± 0.019 | 0.105 ± 0.005 | 0.068 ± 0.012 | **0.061 ± 0.003** | 0.573 ± 0.015 | – |
| tab_maternal_risk | 0.234 ± 0.041 | 0.368 ± 0.017 | 0.352 ± 0.033 | 0.268 ± 0.040 | 0.291 ± 0.089 | 0.291 ± 0.258 | 0.435 ± 0.263 | 0.313 ± 0.029 | 0.144 ± 0.039 | **0.111 ± 0.026** | 0.181 ± 0.021 | 0.454 ± 0.010 | – |
| tab_website_phishing | – | – | – | – | – | – | – | – | – | – | – | – | – |

Table 6: KL divergence (↓) — mean ± 95% CI over seeds

| Dataset | DISCO-TAB | CTGAN | TVAE | GaussCop | TabDDPM | ADS-GAN | PATE-GAN | RTVAE | NFlow | ARF | BayesNet | Marginals | GReaT |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| heart_failure | 0.396 ± 0.118 | 0.417 ± 0.141 | 5.073 ± 2.677 | 0.218 ± 0.021 | 0.889 | 0.739 ± 0.149 | 0.988 ± 1.095 | 6.855 ± 1.418 | 0.261 ± 0.126 | **0.189 ± 0.060** | 0.193 ± 0.014 | 0.548 ± 0.087 | – |
| breast_cancer | 0.499 ± 0.214 | 0.531 ± 0.050 | 1.804 ± 0.285 | 0.260 ± 0.236 | 1.136 | 0.522 ± 0.361 | 6.607 ± 11.311 | 11.996 ± 1.070 | 0.173 ± 0.047 | 0.146 ± 0.014 | **0.071 ± 0.009** | 0.801 ± 0.102 | – |
| liver_disorders | 0.432 ± 0.222 | 0.889 ± 0.385 | 3.393 ± 2.232 | 0.304 ± 0.132 | 0.261 | 1.205 ± 0.209 | 0.703 ± 1.668 | 11.831 ± 0.843 | 0.367 ± 0.232 | **0.191 ± 0.103** | 0.253 ± 0.051 | 0.887 ± 0.088 | – |
| parkinsons | 1.233 ± 0.905 | 1.241 ± 0.611 | 2.603 ± 0.351 | 0.430 ± 0.264 | 1.399 | 1.338 ± 0.547 | 0.973 ± 1.641 | 15.105 ± 2.132 | 0.508 ± 0.330 | 0.536 ± 0.018 | **0.396 ± 0.294** | 0.766 ± 0.056 | – |
| obesity | 0.109 ± 0.106 | 0.122 ± 0.045 | 1.345 ± 0.151 | 0.233 ± 0.141 | 0.124 | 0.360 ± 0.562 | 0.456 ± 0.765 | 4.246 ± 0.378 | 0.101 ± 0.038 | 0.089 ± 0.016 | **0.026 ± 0.000** | 0.568 ± 0.006 | – |
| diabetes | 0.104 ± 0.126 | 0.079 ± 0.006 | 0.360 ± 0.468 | 0.175 ± 0.041 | **0.006 ± 0.001** | 0.380 ± 0.574 | 0.584 ± 0.449 | 2.691 ± 0.966 | 0.093 ± 0.045 | 0.088 ± 0.014 | 0.015 ± 0.003 | 0.570 ± 0.056 | – |
| german_credit | 2.010 ± 2.123 | 0.058 ± 0.035 | 3.982 ± 0.208 | 0.036 ± 0.017 | 0.024 ± 0.005 | 0.142 ± 0.176 | 0.180 ± 0.244 | 1.708 ± 0.256 | 0.076 ± 0.052 | 0.030 ± 0.008 | **0.015 ± 0.011** | 0.368 ± 0.027 | – |
| bank_marketing | 0.180 ± 0.117 | 0.028 ± 0.012 | 0.615 ± 0.457 | 0.111 ± 0.106 | **0.014 ± 0.003** | 0.235 ± 0.192 | 0.300 ± 0.530 | 2.197 ± 1.182 | 0.133 ± 0.087 | 0.039 ± 0.005 | 0.039 ± 0.003 | 0.819 ± 0.139 | – |
| census_income | 0.450 ± 0.102 | 0.062 ± 0.018 | 0.987 ± 0.673 | 0.155 ± 0.086 | **0.031 ± 0.014** | 0.463 ± 0.390 | 1.749 ± 6.259 | 3.233 ± 3.534 | 0.257 ± 0.069 | 0.090 ± 0.017 | 0.132 ± 0.081 | 1.137 ± 0.033 | – |
| surv_melanoma | 0.355 ± 0.307 | 0.665 ± 0.551 | 3.288 ± 3.752 | 0.245 ± 0.161 | 0.227 ± 0.221 | 0.812 ± 1.141 | 1.628 ± 4.267 | 8.892 ± 1.341 | 0.200 ± 0.149 | 0.198 ± 0.144 | **0.162 ± 0.109** | 0.343 ± 0.092 | – |
| surv_pbc | 0.545 ± 1.516 | 0.059 ± 0.049 | 2.044 ± 0.410 | 0.025 ± 0.017 | 0.018 ± 0.008 | 0.183 ± 0.314 | 0.157 ± 0.086 | 1.650 ± 3.313 | 0.087 ± 0.091 | 0.031 ± 0.043 | **0.011 ± 0.008** | 0.175 ± 0.028 | – |
| surv_pbc3 | 0.350 ± 0.042 | 0.323 ± 0.159 | 3.822 ± 1.163 | 0.180 ± 0.084 | 0.727 ± 0.314 | 0.488 ± 0.077 | 1.163 ± 2.106 | 7.724 ± 1.264 | 0.231 ± 0.188 | 0.162 ± 0.110 | **0.092 ± 0.032** | 0.502 ± 0.071 | 0.511 |
| surv_retinopathy | 2.306 ± 9.510 | 0.054 ± 0.073 | 4.137 ± 1.474 | 0.026 ± 0.011 | 0.017 ± 0.015 | 0.106 ± 0.085 | 0.178 ± 0.227 | 2.314 ± 0.104 | 0.059 ± 0.027 | 0.016 ± 0.013 | **0.011 ± 0.008** | 0.092 ± 0.022 | – |
| surv_whas500 | 0.258 ± 0.113 | 0.291 ± 0.234 | 3.193 ± 1.167 | 0.091 ± 0.039 | 0.233 ± 0.085 | 0.241 ± 0.108 | 0.495 ± 0.674 | 5.838 ± 0.587 | 0.106 ± 0.104 | 0.064 ± 0.024 | **0.048 ± 0.029** | 0.339 ± 0.018 | – |
| surv_gbsg2 | 0.311 ± 0.195 | 0.311 ± 0.276 | 4.253 ± 1.033 | **0.093 ± 0.049** | 0.115 ± 0.053 | 0.586 ± 0.792 | 0.609 ± 1.635 | 4.183 ± 1.379 | 0.138 ± 0.071 | 0.094 ± 0.033 | 0.153 ± 0.043 | 0.780 ± 0.034 | 1.170 |
| surv_colon | 0.108 ± 0.095 | 0.104 ± 0.067 | 2.855 ± 0.867 | 0.027 ± 0.006 | **0.016 ± 0.003** | 0.273 ± 0.301 | 0.616 ± 0.901 | 2.297 ± 1.226 | 0.031 ± 0.010 | 0.018 ± 0.016 | 0.023 ± 0.005 | 0.350 ± 0.047 | – |
| surv_grace | 0.289 ± 0.166 | 0.338 ± 0.247 | 1.602 ± 1.417 | 0.041 ± 0.015 | 0.088 ± 0.186 | 0.862 ± 1.509 | 2.834 ± 10.957 | 6.324 ± 1.459 | 0.085 ± 0.060 | **0.032 ± 0.015** | 0.034 ± 0.026 | 0.384 ± 0.060 | – |
| surv_trace | 0.093 ± 0.018 | 0.154 ± 0.095 | 2.590 ± 4.084 | 0.024 ± 0.003 | **0.009 ± 0.010** | 0.304 ± 0.308 | 0.298 ± 0.127 | 3.033 ± 1.847 | 0.059 ± 0.026 | 0.015 ± 0.007 | 0.017 ± 0.004 | 0.331 ± 0.009 | – |
| surv_framingham | 0.312 ± 0.015 | 0.157 ± 0.139 | 1.271 ± 0.872 | 0.044 ± 0.023 | **0.011 ± 0.003** | 0.407 ± 0.393 | 1.058 ± 0.287 | 3.481 ± 6.399 | 0.114 ± 0.039 | 0.029 ± 0.010 | 0.021 ± 0.008 | 0.525 ± 0.027 | – |
| tab_blood_transfusion | 1.269 ± 2.769 | 0.563 ± 0.252 | 3.656 ± 1.451 | **0.162 ± 0.038** | 0.199 ± 0.234 | 2.008 ± 0.824 | 1.265 ± 1.770 | 11.252 ± 1.432 | 0.251 ± 0.073 | 0.167 ± 0.107 | 0.164 ± 0.087 | 0.906 ± 0.074 | – |
| tab_credit_g | 3.869 ± 0.059 | 0.071 ± 0.096 | 4.069 ± 0.691 | 0.026 ± 0.008 | 0.027 ± 0.028 | 0.098 ± 0.075 | 0.366 ± 0.862 | 1.565 ± 0.130 | 0.064 ± 0.038 | 0.023 ± 0.004 | **0.015 ± 0.006** | 0.377 ± 0.013 | – |
| tab_qsar_biodeg | 0.776 ± 0.298 | 0.263 ± 0.076 | 2.156 ± 0.388 | 1.629 ± 1.281 | 10.681 ± 1.055 | 0.296 ± 0.175 | 3.983 ± 4.591 | 3.813 ± 0.579 | 0.180 ± 0.084 | 0.109 ± 0.014 | **0.051 ± 0.012** | 1.125 ± 0.028 | – |
| tab_maternal_risk | 0.715 ± 0.658 | 1.003 ± 0.195 | 5.678 ± 1.994 | 1.110 ± 0.781 | 1.200 ± 0.729 | 1.545 ± 1.718 | 3.575 ± 4.765 | 12.353 ± 0.949 | 0.412 ± 0.222 | **0.335 ± 0.050** | 0.452 ± 0.173 | 1.023 ± 0.059 | – |
| tab_website_phishing | 0.854 ± 3.562 | 5.346 ± 0.005 | 5.952 ± 0.319 | 5.343 ± 0.002 | 0.002 ± 0.002 | 0.013 ± 0.017 | 0.194 ± 0.199 | 0.009 ± 0.014 | 0.022 ± 0.011 | **0.001 ± 0.000** | 0.002 ± 0.002 | 0.074 ± 0.012 | – |

Table 7: Hellinger (↓) — mean ± 95% CI over seeds

| Dataset | DISCO-TAB | CTGAN | TVAE | GaussCop | TabDDPM | ADS-GAN | PATE-GAN | RTVAE | NFlow | ARF | BayesNet | Marginals | GReaT |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| heart_failure | 0.183 ± 0.099 | 0.208 ± 0.010 | 0.419 ± 0.047 | 0.137 ± 0.008 | 0.326 | 0.195 ± 0.025 | 0.319 ± 0.127 | 0.376 ± 0.039 | 0.147 ± 0.013 | 0.137 ± 0.016 | **0.134 ± 0.032** | 0.322 ± 0.018 | – |
| breast_cancer | 0.209 ± 0.015 | 0.313 ± 0.013 | 0.297 ± 0.028 | 0.138 ± 0.015 | 0.490 | 0.219 ± 0.031 | 0.668 ± 0.263 | 0.636 ± 0.019 | 0.132 ± 0.033 | 0.111 ± 0.015 | **0.099 ± 0.006** | 0.477 ± 0.027 | – |
| liver_disorders | 0.219 ± 0.094 | 0.360 ± 0.045 | 0.331 ± 0.090 | 0.147 ± 0.013 | 0.147 | 0.290 ± 0.047 | 0.281 ± 0.104 | 0.633 ± 0.042 | 0.176 ± 0.011 | **0.144 ± 0.014** | 0.150 ± 0.013 | 0.501 ± 0.018 | – |
| parkinsons | 0.238 ± 0.079 | 0.357 ± 0.033 | 0.333 ± 0.018 | 0.208 ± 0.028 | 0.427 | 0.301 ± 0.022 | 0.320 ± 0.186 | 0.718 ± 0.051 | 0.195 ± 0.010 | 0.204 ± 0.030 | **0.178 ± 0.033** | 0.453 ± 0.017 | – |
| obesity | 0.098 ± 0.018 | 0.149 ± 0.022 | 0.245 ± 0.024 | 0.133 ± 0.011 | 0.154 | 0.143 ± 0.126 | 0.241 ± 0.170 | 0.277 ± 0.014 | 0.128 ± 0.028 | 0.103 ± 0.011 | **0.059 ± 0.003** | 0.367 ± 0.007 | – |
| diabetes | 0.118 ± 0.043 | 0.135 ± 0.011 | 0.142 ± 0.028 | 0.148 ± 0.005 | **0.027 ± 0.007** | 0.161 ± 0.056 | 0.302 ± 0.127 | 0.208 ± 0.060 | 0.133 ± 0.038 | 0.116 ± 0.006 | 0.051 ± 0.005 | 0.402 ± 0.020 | – |
| german_credit | 0.309 ± 0.604 | 0.087 ± 0.010 | 0.368 ± 0.015 | 0.043 ± 0.005 | 0.049 ± 0.002 | 0.082 ± 0.009 | 0.126 ± 0.043 | 0.138 ± 0.011 | 0.112 ± 0.016 | 0.040 ± 0.008 | **0.039 ± 0.007** | 0.300 ± 0.010 | – |
| bank_marketing | 0.102 ± 0.019 | 0.069 ± 0.021 | 0.163 ± 0.041 | 0.065 ± 0.010 | **0.046 ± 0.015** | 0.112 ± 0.029 | 0.200 ± 0.145 | 0.178 ± 0.048 | 0.140 ± 0.036 | 0.066 ± 0.002 | 0.065 ± 0.011 | 0.416 ± 0.020 | – |
| census_income | 0.182 ± 0.065 | 0.109 ± 0.020 | 0.186 ± 0.037 | 0.111 ± 0.069 | **0.062 ± 0.010** | 0.194 ± 0.067 | 0.308 ± 0.328 | 0.274 ± 0.092 | 0.213 ± 0.050 | 0.124 ± 0.009 | 0.126 ± 0.017 | 0.491 ± 0.007 | – |
| surv_melanoma | 0.189 ± 0.222 | 0.249 ± 0.022 | 0.337 ± 0.113 | 0.137 ± 0.028 | 0.154 ± 0.042 | 0.200 ± 0.101 | 0.328 ± 0.231 | 0.436 ± 0.036 | 0.144 ± 0.019 | 0.120 ± 0.028 | **0.110 ± 0.017** | 0.269 ± 0.044 | – |
| surv_pbc | 0.164 ± 0.146 | 0.081 ± 0.010 | 0.229 ± 0.021 | 0.045 ± 0.043 | 0.039 ± 0.006 | 0.098 ± 0.019 | 0.157 ± 0.078 | 0.153 ± 0.175 | 0.090 ± 0.024 | 0.036 ± 0.011 | **0.035 ± 0.020** | 0.184 ± 0.017 | – |
| surv_pbc3 | 0.165 ± 0.040 | 0.185 ± 0.018 | 0.326 ± 0.052 | 0.106 ± 0.013 | 0.294 ± 0.069 | 0.170 ± 0.025 | 0.311 ± 0.156 | 0.417 ± 0.041 | 0.142 ± 0.005 | 0.110 ± 0.020 | **0.093 ± 0.003** | 0.332 ± 0.016 | 0.158 |
| surv_retinopathy | 0.212 ± 0.343 | 0.074 ± 0.022 | 0.308 ± 0.049 | 0.050 ± 0.004 | 0.040 ± 0.017 | 0.105 ± 0.051 | 0.140 ± 0.067 | 0.144 ± 0.010 | 0.094 ± 0.005 | 0.042 ± 0.015 | **0.037 ± 0.010** | 0.115 ± 0.013 | – |
| surv_whas500 | 0.131 ± 0.069 | 0.172 ± 0.063 | 0.318 ± 0.037 | 0.067 ± 0.005 | 0.164 ± 0.039 | 0.113 ± 0.023 | 0.233 ± 0.122 | 0.332 ± 0.045 | 0.086 ± 0.015 | 0.065 ± 0.005 | **0.060 ± 0.012** | 0.273 ± 0.006 | – |
| surv_gbsg2 | 0.162 ± 0.127 | 0.167 ± 0.048 | 0.379 ± 0.021 | 0.088 ± 0.009 | 0.114 ± 0.021 | 0.206 ± 0.180 | 0.254 ± 0.261 | 0.294 ± 0.049 | 0.100 ± 0.024 | **0.077 ± 0.012** | 0.116 ± 0.014 | 0.396 ± 0.015 | 0.150 |
| surv_colon | 0.081 ± 0.046 | 0.077 ± 0.014 | 0.274 ± 0.071 | 0.038 ± 0.018 | 0.038 ± 0.008 | 0.145 ± 0.107 | 0.187 ± 0.107 | 0.162 ± 0.024 | 0.062 ± 0.017 | **0.031 ± 0.005** | 0.045 ± 0.005 | 0.262 ± 0.017 | – |
| surv_grace | 0.119 ± 0.038 | 0.185 ± 0.053 | 0.282 ± 0.070 | 0.062 ± 0.007 | 0.085 ± 0.021 | 0.193 ± 0.069 | 0.341 ± 0.442 | 0.383 ± 0.049 | 0.094 ± 0.032 | **0.050 ± 0.004** | 0.062 ± 0.007 | 0.271 ± 0.008 | – |
| surv_trace | 0.069 ± 0.065 | 0.145 ± 0.046 | 0.277 ± 0.114 | 0.048 ± 0.011 | **0.032 ± 0.011** | 0.148 ± 0.045 | 0.218 ± 0.079 | 0.275 ± 0.061 | 0.100 ± 0.021 | 0.036 ± 0.005 | 0.040 ± 0.006 | 0.282 ± 0.004 | – |
| surv_framingham | 0.133 ± 0.003 | 0.191 ± 0.094 | 0.206 ± 0.026 | 0.074 ± 0.014 | **0.049 ± 0.012** | 0.190 ± 0.036 | 0.432 ± 0.041 | 0.382 ± 0.152 | 0.150 ± 0.025 | 0.065 ± 0.008 | 0.058 ± 0.003 | 0.359 ± 0.003 | – |
| tab_blood_transfusion | 0.289 ± 0.247 | 0.303 ± 0.106 | 0.399 ± 0.079 | 0.166 ± 0.041 | 0.149 ± 0.017 | 0.364 ± 0.044 | 0.399 ± 0.075 | 0.609 ± 0.078 | 0.178 ± 0.040 | 0.138 ± 0.038 | **0.132 ± 0.014** | 0.499 ± 0.029 | – |
| tab_credit_g | 0.287 ± 0.015 | 0.081 ± 0.030 | 0.365 ± 0.039 | 0.044 ± 0.002 | 0.058 ± 0.015 | 0.086 ± 0.043 | 0.165 ± 0.131 | 0.135 ± 0.002 | 0.099 ± 0.030 | 0.041 ± 0.003 | **0.037 ± 0.006** | 0.304 ± 0.004 | – |
| tab_qsar_biodeg | 0.208 ± 0.006 | 0.190 ± 0.009 | 0.257 ± 0.018 | 0.200 ± 0.021 | 0.602 ± 0.008 | 0.144 ± 0.027 | 0.401 ± 0.195 | 0.290 ± 0.020 | 0.133 ± 0.013 | 0.095 ± 0.008 | **0.069 ± 0.002** | 0.550 ± 0.008 | – |
| tab_maternal_risk | **0.175 ± 0.057** | 0.471 ± 0.056 | 0.470 ± 0.022 | 0.427 ± 0.009 | 0.450 ± 0.103 | 0.397 ± 0.119 | 0.513 ± 0.162 | 0.618 ± 0.026 | 0.304 ± 0.025 | 0.288 ± 0.014 | 0.287 ± 0.019 | 0.538 ± 0.010 | – |
| tab_website_phishing | 0.167 ± 0.437 | 0.247 ± 0.012 | 0.358 ± 0.005 | 0.236 ± 0.007 | 0.019 ± 0.009 | 0.050 ± 0.023 | 0.184 ± 0.070 | 0.039 ± 0.033 | 0.066 ± 0.009 | **0.015 ± 0.004** | 0.020 ± 0.008 | 0.121 ± 0.010 | – |

Table 8: Pearson corr. preservation (↑) — mean ± 95% CI over seeds

| Dataset | DISCO-TAB | CTGAN | TVAE | GaussCop | TabDDPM | ADS-GAN | PATE-GAN | RTVAE | NFlow | ARF | BayesNet | Marginals | GReaT |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| heart_failure | 0.927 ± 0.008 | 0.904 ± 0.030 | 0.926 ± 0.013 | 0.930 ± 0.012 | 0.822 | 0.912 ± 0.018 | 0.844 ± 0.042 | 0.918 ± 0.012 | 0.908 ± 0.011 | **0.939 ± 0.019** | 0.907 ± 0.037 | 0.900 ± 0.005 | – |
| breast_cancer | 0.868 ± 0.033 | 0.606 ± 0.006 | 0.799 ± 0.009 | **0.948 ± 0.018** | 0.773 | 0.887 ± 0.039 | 0.566 ± 0.047 | 0.753 ± 0.028 | 0.694 ± 0.048 | 0.899 ± 0.008 | 0.945 ± 0.028 | 0.607 ± 0.012 | – |
| liver_disorders | 0.917 ± 0.103 | 0.744 ± 0.034 | 0.831 ± 0.125 | 0.948 ± 0.005 | 0.912 | 0.862 ± 0.092 | 0.640 ± 0.144 | 0.734 ± 0.068 | 0.814 ± 0.021 | **0.949 ± 0.010** | 0.899 ± 0.063 | 0.743 ± 0.034 | – |
| parkinsons | 0.845 ± 0.085 | 0.487 ± 0.042 | 0.653 ± 0.012 | 0.936 ± 0.010 | 0.739 | 0.814 ± 0.054 | 0.481 ± 0.057 | 0.606 ± 0.043 | 0.610 ± 0.052 | 0.863 ± 0.021 | **0.950 ± 0.006** | 0.486 ± 0.030 | – |
| obesity | 0.945 ± 0.024 | 0.871 ± 0.007 | 0.895 ± 0.014 | 0.941 ± 0.014 | 0.847 | 0.953 ± 0.088 | 0.856 ± 0.053 | 0.923 ± 0.004 | 0.892 ± 0.022 | 0.964 ± 0.021 | **0.974 ± 0.008** | 0.874 ± 0.010 | – |
| diabetes | 0.928 ± 0.102 | 0.898 ± 0.033 | 0.934 ± 0.037 | 0.916 ± 0.011 | **0.986 ± 0.004** | 0.968 ± 0.008 | 0.882 ± 0.039 | 0.905 ± 0.015 | 0.918 ± 0.032 | 0.982 ± 0.005 | 0.956 ± 0.005 | 0.868 ± 0.013 | – |
| german_credit | 0.926 ± 0.061 | 0.896 ± 0.015 | 0.885 ± 0.052 | 0.933 ± 0.020 | 0.923 ± 0.057 | 0.933 ± 0.023 | 0.879 ± 0.022 | 0.927 ± 0.029 | 0.905 ± 0.004 | 0.953 ± 0.012 | **0.961 ± 0.022** | 0.897 ± 0.008 | – |
| bank_marketing | 0.952 ± 0.025 | 0.943 ± 0.028 | 0.896 ± 0.062 | 0.966 ± 0.022 | 0.824 ± 0.174 | 0.969 ± 0.016 | 0.922 ± 0.008 | 0.969 ± 0.031 | 0.947 ± 0.006 | **0.979 ± 0.014** | 0.965 ± 0.011 | 0.940 ± 0.002 | – |
| census_income | 0.951 ± 0.029 | 0.953 ± 0.018 | 0.942 ± 0.004 | 0.980 ± 0.005 | 0.864 ± 0.038 | 0.970 ± 0.013 | 0.914 ± 0.030 | 0.961 ± 0.014 | 0.946 ± 0.021 | 0.980 ± 0.007 | **0.982 ± 0.003** | 0.941 ± 0.005 | – |
| surv_melanoma | 0.910 ± 0.054 | 0.832 ± 0.032 | 0.888 ± 0.035 | 0.872 ± 0.060 | 0.889 ± 0.034 | 0.855 ± 0.119 | 0.794 ± 0.028 | 0.848 ± 0.022 | 0.865 ± 0.109 | **0.935 ± 0.015** | 0.924 ± 0.027 | 0.827 ± 0.065 | – |
| surv_pbc | 0.877 ± 0.148 | 0.859 ± 0.043 | 0.709 ± 0.220 | 0.897 ± 0.048 | 0.941 ± 0.015 | 0.924 ± 0.023 | 0.862 ± 0.055 | 0.887 ± 0.034 | 0.897 ± 0.063 | 0.938 ± 0.050 | **0.943 ± 0.037** | 0.861 ± 0.029 | – |
| surv_pbc3 | 0.912 ± 0.031 | 0.864 ± 0.011 | 0.883 ± 0.051 | 0.921 ± 0.017 | 0.860 ± 0.027 | 0.916 ± 0.030 | 0.857 ± 0.023 | 0.889 ± 0.007 | 0.882 ± 0.014 | 0.939 ± 0.011 | **0.942 ± 0.010** | 0.863 ± 0.021 | 0.857 |
| surv_retinopathy | 0.958 ± 0.041 | 0.781 ± 0.063 | 0.820 ± 0.177 | 0.824 ± 0.110 | 0.959 ± 0.060 | 0.868 ± 0.017 | 0.784 ± 0.037 | 0.852 ± 0.077 | 0.813 ± 0.027 | 0.899 ± 0.040 | **0.964 ± 0.079** | 0.779 ± 0.034 | – |
| surv_whas500 | 0.918 ± 0.025 | 0.891 ± 0.015 | 0.903 ± 0.044 | 0.931 ± 0.010 | 0.891 ± 0.039 | 0.931 ± 0.011 | 0.870 ± 0.023 | 0.917 ± 0.003 | 0.914 ± 0.010 | 0.942 ± 0.003 | **0.944 ± 0.011** | 0.897 ± 0.009 | – |
| surv_gbsg2 | 0.902 ± 0.088 | 0.853 ± 0.040 | 0.871 ± 0.062 | 0.920 ± 0.042 | 0.820 ± 0.186 | 0.882 ± 0.116 | 0.803 ± 0.059 | 0.884 ± 0.081 | 0.887 ± 0.048 | **0.949 ± 0.054** | 0.921 ± 0.036 | 0.848 ± 0.018 | 0.886 |
| surv_colon | 0.957 ± 0.026 | 0.921 ± 0.013 | 0.807 ± 0.061 | 0.942 ± 0.032 | 0.961 ± 0.015 | 0.915 ± 0.062 | 0.911 ± 0.004 | 0.947 ± 0.005 | 0.931 ± 0.009 | **0.967 ± 0.018** | 0.940 ± 0.006 | 0.917 ± 0.013 | – |
| surv_grace | 0.951 ± 0.031 | 0.916 ± 0.036 | 0.873 ± 0.061 | 0.958 ± 0.006 | 0.947 ± 0.061 | 0.942 ± 0.031 | 0.883 ± 0.074 | 0.918 ± 0.014 | 0.950 ± 0.036 | **0.978 ± 0.009** | 0.923 ± 0.038 | 0.919 ± 0.041 | – |
| surv_trace | 0.965 ± 0.018 | 0.870 ± 0.003 | 0.907 ± 0.023 | 0.930 ± 0.005 | **0.982 ± 0.007** | 0.931 ± 0.058 | 0.853 ± 0.021 | 0.877 ± 0.046 | 0.924 ± 0.057 | 0.977 ± 0.017 | 0.978 ± 0.014 | 0.876 ± 0.009 | – |
| surv_framingham | 0.969 ± 0.012 | 0.846 ± 0.031 | 0.740 ± 0.050 | **0.983 ± 0.004** | 0.968 ± 0.026 | 0.964 ± 0.004 | 0.792 ± 0.016 | 0.887 ± 0.017 | 0.883 ± 0.025 | 0.980 ± 0.005 | 0.921 ± 0.015 | 0.833 ± 0.006 | – |
| tab_blood_transfusion | 0.683 ± 0.613 | 0.548 ± 0.016 | 0.729 ± 0.060 | 0.931 ± 0.038 | **0.969 ± 0.051** | 0.880 ± 0.110 | 0.576 ± 0.145 | 0.680 ± 0.163 | 0.731 ± 0.142 | 0.934 ± 0.039 | 0.864 ± 0.076 | 0.549 ± 0.092 | – |
| tab_credit_g | 0.959 ± 0.033 | 0.892 ± 0.024 | 0.904 ± 0.027 | 0.933 ± 0.021 | 0.905 ± 0.079 | 0.930 ± 0.033 | 0.889 ± 0.033 | 0.930 ± 0.002 | 0.903 ± 0.016 | 0.954 ± 0.025 | **0.966 ± 0.006** | 0.888 ± 0.009 | – |

| Dataset | DISCO-TAB | CTGAN | TVAE | GaussCop | TabDDPM | ADS-GAN | PATE-GAN | RTVAE | NFlow | ARF | BayesNet | Marginals | GReaT |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| tab_qsar_biodeg | 0.791 ± 0.045 | 0.829 ± 0.002 | 0.856 ± 0.036 | 0.912 ± 0.013 | 0.822 ± 0.026 | 0.906 ± 0.010 | 0.809 ± 0.025 | 0.873 ± 0.002 | 0.837 ± 0.014 | 0.933 ± 0.009 | **0.966 ± 0.006** | 0.829 ± 0.003 | – |
| tab_maternal_risk | 0.916 ± 0.059 | 0.755 ± 0.008 | 0.832 ± 0.035 | 0.929 ± 0.017 | 0.794 ± 0.155 | 0.906 ± 0.081 | 0.751 ± 0.036 | 0.882 ± 0.039 | 0.822 ± 0.033 | **0.957 ± 0.009** | 0.910 ± 0.048 | 0.750 ± 0.035 | – |
| tab_website_phishing | – | – | – | – | – | – | – | – | – | – | – | – | – |

Table 9: JS numeric (↓) — mean ± 95% CI over seeds

| Dataset | DISCO-TAB | CTGAN | TVAE | GaussCop | TabDDPM | ADS-GAN | PATE-GAN | RTVAE | NFlow | ARF | BayesNet | Marginals | GReaT |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| heart_failure | 0.054 ± 0.054 | 0.094 ± 0.020 | 0.200 ± 0.033 | 0.042 ± 0.002 | 0.194 | 0.070 ± 0.006 | 0.192 ± 0.132 | 0.265 ± 0.048 | 0.040 ± 0.004 | 0.041 ± 0.007 | **0.034 ± 0.007** | 0.189 ± 0.014 | – |
| breast_cancer | 0.062 ± 0.010 | 0.143 ± 0.007 | 0.111 ± 0.022 | 0.027 ± 0.011 | 0.294 | 0.072 ± 0.019 | 0.568 ± 0.357 | 0.466 ± 0.024 | 0.024 ± 0.014 | 0.016 ± 0.005 | **0.013 ± 0.002** | 0.293 ± 0.032 | – |
| liver_disorders | 0.068 ± 0.052 | 0.184 ± 0.038 | 0.129 ± 0.058 | 0.028 ± 0.005 | 0.028 | 0.113 ± 0.031 | 0.118 ± 0.095 | 0.450 ± 0.074 | 0.040 ± 0.004 | **0.026 ± 0.002** | 0.029 ± 0.001 | 0.316 ± 0.019 | – |
| parkinsons | 0.082 ± 0.044 | 0.175 ± 0.029 | 0.136 ± 0.013 | 0.056 ± 0.013 | 0.227 | 0.120 ± 0.015 | 0.143 ± 0.175 | 0.578 ± 0.075 | 0.049 ± 0.004 | 0.053 ± 0.016 | **0.039 ± 0.013** | 0.269 ± 0.016 | – |
| obesity | 0.040 ± 0.003 | 0.081 ± 0.036 | 0.136 ± 0.030 | 0.114 ± 0.022 | 0.072 | 0.093 ± 0.144 | 0.237 ± 0.334 | 0.313 ± 0.013 | 0.052 ± 0.014 | 0.059 ± 0.008 | **0.019 ± 0.002** | 0.206 ± 0.008 | – |
| diabetes | 0.028 ± 0.022 | 0.033 ± 0.003 | 0.030 ± 0.011 | 0.089 ± 0.009 | **0.002 ± 0.001** | 0.082 ± 0.054 | 0.216 ± 0.114 | 0.125 ± 0.072 | 0.036 ± 0.022 | 0.052 ± 0.007 | 0.009 ± 0.002 | 0.248 ± 0.024 | – |
| german_credit | 0.050 ± 0.076 | 0.047 ± 0.021 | 0.202 ± 0.060 | 0.023 ± 0.009 | 0.017 ± 0.001 | 0.044 ± 0.004 | 0.111 ± 0.054 | 0.181 ± 0.026 | 0.031 ± 0.014 | 0.019 ± 0.010 | **0.013 ± 0.011** | 0.160 ± 0.013 | – |
| bank_marketing | 0.019 ± 0.006 | 0.010 ± 0.006 | 0.043 ± 0.030 | 0.035 ± 0.013 | 0.007 ± 0.007 | 0.027 ± 0.013 | 0.098 ± 0.163 | 0.123 ± 0.003 | 0.025 ± 0.015 | 0.008 ± 0.002 | **0.004 ± 0.002** | 0.454 ± 0.076 | – |
| census_income | 0.052 ± 0.049 | 0.020 ± 0.008 | 0.071 ± 0.036 | 0.115 ± 0.139 | **0.010 ± 0.003** | 0.069 ± 0.034 | 0.188 ± 0.498 | 0.139 ± 0.098 | 0.046 ± 0.023 | 0.033 ± 0.004 | 0.019 ± 0.002 | 0.445 ± 0.025 | – |
| surv_melanoma | 0.064 ± 0.119 | 0.118 ± 0.005 | 0.147 ± 0.076 | 0.038 ± 0.013 | 0.044 ± 0.022 | 0.077 ± 0.079 | 0.187 ± 0.239 | 0.331 ± 0.072 | 0.033 ± 0.005 | 0.027 ± 0.016 | **0.023 ± 0.005** | 0.127 ± 0.033 | – |
| surv_pbc | 0.039 ± 0.074 | 0.026 ± 0.022 | 0.074 ± 0.014 | 0.006 ± 0.008 | 0.007 ± 0.003 | 0.025 ± 0.014 | 0.062 ± 0.012 | 0.111 ± 0.198 | 0.016 ± 0.012 | 0.005 ± 0.001 | **0.005 ± 0.003** | 0.058 ± 0.005 | – |
| surv_pbc3 | 0.042 ± 0.028 | 0.076 ± 0.006 | 0.126 ± 0.042 | 0.024 ± 0.011 | 0.166 ± 0.064 | 0.055 ± 0.010 | 0.197 ± 0.181 | 0.330 ± 0.047 | 0.030 ± 0.003 | 0.024 ± 0.009 | **0.017 ± 0.002** | 0.194 ± 0.016 | 0.043 |
| surv_retinopathy | 0.018 ± 0.007 | 0.032 ± 0.042 | 0.269 ± 0.021 | 0.016 ± 0.007 | 0.009 ± 0.007 | 0.036 ± 0.023 | 0.065 ± 0.082 | 0.159 ± 0.013 | 0.031 ± 0.020 | 0.009 ± 0.008 | **0.007 ± 0.005** | 0.054 ± 0.012 | – |
| surv_whas500 | 0.034 ± 0.030 | 0.075 ± 0.051 | 0.131 ± 0.032 | 0.011 ± 0.001 | 0.063 ± 0.029 | 0.030 ± 0.013 | 0.129 ± 0.116 | 0.244 ± 0.020 | 0.015 ± 0.005 | 0.010 ± 0.001 | **0.008 ± 0.003** | 0.129 ± 0.010 | – |
| surv_gbsg2 | 0.038 ± 0.057 | 0.098 ± 0.071 | 0.193 ± 0.039 | 0.023 ± 0.010 | 0.035 ± 0.012 | 0.109 ± 0.127 | 0.198 ± 0.306 | 0.260 ± 0.092 | 0.022 ± 0.004 | 0.017 ± 0.005 | **0.014 ± 0.006** | 0.392 ± 0.027 | 0.069 |
| surv_colon | 0.011 ± 0.007 | 0.026 ± 0.016 | 0.103 ± 0.054 | 0.004 ± 0.001 | 0.006 ± 0.002 | 0.056 ± 0.077 | 0.128 ± 0.102 | 0.110 ± 0.024 | 0.009 ± 0.006 | **0.004 ± 0.000** | 0.006 ± 0.000 | 0.145 ± 0.019 | – |
| surv_grace | 0.023 ± 0.008 | 0.085 ± 0.052 | 0.120 ± 0.043 | 0.008 ± 0.001 | 0.013 ± 0.005 | 0.069 ± 0.042 | 0.225 ± 0.495 | 0.269 ± 0.055 | 0.015 ± 0.012 | **0.005 ± 0.001** | 0.007 ± 0.001 | 0.149 ± 0.019 | – |
| surv_trace | 0.010 ± 0.012 | 0.055 ± 0.032 | 0.111 ± 0.078 | 0.009 ± 0.001 | **0.003 ± 0.002** | 0.058 ± 0.041 | 0.100 ± 0.047 | 0.180 ± 0.130 | 0.020 ± 0.009 | 0.005 ± 0.000 | 0.006 ± 0.001 | 0.134 ± 0.003 | – |
| surv_framingham | 0.025 ± 0.005 | 0.059 ± 0.052 | 0.095 ± 0.012 | 0.010 ± 0.003 | **0.004 ± 0.002** | 0.060 ± 0.020 | 0.297 ± 0.057 | 0.226 ± 0.165 | 0.035 ± 0.011 | 0.009 ± 0.002 | 0.007 ± 0.002 | 0.205 ± 0.008 | – |
| tab_blood_transfusion | 0.123 ± 0.173 | 0.135 ± 0.061 | 0.193 ± 0.072 | 0.037 ± 0.018 | 0.029 ± 0.004 | 0.173 ± 0.049 | 0.228 ± 0.092 | 0.425 ± 0.105 | 0.043 ± 0.020 | 0.025 ± 0.014 | **0.024 ± 0.002** | 0.333 ± 0.025 | – |
| tab_credit_g | 0.026 ± 0.012 | 0.046 ± 0.025 | 0.193 ± 0.072 | 0.019 ± 0.002 | 0.017 ± 0.002 | 0.043 ± 0.037 | 0.145 ± 0.226 | 0.170 ± 0.018 | 0.029 ± 0.008 | 0.019 ± 0.002 | **0.010 ± 0.002** | 0.166 ± 0.014 | – |
| tab_qsar_biodeg | 0.071 ± 0.009 | 0.065 ± 0.014 | 0.090 ± 0.010 | 0.094 ± 0.027 | 0.509 ± 0.001 | 0.039 ± 0.016 | 0.272 ± 0.226 | 0.144 ± 0.022 | 0.029 ± 0.005 | 0.016 ± 0.003 | **0.008 ± 0.001** | 0.391 ± 0.011 | – |
| tab_maternal_risk | **0.043 ± 0.035** | 0.289 ± 0.069 | 0.264 ± 0.025 | 0.249 ± 0.020 | 0.275 ± 0.119 | 0.235 ± 0.132 | 0.362 ± 0.175 | 0.476 ± 0.047 | 0.140 ± 0.020 | 0.125 ± 0.022 | 0.125 ± 0.015 | 0.360 ± 0.014 | – |
| tab_website_phishing | – | – | – | – | – | – | – | – | – | – | – | – | – |

Table 10: JS categorical (↓) — mean ± 95% CI over seeds

| Dataset | DISCO-TAB | CTGAN | TVAE | GaussCop | TabDDPM | ADS-GAN | PATE-GAN | RTVAE | NFlow | ARF | BayesNet | Marginals | GReaT |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| heart_failure | – | – | – | – | – | – | – | – | – | – | – | – | – |
| breast_cancer | – | – | – | – | – | – | – | – | – | – | – | – | – |
| liver_disorders | – | – | – | – | – | – | – | – | – | – | – | – | – |
| parkinsons | – | – | – | – | – | – | – | – | – | – | – | – | – |
| obesity | 0.004 ± 0.003 | 0.006 ± 0.003 | 0.044 ± 0.001 | **0.000 ± 0.000** | 0.015 | 0.002 ± 0.005 | 0.015 ± 0.018 | 0.002 ± 0.003 | 0.014 ± 0.018 | 0.001 ± 0.001 | 0.000 ± 0.001 | 0.195 ± 0.012 | – |
| diabetes | 0.010 ± 0.015 | 0.020 ± 0.013 | 0.049 ± 0.025 | **0.000 ± 0.000** | 0.000 ± 0.000 | 0.003 ± 0.005 | 0.047 ± 0.104 | 0.002 ± 0.005 | 0.022 ± 0.028 | 0.000 ± 0.000 | 0.000 ± 0.000 | 0.155 ± 0.009 | – |
| german_credit | 0.290 ± 0.844 | 0.005 ± 0.001 | 0.153 ± 0.023 | 0.001 ± 0.000 | 0.001 ± 0.000 | 0.005 ± 0.000 | 0.011 ± 0.010 | 0.006 ± 0.002 | 0.019 ± 0.005 | **0.001 ± 0.000** | 0.001 ± 0.000 | 0.129 ± 0.007 | – |
| bank_marketing | 0.014 ± 0.005 | 0.008 ± 0.004 | 0.038 ± 0.012 | **0.000 ± 0.000** | 0.003 ± 0.001 | 0.050 ± 0.027 | 0.079 ± 0.070 | 0.052 ± 0.061 | 0.051 ± 0.040 | 0.018 ± 0.002 | 0.023 ± 0.000 | 0.135 ± 0.003 | – |
| census_income | 0.049 ± 0.032 | 0.018 ± 0.014 | 0.043 ± 0.014 | **0.000 ± 0.000** | 0.006 ± 0.003 | 0.092 ± 0.109 | 0.177 ± 0.239 | 0.150 ± 0.195 | 0.114 ± 0.036 | 0.040 ± 0.009 | 0.076 ± 0.034 | 0.274 ± 0.003 | – |
| surv_melanoma | – | – | – | – | – | – | – | – | – | – | – | – | – |
| surv_pbc | 0.129 ± 0.460 | 0.004 ± 0.004 | 0.058 ± 0.008 | 0.002 ± 0.004 | 0.000 ± 0.001 | 0.004 ± 0.006 | 0.012 ± 0.019 | 0.007 ± 0.016 | 0.005 ± 0.011 | **0.000 ± 0.001** | 0.001 ± 0.002 | 0.129 ± 0.048 | – |
| surv_pbc3 | 0.077 ± 0.039 | 0.003 ± 0.005 | 0.307 ± 0.035 | **0.001 ± 0.002** | 0.028 ± 0.051 | 0.008 ± 0.016 | 0.023 ± 0.037 | 0.008 ± 0.013 | 0.050 ± 0.013 | 0.005 ± 0.003 | 0.003 ± 0.002 | 0.007 ± 0.007 | 0.014 |
| surv_retinopathy | 0.270 ± 0.868 | 0.001 ± 0.001 | 0.034 ± 0.051 | **0.000 ± 0.000** | 0.001 ± 0.001 | 0.011 ± 0.025 | 0.013 ± 0.017 | 0.001 ± 0.005 | 0.006 ± 0.006 | 0.001 ± 0.001 | 0.001 ± 0.001 | 0.003 ± 0.003 | – |
| surv_whas500 | – | – | – | – | – | – | – | – | – | – | – | – | – |
| surv_gbsg2 | 0.061 ± 0.202 | 0.002 ± 0.004 | 0.139 ± 0.009 | 0.001 ± 0.001 | 0.001 ± 0.001 | 0.033 ± 0.083 | 0.018 ± 0.070 | 0.001 ± 0.001 | 0.005 ± 0.013 | **0.000 ± 0.001** | 0.066 ± 0.004 | 0.035 ± 0.016 | 0.007 |
| surv_colon | 0.012 ± 0.022 | 0.002 ± 0.001 | 0.108 ± 0.045 | 0.001 ± 0.001 | **0.001 ± 0.001** | 0.032 ± 0.034 | 0.018 ± 0.019 | 0.006 ± 0.006 | 0.007 ± 0.010 | 0.001 ± 0.001 | 0.002 ± 0.002 | 0.087 ± 0.010 | – |
| surv_grace | – | – | – | – | – | – | – | – | – | – | – | – | – |
| surv_trace | – | – | – | – | – | – | – | – | – | – | – | – | – |
| surv_framingham | – | – | – | – | – | – | – | – | – | – | – | – | – |
| tab_blood_transfusion | – | – | – | – | – | – | – | – | – | – | – | – | – |
| tab_credit_g | 0.271 ± 0.003 | 0.006 ± 0.007 | 0.151 ± 0.015 | 0.001 ± 0.000 | 0.003 ± 0.002 | 0.006 ± 0.007 | 0.025 ± 0.044 | 0.005 ± 0.001 | 0.017 ± 0.016 | **0.001 ± 0.000** | 0.001 ± 0.001 | 0.131 ± 0.001 | – |
| tab_qsar_biodeg | – | – | – | – | – | – | – | – | – | – | – | – | – |
| tab_maternal_risk | – | – | – | – | – | – | – | – | – | – | – | – | – |
| tab_website_phishing | 0.100 ± 0.394 | 0.224 ± 0.001 | 0.259 ± 0.003 | 0.223 ± 0.000 | 0.001 ± 0.001 | 0.005 ± 0.006 | 0.060 ± 0.052 | 0.003 ± 0.005 | 0.008 ± 0.005 | **0.000 ± 0.000** | 0.001 ± 0.001 | 0.028 ± 0.004 | – |

## Privacy / disclosure

Table 11: DCR median (↑=less memorization) — mean ± 95% CI over seeds

| Dataset | DISCO-TAB | CTGAN | TVAE | GaussCop | TabDDPM | ADS-GAN | PATE-GAN | RTVAE | NFlow | ARF | BayesNet | Marginals | GReaT |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| heart_failure | 0.731 ± 0.064 | 0.869 ± 0.158 | 0.553 ± 0.234 | 0.736 ± 0.007 | **1.971** | 0.703 ± 0.090 | 1.043 ± 0.312 | 0.617 ± 0.114 | 0.715 ± 0.050 | 0.737 ± 0.022 | 0.722 ± 0.072 | 1.873 ± 0.111 | – |
| breast_cancer | 0.550 ± 0.162 | 1.245 ± 0.120 | 0.476 ± 0.047 | 0.555 ± 0.037 | **2.769** | 0.672 ± 0.083 | 1.726 ± 0.372 | 0.433 ± 0.045 | 0.850 ± 0.176 | 0.607 ± 0.027 | 0.488 ± 0.083 | 2.494 ± 0.208 | – |
| liver_disorders | 0.412 ± 0.158 | 0.682 ± 0.169 | 0.240 ± 0.013 | 0.382 ± 0.048 | 0.307 | 0.316 ± 0.036 | 0.570 ± 0.144 | 0.379 ± 0.087 | 0.376 ± 0.124 | 0.355 ± 0.033 | 0.359 ± 0.060 | **1.677 ± 0.319** | – |
| parkinsons | 0.483 ± 0.222 | 1.033 ± 0.097 | 0.510 ± 0.052 | 0.538 ± 0.063 | 0.928 | 0.625 ± 0.124 | 1.019 ± 0.193 | 0.457 ± 0.028 | 0.742 ± 0.102 | 0.570 ± 0.060 | 0.437 ± 0.022 | **1.923 ± 0.271** | – |
| obesity | 0.277 ± 0.035 | 0.437 ± 0.042 | 0.247 ± 0.025 | 0.403 ± 0.032 | 0.325 | 0.342 ± 0.088 | 0.471 ± 0.041 | 0.349 ± 0.023 | 0.426 ± 0.019 | 0.305 ± 0.007 | 0.269 ± 0.015 | **0.706 ± 0.031** | – |
| diabetes | 0.053 ± 0.042 | 0.082 ± 0.021 | 0.038 ± 0.009 | 0.116 ± 0.021 | 0.056 ± 0.001 | 0.065 ± 0.008 | 0.118 ± 0.169 | 0.068 ± 0.041 | 0.081 ± 0.021 | 0.079 ± 0.001 | 0.060 ± 0.003 | **0.545 ± 0.125** | – |
| german_credit | 0.432 ± 0.113 | 0.456 ± 0.004 | 0.308 ± 0.040 | 0.426 ± 0.008 | 0.392 ± 0.005 | 0.418 ± 0.023 | 0.431 ± 0.001 | 0.402 ± 0.022 | 0.446 ± 0.024 | 0.415 ± 0.004 | 0.399 ± 0.007 | **0.642 ± 0.028** | – |
| bank_marketing | 0.160 ± 0.082 | 0.254 ± 0.020 | 0.096 ± 0.006 | 0.269 ± 0.015 | 0.211 ± 0.002 | 0.250 ± 0.039 | 0.331 ± 0.128 | 0.226 ± 0.033 | 0.289 ± 0.037 | 0.247 ± 0.008 | 0.243 ± 0.009 | **2.937 ± 5.289** | – |

| Dataset | DISCO-TAB | CTGAN | TVAE | GaussCop | TabDDPM | ADS-GAN | PATE-GAN | RTVAE | NFlow | ARF | BayesNet | Marginals | GReaT |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| census_income | $0.087 \pm 0.058$ | $0.195 \pm 0.012$ | $0.044 \pm 0.007$ | $0.254 \pm 0.137$ | $0.140 \pm 0.000$ | $0.188 \pm 0.056$ | $0.247 \pm 0.050$ | $0.189 \pm 0.077$ | $0.222 \pm 0.039$ | $0.195 \pm 0.005$ | $0.196 \pm 0.001$ | $\mathbf{0.683 \pm 0.097}$ | – |
| surv_melanoma | $0.364 \pm 0.093$ | $0.655 \pm 0.209$ | $0.322 \pm 0.130$ | $0.404 \pm 0.106$ | $0.360 \pm 0.028$ | $0.375 \pm 0.047$ | $0.748 \pm 0.091$ | $0.348 \pm 0.097$ | $0.419 \pm 0.084$ | $0.376 \pm 0.051$ | $0.379 \pm 0.035$ | $\mathbf{0.891 \pm 0.076}$ | – |
| surv_pbc | $0.247 \pm 0.707$ | $0.201 \pm 0.064$ | $0.080 \pm 0.065$ | $0.130 \pm 0.105$ | $0.080 \pm 0.006$ | $0.075 \pm 0.031$ | $0.158 \pm 0.152$ | $0.242 \pm 0.773$ | $0.084 \pm 0.007$ | $0.092 \pm 0.021$ | $0.087 \pm 0.011$ | $\mathbf{0.571 \pm 0.057}$ | – |
| surv_pbc3 | $0.568 \pm 0.107$ | $0.722 \pm 0.037$ | $0.457 \pm 0.013$ | $0.602 \pm 0.006$ | $1.112 \pm 0.474$ | $0.600 \pm 0.044$ | $0.809 \pm 0.259$ | $0.464 \pm 0.025$ | $0.575 \pm 0.060$ | $0.606 \pm 0.042$ | $0.598 \pm 0.033$ | $\mathbf{1.414 \pm 0.109}$ | 0.524 |
| surv_retinopathy | $0.342 \pm 0.315$ | $0.472 \pm 0.055$ | $0.111 \pm 0.107$ | $0.372 \pm 0.134$ | $0.259 \pm 0.114$ | $0.300 \pm 0.012$ | $0.454 \pm 0.134$ | $0.321 \pm 0.021$ | $0.373 \pm 0.161$ | $0.285 \pm 0.047$ | $0.230 \pm 0.050$ | $\mathbf{0.542 \pm 0.036}$ | – |
| surv_whas500 | $0.643 \pm 0.102$ | $0.934 \pm 0.105$ | $0.480 \pm 0.138$ | $0.722 \pm 0.026$ | $0.725 \pm 0.036$ | $0.678 \pm 0.070$ | $1.125 \pm 0.123$ | $0.571 \pm 0.116$ | $0.702 \pm 0.018$ | $0.701 \pm 0.039$ | $0.692 \pm 0.003$ | $\mathbf{1.822 \pm 0.304}$ | – |
| surv_gbsg2 | $0.311 \pm 0.142$ | $0.450 \pm 0.042$ | $0.124 \pm 0.012$ | $0.325 \pm 0.025$ | $0.251 \pm 0.027$ | $0.324 \pm 0.187$ | $0.534 \pm 0.309$ | $0.261 \pm 0.049$ | $0.334 \pm 0.046$ | $0.306 \pm 0.024$ | $0.371 \pm 0.052$ | $\mathbf{1.615 \pm 0.146}$ | 0.242 |
| surv_colon | $0.185 \pm 0.081$ | $0.368 \pm 0.082$ | $0.074 \pm 0.006$ | $0.258 \pm 0.026$ | $0.193 \pm 0.051$ | $0.271 \pm 0.292$ | $0.430 \pm 0.059$ | $0.142 \pm 0.079$ | $0.206 \pm 0.107$ | $0.215 \pm 0.049$ | $0.202 \pm 0.019$ | $\mathbf{1.057 \pm 0.203}$ | – |
| surv_grace | $0.228 \pm 0.035$ | $0.354 \pm 0.236$ | $0.177 \pm 0.029$ | $0.233 \pm 0.028$ | $0.221 \pm 0.019$ | $0.207 \pm 0.032$ | $0.676 \pm 0.890$ | $0.191 \pm 0.019$ | $0.209 \pm 0.014$ | $0.231 \pm 0.022$ | $0.236 \pm 0.010$ | $\mathbf{0.975 \pm 0.106}$ | – |
| surv_trace | $0.081 \pm 0.036$ | $0.233 \pm 0.031$ | $0.069 \pm 0.073$ | $0.101 \pm 0.001$ | $0.087 \pm 0.009$ | $0.072 \pm 0.025$ | $0.230 \pm 0.356$ | $0.238 \pm 0.562$ | $0.085 \pm 0.017$ | $0.083 \pm 0.009$ | $0.079 \pm 0.002$ | $\mathbf{0.768 \pm 0.023}$ | – |
| surv_framingham | $0.306 \pm 0.017$ | $0.467 \pm 0.003$ | $0.276 \pm 0.016$ | $0.325 \pm 0.008$ | $0.308 \pm 0.010$ | $0.330 \pm 0.051$ | $0.724 \pm 0.431$ | $0.466 \pm 0.464$ | $0.342 \pm 0.036$ | $0.324 \pm 0.009$ | $0.329 \pm 0.006$ | $\mathbf{1.356 \pm 0.084}$ | – |
| tab_blood_transfusion | $0.260 \pm 0.312$ | $0.542 \pm 0.116$ | $0.146 \pm 0.025$ | $0.196 \pm 0.042$ | $0.153 \pm 0.017$ | $0.172 \pm 0.069$ | $0.313 \pm 0.135$ | $0.244 \pm 0.035$ | $0.230 \pm 0.058$ | $0.196 \pm 0.001$ | $0.160 \pm 0.032$ | $\mathbf{2.031 \pm 0.130}$ | – |
| tab_credit_g | $0.422 \pm 0.022$ | $0.446 \pm 0.030$ | $0.324 \pm 0.066$ | $0.425 \pm 0.009$ | $0.389 \pm 0.009$ | $0.405 \pm 0.033$ | $0.453 \pm 0.097$ | $0.395 \pm 0.013$ | $0.432 \pm 0.007$ | $0.416 \pm 0.001$ | $0.395 \pm 0.011$ | $\mathbf{0.642 \pm 0.025}$ | – |
| tab_qsar_biodeg | $0.474 \pm 0.102$ | $1.089 \pm 0.089$ | $0.389 \pm 0.022$ | $0.995 \pm 0.356$ | $\mathbf{5.396 \pm 1.573}$ | $0.591 \pm 0.164$ | $1.033 \pm 0.515$ | $0.482 \pm 0.014$ | $0.829 \pm 0.365$ | $0.473 \pm 0.013$ | $0.376 \pm 0.007$ | $4.355 \pm 0.251$ | – |
| tab_maternal_risk | $0.260 \pm 0.125$ | $0.653 \pm 0.325$ | $0.256 \pm 0.083$ | $0.442 \pm 0.011$ | $1.012 \pm 0.821$ | $0.353 \pm 0.079$ | $0.486 \pm 0.099$ | $0.307 \pm 0.084$ | $0.440 \pm 0.081$ | $0.359 \pm 0.016$ | $0.373 \pm 0.056$ | $\mathbf{1.381 \pm 0.189}$ | – |
| tab_website_phishing | $0.322 \pm 0.168$ | $\mathbf{0.400 \pm 0.000}$ | $\mathbf{0.400 \pm 0.000}$ | $\mathbf{0.400 \pm 0.000}$ | $0.283 \pm 0.000$ | $0.283 \pm 0.000$ | $\mathbf{0.400 \pm 0.000}$ | $0.283 \pm 0.000$ | $0.322 \pm 0.168$ | $0.283 \pm 0.000$ | $0.283 \pm 0.000$ | $\mathbf{0.400 \pm 0.000}$ | – |

Table 12: DCR real–real reference — mean ± 95% CI over seeds

| Dataset | DISCO-TAB | CTGAN | TVAE | GaussCop | TabDDPM | ADS-GAN | PATE-GAN | RTVAE | NFlow | ARF | BayesNet | Marginals | GReaT |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| heart_failure | $0.801 \pm 0.024$ | $0.801 \pm 0.024$ | $0.801 \pm 0.024$ | $0.801 \pm 0.024$ | 0.796 | $0.801 \pm 0.024$ | $0.801 \pm 0.024$ | $0.801 \pm 0.024$ | $0.801 \pm 0.024$ | $0.801 \pm 0.024$ | $0.801 \pm 0.024$ | $0.801 \pm 0.024$ | – |
| breast_cancer | $0.465 \pm 0.057$ | $0.465 \pm 0.057$ | $0.465 \pm 0.057$ | $0.465 \pm 0.057$ | 0.468 | $0.465 \pm 0.057$ | $0.465 \pm 0.057$ | $0.465 \pm 0.057$ | $0.465 \pm 0.057$ | $0.465 \pm 0.057$ | $0.465 \pm 0.057$ | $0.465 \pm 0.057$ | – |
| liver_disorders | $0.380 \pm 0.131$ | $0.380 \pm 0.131$ | $0.380 \pm 0.131$ | $0.380 \pm 0.131$ | 0.342 | $0.380 \pm 0.131$ | $0.380 \pm 0.131$ | $0.380 \pm 0.131$ | $0.380 \pm 0.131$ | $0.380 \pm 0.131$ | $0.380 \pm 0.131$ | $0.380 \pm 0.131$ | – |
| parkinsons | $0.524 \pm 0.058$ | $0.524 \pm 0.058$ | $0.524 \pm 0.058$ | $0.524 \pm 0.058$ | 0.538 | $0.524 \pm 0.058$ | $0.524 \pm 0.058$ | $0.524 \pm 0.058$ | $0.524 \pm 0.058$ | $0.524 \pm 0.058$ | $0.524 \pm 0.058$ | $0.524 \pm 0.058$ | – |
| obesity | $0.279 \pm 0.028$ | $0.279 \pm 0.028$ | $0.279 \pm 0.028$ | $0.279 \pm 0.028$ | 0.284 | $0.276 \pm 0.129$ | $0.279 \pm 0.028$ | $0.279 \pm 0.028$ | $0.279 \pm 0.028$ | $0.279 \pm 0.028$ | $0.279 \pm 0.028$ | $0.279 \pm 0.028$ | – |
| diabetes | $0.071 \pm 0.000$ | $0.071 \pm 0.000$ | $0.071 \pm 0.000$ | $0.071 \pm 0.000$ | $0.071 \pm 0.000$ | $0.071 \pm 0.000$ | $0.071 \pm 0.000$ | $0.071 \pm 0.000$ | $0.071 \pm 0.000$ | $0.071 \pm 0.000$ | $0.071 \pm 0.000$ | $0.071 \pm 0.000$ | – |
| german_credit | $0.419 \pm 0.015$ | $0.419 \pm 0.015$ | $0.419 \pm 0.015$ | $0.419 \pm 0.015$ | $0.419 \pm 0.015$ | $0.419 \pm 0.015$ | $0.419 \pm 0.015$ | $0.419 \pm 0.015$ | $0.419 \pm 0.015$ | $0.419 \pm 0.015$ | $0.419 \pm 0.015$ | $0.419 \pm 0.015$ | – |
| bank_marketing | $0.209 \pm 0.002$ | $0.209 \pm 0.002$ | $0.209 \pm 0.002$ | $0.209 \pm 0.002$ | $0.209 \pm 0.002$ | $0.209 \pm 0.002$ | $0.209 \pm 0.002$ | $0.209 \pm 0.002$ | $0.209 \pm 0.002$ | $0.209 \pm 0.002$ | $0.209 \pm 0.002$ | $0.209 \pm 0.002$ | – |
| census_income | $0.136 \pm 0.001$ | $0.136 \pm 0.001$ | $0.136 \pm 0.001$ | $0.136 \pm 0.001$ | $0.136 \pm 0.001$ | $0.136 \pm 0.001$ | $0.136 \pm 0.001$ | $0.136 \pm 0.001$ | $0.136 \pm 0.001$ | $0.136 \pm 0.001$ | $0.136 \pm 0.001$ | $0.136 \pm 0.001$ | – |
| surv_melanoma | $0.524 \pm 0.041$ | $0.524 \pm 0.041$ | $0.524 \pm 0.041$ | $0.524 \pm 0.041$ | $0.524 \pm 0.041$ | $0.524 \pm 0.041$ | $0.524 \pm 0.041$ | $0.524 \pm 0.041$ | $0.524 \pm 0.041$ | $0.524 \pm 0.041$ | $0.524 \pm 0.041$ | $0.524 \pm 0.041$ | – |
| surv_pbc | $0.196 \pm 0.163$ | $0.196 \pm 0.163$ | $0.196 \pm 0.163$ | $0.196 \pm 0.163$ | $0.196 \pm 0.163$ | $0.196 \pm 0.163$ | $0.196 \pm 0.163$ | $0.196 \pm 0.163$ | $0.196 \pm 0.163$ | $0.196 \pm 0.163$ | $0.196 \pm 0.163$ | $0.196 \pm 0.163$ | – |
| surv_pbc3 | $0.626 \pm 0.053$ | $0.626 \pm 0.053$ | $0.626 \pm 0.053$ | $0.626 \pm 0.053$ | $0.626 \pm 0.053$ | $0.626 \pm 0.053$ | $0.626 \pm 0.053$ | $0.626 \pm 0.053$ | $0.626 \pm 0.053$ | $0.626 \pm 0.053$ | $0.626 \pm 0.053$ | $0.626 \pm 0.053$ | 0.629 |
| surv_retinopathy | $0.318 \pm 0.016$ | $0.318 \pm 0.016$ | $0.318 \pm 0.016$ | $0.318 \pm 0.016$ | $0.318 \pm 0.016$ | $0.318 \pm 0.016$ | $0.318 \pm 0.016$ | $0.318 \pm 0.016$ | $0.318 \pm 0.016$ | $0.318 \pm 0.016$ | $0.318 \pm 0.016$ | $0.318 \pm 0.016$ | – |
| surv_whas500 | $0.745 \pm 0.023$ | $0.745 \pm 0.023$ | $0.745 \pm 0.023$ | $0.745 \pm 0.023$ | $0.745 \pm 0.023$ | $0.745 \pm 0.023$ | $0.745 \pm 0.023$ | $0.745 \pm 0.023$ | $0.745 \pm 0.023$ | $0.745 \pm 0.023$ | $0.745 \pm 0.023$ | $0.745 \pm 0.023$ | – |
| surv_gbsg2 | $0.323 \pm 0.105$ | $0.323 \pm 0.105$ | $0.323 \pm 0.105$ | $0.323 \pm 0.105$ | $0.323 \pm 0.105$ | $0.323 \pm 0.105$ | $0.323 \pm 0.105$ | $0.323 \pm 0.105$ | $0.323 \pm 0.105$ | $0.323 \pm 0.105$ | $0.323 \pm 0.105$ | $0.323 \pm 0.105$ | 0.371 |
| surv_colon | $0.337 \pm 0.202$ | $0.337 \pm 0.202$ | $0.337 \pm 0.202$ | $0.337 \pm 0.202$ | $0.337 \pm 0.202$ | $0.337 \pm 0.202$ | $0.337 \pm 0.202$ | $0.337 \pm 0.202$ | $0.337 \pm 0.202$ | $0.337 \pm 0.202$ | $0.337 \pm 0.202$ | $0.337 \pm 0.202$ | – |
| surv_grace | $0.290 \pm 0.029$ | $0.290 \pm 0.029$ | $0.290 \pm 0.029$ | $0.290 \pm 0.029$ | $0.290 \pm 0.029$ | $0.290 \pm 0.029$ | $0.290 \pm 0.029$ | $0.290 \pm 0.029$ | $0.290 \pm 0.029$ | $0.290 \pm 0.029$ | $0.290 \pm 0.029$ | $0.290 \pm 0.029$ | – |
| surv_trace | $0.109 \pm 0.006$ | $0.109 \pm 0.006$ | $0.109 \pm 0.006$ | $0.109 \pm 0.006$ | $0.109 \pm 0.006$ | $0.109 \pm 0.006$ | $0.109 \pm 0.006$ | $0.109 \pm 0.006$ | $0.109 \pm 0.006$ | $0.109 \pm 0.006$ | $0.109 \pm 0.006$ | $0.109 \pm 0.006$ | – |
| surv_framingham | $0.352 \pm 0.009$ | $0.352 \pm 0.009$ | $0.352 \pm 0.009$ | $0.352 \pm 0.009$ | $0.352 \pm 0.009$ | $0.352 \pm 0.009$ | $0.352 \pm 0.009$ | $0.352 \pm 0.009$ | $0.352 \pm 0.009$ | $0.352 \pm 0.009$ | $0.352 \pm 0.009$ | $0.352 \pm 0.009$ | – |
| tab_blood_transfusion | $0.201 \pm 0.026$ | $0.201 \pm 0.026$ | $0.201 \pm 0.026$ | $0.201 \pm 0.026$ | $0.201 \pm 0.026$ | $0.201 \pm 0.026$ | $0.201 \pm 0.026$ | $0.201 \pm 0.026$ | $0.201 \pm 0.026$ | $0.201 \pm 0.026$ | $0.201 \pm 0.026$ | $0.201 \pm 0.026$ | – |
| tab_credit_g | $0.410 \pm 0.019$ | $0.410 \pm 0.019$ | $0.410 \pm 0.019$ | $0.410 \pm 0.019$ | $0.410 \pm 0.019$ | $0.410 \pm 0.019$ | $0.410 \pm 0.019$ | $0.410 \pm 0.019$ | $0.410 \pm 0.019$ | $0.410 \pm 0.019$ | $0.410 \pm 0.019$ | $0.410 \pm 0.019$ | – |
| tab_qsar_biodeg | $0.436 \pm 0.030$ | $0.436 \pm 0.030$ | $0.436 \pm 0.030$ | $0.436 \pm 0.030$ | $0.436 \pm 0.030$ | $0.436 \pm 0.030$ | $0.436 \pm 0.030$ | $0.436 \pm 0.030$ | $0.436 \pm 0.030$ | $0.436 \pm 0.030$ | $0.436 \pm 0.030$ | $0.436 \pm 0.030$ | – |
| tab_maternal_risk | $0.357 \pm 0.174$ | $0.357 \pm 0.174$ | $0.357 \pm 0.174$ | $0.357 \pm 0.174$ | $0.357 \pm 0.174$ | $0.357 \pm 0.174$ | $0.357 \pm 0.174$ | $0.357 \pm 0.174$ | $0.357 \pm 0.174$ | $0.357 \pm 0.174$ | $0.357 \pm 0.174$ | $0.357 \pm 0.174$ | – |
| tab_website_phishing | $0.400 \pm 0.000$ | $0.400 \pm 0.000$ | $0.400 \pm 0.000$ | $0.400 \pm 0.000$ | $0.400 \pm 0.000$ | $0.400 \pm 0.000$ | $0.400 \pm 0.000$ | $0.400 \pm 0.000$ | $0.400 \pm 0.000$ | $0.400 \pm 0.000$ | $0.400 \pm 0.000$ | $0.400 \pm 0.000$ | – |

Table 13: DCR encoded (↑) — mean ± 95% CI over seeds

| Dataset | DISCO-TAB | CTGAN | TVAE | GaussCop | TabDDPM | ADS-GAN | PATE-GAN | RTVAE | NFlow | ARF | BayesNet | Marginals | GReaT |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| heart_failure | $2.153 \pm 0.553$ | $2.316 \pm 0.299$ | $3.165 \pm 0.177$ | $1.999 \pm 0.143$ | 2.022 | $2.025 \pm 0.662$ | $2.880 \pm 0.566$ | $1.582 \pm 0.178$ | $1.918 \pm 0.340$ | $1.946 \pm 0.085$ | $2.079 \pm 0.405$ | $\mathbf{3.810 \pm 0.402}$ | – |
| breast_cancer | $3.280 \pm 0.411$ | $5.484 \pm 0.782$ | $3.199 \pm 0.489$ | $2.429 \pm 0.233$ | 2.308 | $3.052 \pm 0.345$ | $6.717 \pm 3.045$ | $2.951 \pm 0.615$ | $3.495 \pm 0.493$ | $2.793 \pm 0.561$ | $2.273 \pm 0.426$ | $\mathbf{12.169 \pm 0.265}$ | – |
| liver_disorders | $0.641 \pm 0.058$ | $0.989 \pm 0.150$ | $0.570 \pm 0.047$ | $0.560 \pm 0.121$ | 0.470 | $0.676 \pm 0.243$ | $0.797 \pm 0.059$ | $0.937 \pm 0.602$ | $0.594 \pm 0.033$ | $0.551 \pm 0.035$ | $0.577 \pm 0.075$ | $\mathbf{1.627 \pm 0.322}$ | – |
| parkinsons | $2.372 \pm 0.729$ | $3.982 \pm 0.120$ | $2.595 \pm 0.205$ | $1.978 \pm 0.149$ | 1.756 | $2.676 \pm 0.238$ | $3.538 \pm 0.665$ | $2.529 \pm 0.171$ | $2.767 \pm 0.446$ | $2.100 \pm 0.217$ | $1.513 \pm 0.310$ | $\mathbf{8.250 \pm 1.029}$ | – |
| obesity | $1.166 \pm 0.222$ | $1.909 \pm 0.085$ | $1.959 \pm 0.111$ | $1.860 \pm 0.310$ | 1.273 | $1.516 \pm 0.066$ | $2.083 \pm 0.472$ | $1.459 \pm 0.129$ | $1.770 \pm 0.063$ | $1.264 \pm 0.182$ | $1.023 \pm 0.019$ | $\mathbf{3.013 \pm 0.145}$ | – |
| diabetes | $0.498 \pm 0.059$ | $0.558 \pm 0.027$ | $0.546 \pm 0.076$ | $0.660 \pm 0.105$ | – | $0.517 \pm 0.052$ | $0.822 \pm 0.185$ | $0.524 \pm 0.106$ | $0.568 \pm 0.060$ | $0.501 \pm 0.007$ | $0.484 \pm 0.011$ | $\mathbf{1.604 \pm 0.012}$ | – |
| german_credit | $2.974 \pm 0.742$ | $3.107 \pm 0.154$ | $3.516 \pm 0.132$ | $3.007 \pm 0.024$ | – | $2.914 \pm 0.062$ | $3.187 \pm 0.138$ | $2.880 \pm 0.062$ | $3.105 \pm 0.140$ | $2.916 \pm 0.123$ | $2.866 \pm 0.062$ | $\mathbf{4.129 \pm 0.139}$ | – |
| bank_marketing | $1.652 \pm 0.047$ | $1.951 \pm 0.121$ | $1.893 \pm 0.243$ | $2.006 \pm 0.020$ | – | $1.994 \pm 0.168$ | $2.382 \pm 0.464$ | $1.959 \pm 0.405$ | $2.111 \pm 0.111$ | $1.841 \pm 0.051$ | $1.855 \pm 0.009$ | $\mathbf{5.609 \pm 1.075}$ | – |
| census_income | $1.640 \pm 0.098$ | $2.048 \pm 0.035$ | $1.781 \pm 0.231$ | $2.226 \pm 0.404$ | – | $2.049 \pm 0.320$ | $2.686 \pm 0.876$ | $2.221 \pm 0.626$ | $2.212 \pm 0.155$ | $1.902 \pm 0.051$ | $1.786 \pm 0.285$ | $\mathbf{4.370 \pm 0.304}$ | – |
| surv_melanoma | $0.587 \pm 0.084$ | – | – | – | – | $0.597 \pm 0.283$ | $\mathbf{1.072 \pm 0.963}$ | $0.865 \pm 0.360$ | $0.525 \pm 0.056$ | $0.521 \pm 0.081$ | $0.514 \pm 0.062$ | $0.904 \pm 0.178$ | – |
| surv_pbc | $\mathbf{0.630 \pm 1.356}$ | – | – | – | – | $0.075 \pm 0.041$ | $0.171 \pm 0.108$ | $0.574 \pm 1.595$ | $0.068 \pm 0.032$ | $0.060 \pm 0.023$ | $0.064 \pm 0.011$ | $0.193 \pm 0.108$ | – |
| surv_pbc3 | $2.000 \pm 0.207$ | – | – | – | – | $2.127 \pm 0.176$ | $2.730 \pm 0.463$ | $1.971 \pm 0.114$ | $2.029 \pm 0.322$ | $1.983 \pm 0.100$ | $1.981 \pm 0.055$ | $\mathbf{3.829 \pm 0.484}$ | – |
| surv_retinopathy | $0.628 \pm 1.706$ | – | – | – | – | $0.352 \pm 0.184$ | $0.820 \pm 0.071$ | $0.507 \pm 0.241$ | $0.617 \pm 0.261$ | $0.253 \pm 0.121$ | $0.132 \pm 0.003$ | $\mathbf{0.852 \pm 0.202}$ | – |
| surv_whas500 | $2.270 \pm 0.345$ | – | – | – | – | $2.260 \pm 0.094$ | $3.447 \pm 0.792$ | $2.218 \pm 0.077$ | $2.387 \pm 0.362$ | $2.251 \pm 0.158$ | $2.356 \pm 0.238$ | $\mathbf{4.183 \pm 0.777}$ | – |
| surv_gbsg2 | $0.789 \pm 0.403$ | – | – | – | – | $0.868 \pm 0.408$ | $1.240 \pm 0.741$ | $0.742 \pm 0.116$ | $0.692 \pm 0.103$ | $0.663 \pm 0.073$ | $0.964 \pm 0.210$ | $\mathbf{2.470 \pm 0.172}$ | – |
| surv_colon | $0.341 \pm 0.356$ | – | – | – | – | $0.513 \pm 0.233$ | $1.220 \pm 1.010$ | $0.433 \pm 0.005$ | $0.341 \pm 0.016$ | $0.334 \pm 0.067$ | $0.368 \pm 0.088$ | $\mathbf{1.772 \pm 0.360}$ | – |
| surv_grace | $0.354 \pm 0.049$ | – | – | – | – | $0.395 \pm 0.162$ | $\mathbf{1.192 \pm 2.913}$ | $0.603 \pm 0.047$ | $0.321 \pm 0.026$ | $0.329 \pm 0.025$ | $0.332 \pm 0.026$ | $0.666 \pm 0.091$ | – |
| surv_trace | $0.089 \pm 0.027$ | – | – | – | – | $0.177 \pm 0.106$ | $0.238 \pm 0.079$ | $\mathbf{0.737 \pm 1.768}$ | $0.108 \pm 0.013$ | $0.090 \pm 0.002$ | $0.101 \pm 0.013$ | $0.294 \pm 0.008$ | – |
| surv_framingham | $0.726 \pm 0.014$ | – | – | – | – | $0.780 \pm 0.126$ | $1.368 \pm 0.122$ | $1.252 \pm 1.082$ | $0.786 \pm 0.089$ | $0.739 \pm 0.008$ | $0.754 \pm 0.042$ | $\mathbf{1.567 \pm 0.089}$ | – |
| tab_blood_transfusion | $0.554 \pm 0.418$ | – | – | – | – | $0.353 \pm 0.125$ | $0.569 \pm 0.578$ | $0.542 \pm 0.148$ | $0.276 \pm 0.073$ | $0.245 \pm 0.012$ | $0.225 \pm 0.047$ | $\mathbf{0.975 \pm 0.200}$ | – |
| tab_credit_g | $3.156 \pm 0.053$ | – | – | – | – | $2.866 \pm 0.088$ | $3.297 \pm 0.324$ | $2.866 \pm 0.170$ | $3.038 \pm 0.155$ | $2.870 \pm 0.104$ | $2.859 \pm 0.126$ | $\mathbf{4.188 \pm 0.121}$ | – |
| tab_qsar_biodeg | $3.690 \pm 0.252$ | – | – | – | – | $3.096 \pm 0.058$ | $6.437 \pm 2.976$ | $3.072 \pm 0.274$ | $3.949 \pm 0.307$ | $2.546 \pm 0.177$ | $1.938 \pm 0.077$ | $\mathbf{19.399 \pm 7.340}$ | – |
| tab_maternal_risk | $0.452 \pm 0.309$ | – | – | – | – | $0.767 \pm 0.248$ | $1.263 \pm 0.936$ | $0.714 \pm 0.079$ | $0.651 \pm 0.015$ | $0.549 \pm 0.196$ | $0.569 \pm 0.105$ | $\mathbf{1.582 \pm 0.269}$ | – |
| tab_website_phishing | $1.520 \pm 0.456$ | – | – | – | – | $1.414 \pm 0.000$ | $\mathbf{1.609 \pm 0.840}$ | $1.414 \pm 0.000$ | $1.414 \pm 0.000$ | $1.414 \pm 0.000$ | $1.414 \pm 0.000$ | $1.414 \pm 0.000$ | – |

Table 14: NNDR (↑) — mean ± 95% CI over seeds

| Dataset | DISCO-TAB | CTGAN | TVAE | GaussCop | TabDDPM | ADS-GAN | PATE-GAN | RTVAE | NFlow | ARF | BayesNet | Marginals | GReaT |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| heart_failure | 0.892 ± 0.046 | 0.918 ± 0.017 | **0.983 ± 0.016** | 0.880 ± 0.024 | 0.912 | 0.906 ± 0.089 | 0.930 ± 0.014 | 0.909 ± 0.067 | 0.891 ± 0.037 | 0.897 ± 0.050 | 0.922 ± 0.039 | 0.927 ± 0.061 | – |
| breast_cancer | **0.980 ± 0.004** | 0.958 ± 0.022 | 0.965 ± 0.003 | 0.931 ± 0.012 | 0.929 | 0.949 ± 0.033 | 0.877 ± 0.241 | 0.975 ± 0.010 | 0.956 ± 0.004 | 0.953 ± 0.006 | 0.950 ± 0.001 | 0.972 ± 0.025 | – |
| liver_disorders | 0.922 ± 0.075 | 0.877 ± 0.049 | 0.916 ± 0.077 | 0.834 ± 0.034 | 0.867 | 0.897 ± 0.053 | 0.860 ± 0.027 | **1.000 ± 0.000** | 0.856 ± 0.105 | 0.851 ± 0.052 | 0.876 ± 0.016 | 0.864 ± 0.059 | – |
| parkinsons | 0.951 ± 0.033 | 0.956 ± 0.040 | 0.953 ± 0.019 | 0.907 ± 0.063 | 0.828 | 0.935 ± 0.012 | 0.944 ± 0.036 | **0.992 ± 0.016** | 0.935 ± 0.010 | 0.916 ± 0.057 | 0.922 ± 0.012 | 0.969 ± 0.022 | – |
| obesity | 0.902 ± 0.023 | 0.929 ± 0.014 | **0.963 ± 0.010** | 0.925 ± 0.020 | 0.903 | 0.910 ± 0.019 | 0.935 ± 0.023 | 0.910 ± 0.014 | 0.923 ± 0.002 | 0.893 ± 0.014 | 0.911 ± 0.020 | 0.956 ± 0.013 | – |
| diabetes | 0.831 ± 0.019 | 0.834 ± 0.010 | 0.897 ± 0.027 | 0.847 ± 0.012 | – | 0.846 ± 0.026 | 0.871 ± 0.034 | 0.848 ± 0.056 | 0.831 ± 0.013 | 0.832 ± 0.003 | 0.821 ± 0.011 | **0.905 ± 0.021** | – |
| german_credit | 0.953 ± 0.003 | 0.960 ± 0.001 | **0.986 ± 0.006** | 0.957 ± 0.017 | – | 0.951 ± 0.015 | 0.956 ± 0.007 | 0.949 ± 0.010 | 0.953 ± 0.017 | 0.959 ± 0.024 | 0.973 ± 0.004 | 0.962 ± 0.005 | – |
| bank_marketing | 0.927 ± 0.008 | 0.938 ± 0.008 | **0.966 ± 0.010** | 0.945 ± 0.006 | – | 0.943 ± 0.003 | 0.955 ± 0.010 | 0.949 ± 0.018 | 0.945 ± 0.008 | 0.933 ± 0.003 | 0.939 ± 0.006 | 0.892 ± 0.164 | – |
| census_income | 0.931 ± 0.008 | 0.944 ± 0.007 | **0.970 ± 0.009** | 0.948 ± 0.005 | – | 0.947 ± 0.005 | 0.962 ± 0.010 | 0.966 ± 0.025 | 0.952 ± 0.004 | 0.936 ± 0.001 | 0.932 ± 0.024 | 0.957 ± 0.028 | – |
| surv_melanoma | 0.857 ± 0.153 | – | – | – | – | 0.830 ± 0.053 | 0.832 ± 0.049 | **1.000 ± 0.000** | 0.805 ± 0.071 | 0.777 ± 0.025 | 0.851 ± 0.118 | 0.759 ± 0.029 | – |
| surv_pbc | 0.712 ± 0.379 | – | – | – | – | 0.542 ± 0.145 | 0.638 ± 0.298 | **0.817 ± 0.409** | 0.538 ± 0.026 | 0.497 ± 0.035 | 0.542 ± 0.143 | 0.493 ± 0.096 | – |
| surv_pbc3 | 0.917 ± 0.035 | – | – | – | – | 0.911 ± 0.033 | 0.918 ± 0.024 | **0.940 ± 0.030** | 0.916 ± 0.034 | 0.900 ± 0.029 | 0.926 ± 0.056 | 0.932 ± 0.029 | – |
| surv_retinopathy | 0.725 ± 0.617 | – | – | – | – | 0.688 ± 0.059 | 0.810 ± 0.093 | **1.000 ± 0.000** | 0.680 ± 0.274 | 0.588 ± 0.195 | 0.490 ± 0.177 | 0.772 ± 0.036 | – |
| surv_whas500 | 0.933 ± 0.027 | – | – | – | – | 0.897 ± 0.072 | 0.926 ± 0.026 | 0.929 ± 0.013 | 0.907 ± 0.056 | 0.906 ± 0.032 | **0.938 ± 0.046** | 0.910 ± 0.073 | – |
| surv_gbsg2 | 0.871 ± 0.085 | – | – | – | – | 0.887 ± 0.038 | 0.888 ± 0.088 | **0.973 ± 0.055** | 0.850 ± 0.019 | 0.851 ± 0.028 | 0.870 ± 0.053 | 0.847 ± 0.087 | – |
| surv_colon | 0.734 ± 0.207 | – | – | – | – | 0.791 ± 0.091 | 0.841 ± 0.085 | **1.000 ± 0.000** | 0.703 ± 0.028 | 0.691 ± 0.097 | 0.739 ± 0.018 | 0.884 ± 0.055 | – |
| surv_grace | 0.814 ± 0.058 | – | – | – | – | 0.824 ± 0.102 | 0.815 ± 0.068 | **1.000 ± 0.000** | 0.787 ± 0.066 | 0.787 ± 0.068 | 0.808 ± 0.033 | 0.789 ± 0.083 | – |
| surv_trace | 0.720 ± 0.084 | – | – | – | – | 0.786 ± 0.114 | 0.727 ± 0.024 | **0.980 ± 0.086** | 0.734 ± 0.056 | 0.679 ± 0.083 | 0.718 ± 0.039 | 0.687 ± 0.056 | – |
| surv_framingham | 0.895 ± 0.013 | – | – | – | – | 0.898 ± 0.021 | 0.898 ± 0.028 | **0.919 ± 0.037** | 0.895 ± 0.004 | 0.890 ± 0.005 | 0.888 ± 0.024 | 0.883 ± 0.014 | – |
| tab_blood_transfusion | 0.946 ± 0.081 | – | – | – | – | 0.870 ± 0.109 | 0.846 ± 0.037 | **1.000 ± 0.000** | 0.823 ± 0.019 | 0.822 ± 0.012 | 0.824 ± 0.037 | 0.805 ± 0.133 | – |
| tab_credit_g | 0.964 ± 0.005 | – | – | – | – | 0.948 ± 0.020 | 0.960 ± 0.010 | 0.948 ± 0.008 | 0.954 ± 0.009 | 0.948 ± 0.018 | **0.976 ± 0.001** | 0.963 ± 0.021 | – |
| tab_qsar_biodeg | **0.970 ± 0.009** | – | – | – | – | 0.949 ± 0.009 | 0.970 ± 0.028 | 0.949 ± 0.006 | 0.961 ± 0.011 | 0.930 ± 0.007 | 0.922 ± 0.010 | 0.915 ± 0.284 | – |
| tab_maternal_risk | 0.964 ± 0.146 | – | – | – | – | 0.876 ± 0.082 | 0.889 ± 0.054 | **1.000 ± 0.000** | 0.861 ± 0.043 | 0.823 ± 0.069 | 0.851 ± 0.032 | 0.898 ± 0.031 | – |
| tab_website_phishing | **1.000 ± 0.000** | – | – | – | – | **1.000 ± 0.000** | **1.000 ± 0.000** | 0.902 ± 0.420 | **1.000 ± 0.000** | **1.000 ± 0.000** | **1.000 ± 0.000** | **1.000 ± 0.000** | – |

Table 15: NNAA (0.5 ideal) — mean ± 95% CI over seeds

| Dataset | DISCO-TAB | CTGAN | TVAE | GaussCop | TabDDPM | ADS-GAN | PATE-GAN | RTVAE | NFlow | ARF | BayesNet | Marginals | GReaT |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| heart_failure | 0.604 ± 0.174 | 0.738 ± 0.052 | 0.967 ± 0.038 | 0.508 ± 0.131 | 0.655 | 0.600 ± 0.050 | 0.875 ± 0.151 | 0.629 ± 0.086 | 0.536 ± 0.023 | 0.455 ± 0.056 | 0.472 ± 0.109 | 0.927 ± 0.014 | – |
| breast_cancer | 0.975 ± 0.000 | 0.958 ± 0.029 | 0.944 ± 0.017 | 0.639 ± 0.058 | 0.713 | 0.882 ± 0.055 | 0.996 ± 0.007 | 0.932 ± 0.030 | 0.753 ± 0.032 | 0.564 ± 0.030 | 0.311 ± 0.048 | 0.951 ± 0.003 | – |
| liver_disorders | 0.361 ± 0.076 | 0.850 ± 0.037 | 0.673 ± 0.145 | 0.505 ± 0.080 | 0.458 | 0.711 ± 0.060 | 0.772 ± 0.160 | 0.905 ± 0.077 | 0.562 ± 0.070 | 0.470 ± 0.039 | 0.511 ± 0.054 | 0.891 ± 0.043 | – |
| parkinsons | 0.903 ± 0.072 | 0.953 ± 0.017 | 0.934 ± 0.059 | 0.637 ± 0.053 | 0.638 | 0.907 ± 0.029 | 0.886 ± 0.122 | 0.936 ± 0.036 | 0.740 ± 0.072 | 0.619 ± 0.063 | 0.386 ± 0.024 | 0.893 ± 0.056 | – |
| obesity | 0.591 ± 0.013 | 0.844 ± 0.027 | 0.935 ± 0.010 | 0.864 ± 0.088 | 0.726 | 0.775 ± 0.079 | 0.900 ± 0.109 | 0.761 ± 0.014 | 0.819 ± 0.062 | 0.614 ± 0.020 | 0.452 ± 0.015 | 0.950 ± 0.009 | – |
| diabetes | 0.585 ± 0.036 | 0.628 ± 0.015 | 0.682 ± 0.042 | 0.691 ± 0.060 | – | 0.630 ± 0.065 | 0.828 ± 0.124 | 0.624 ± 0.155 | 0.636 ± 0.059 | 0.541 ± 0.010 | 0.537 ± 0.036 | 0.920 ± 0.013 | – |
| german_credit | 0.790 ± 0.393 | 0.703 ± 0.112 | 0.948 ± 0.003 | 0.614 ± 0.019 | – | 0.578 ± 0.048 | 0.775 ± 0.076 | 0.550 ± 0.011 | 0.686 ± 0.039 | 0.463 ± 0.010 | 0.303 ± 0.029 | 0.910 ± 0.014 | – |
| bank_marketing | 0.612 ± 0.041 | 0.703 ± 0.034 | 0.806 ± 0.097 | 0.750 ± 0.033 | – | 0.775 ± 0.084 | 0.903 ± 0.087 | 0.784 ± 0.220 | 0.791 ± 0.040 | 0.626 ± 0.023 | 0.693 ± 0.024 | 0.995 ± 0.005 | – |
| census_income | 0.663 ± 0.068 | 0.734 ± 0.034 | 0.789 ± 0.078 | 0.840 ± 0.163 | – | 0.828 ± 0.179 | 0.944 ± 0.119 | 0.908 ± 0.165 | 0.868 ± 0.029 | 0.658 ± 0.044 | 0.723 ± 0.171 | 0.990 ± 0.003 | – |
| surv_melanoma | 0.536 ± 0.098 | – | – | – | – | 0.641 ± 0.203 | 0.797 ± 0.225 | 0.884 ± 0.062 | 0.534 ± 0.160 | 0.439 ± 0.098 | 0.527 ± 0.033 | 0.763 ± 0.024 | – |
| surv_pbc | 0.679 ± 0.631 | – | – | – | – | 0.562 ± 0.055 | 0.673 ± 0.064 | 0.802 ± 0.468 | 0.533 ± 0.055 | 0.477 ± 0.030 | 0.440 ± 0.010 | 0.701 ± 0.036 | – |
| surv_pbc3 | 0.556 ± 0.102 | – | – | – | – | 0.643 ± 0.049 | 0.880 ± 0.203 | 0.719 ± 0.035 | 0.624 ± 0.062 | 0.483 ± 0.064 | 0.290 ± 0.065 | 0.912 ± 0.005 | – |
| surv_retinopathy | 0.571 ± 0.897 | – | – | – | – | 0.550 ± 0.178 | 0.799 ± 0.084 | 0.691 ± 0.076 | 0.661 ± 0.089 | 0.405 ± 0.026 | 0.327 ± 0.057 | 0.811 ± 0.057 | – |
| surv_whas500 | 0.674 ± 0.034 | – | – | – | – | 0.586 ± 0.020 | 0.898 ± 0.169 | 0.654 ± 0.125 | 0.594 ± 0.080 | 0.482 ± 0.045 | 0.337 ± 0.065 | 0.912 ± 0.017 | – |
| surv_gbsg2 | 0.634 ± 0.319 | – | – | – | – | 0.750 ± 0.244 | 0.852 ± 0.164 | 0.749 ± 0.073 | 0.593 ± 0.071 | 0.496 ± 0.063 | 0.701 ± 0.034 | 0.959 ± 0.009 | – |
| surv_colon | 0.449 ± 0.206 | – | – | – | – | 0.699 ± 0.188 | 0.872 ± 0.220 | 0.762 ± 0.054 | 0.568 ± 0.083 | 0.412 ± 0.071 | 0.547 ± 0.040 | 0.875 ± 0.030 | – |
| surv_grace | 0.504 ± 0.022 | – | – | – | – | 0.672 ± 0.141 | 0.821 ± 0.399 | 0.919 ± 0.027 | 0.536 ± 0.031 | 0.480 ± 0.056 | 0.516 ± 0.015 | 0.831 ± 0.029 | – |
| surv_trace | 0.536 ± 0.063 | – | – | – | – | 0.761 ± 0.060 | 0.830 ± 0.033 | 0.946 ± 0.048 | 0.645 ± 0.028 | 0.529 ± 0.060 | 0.539 ± 0.024 | 0.830 ± 0.003 | – |
| surv_framingham | 0.556 ± 0.039 | – | – | – | – | 0.644 ± 0.165 | 0.936 ± 0.015 | 0.839 ± 0.402 | 0.636 ± 0.043 | 0.490 ± 0.011 | 0.546 ± 0.023 | 0.924 ± 0.013 | – |
| tab_blood_transfusion | 0.862 ± 0.269 | – | – | – | – | 0.814 ± 0.057 | 0.912 ± 0.133 | 0.975 ± 0.007 | 0.718 ± 0.076 | 0.574 ± 0.035 | 0.623 ± 0.026 | 0.956 ± 0.009 | – |
| tab_credit_g | 0.911 ± 0.015 | – | – | – | – | 0.594 ± 0.060 | 0.828 ± 0.226 | 0.540 ± 0.036 | 0.691 ± 0.028 | 0.461 ± 0.056 | 0.297 ± 0.019 | 0.916 ± 0.010 | – |
| tab_qsar_biodeg | 0.980 ± 0.015 | – | – | – | – | 0.890 ± 0.067 | 0.997 ± 0.002 | 0.897 ± 0.022 | 0.893 ± 0.024 | 0.639 ± 0.007 | 0.435 ± 0.016 | 0.995 ± 0.006 | – |
| tab_maternal_risk | 0.668 ± 0.083 | – | – | – | – | 0.860 ± 0.144 | 0.929 ± 0.089 | 0.890 ± 0.007 | 0.759 ± 0.016 | 0.636 ± 0.007 | 0.765 ± 0.039 | 0.911 ± 0.033 | – |
| tab_website_phishing | 0.368 ± 0.919 | – | – | – | – | 0.140 ± 0.047 | 0.520 ± 0.162 | 0.138 ± 0.026 | 0.174 ± 0.046 | 0.071 ± 0.016 | 0.130 ± 0.007 | 0.409 ± 0.049 | – |

Table 16: MIA AUROC (0.5 safe) — mean ± 95% CI over seeds

| Dataset | DISCO-TAB | CTGAN | TVAE | GaussCop | TabDDPM | ADS-GAN | PATE-GAN | RTVAE | NFlow | ARF | BayesNet | Marginals | GReaT |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| heart_failure | 0.499 ± 0.105 | 0.470 ± 0.077 | 0.477 ± 0.120 | 0.493 ± 0.061 | 0.486 | 0.515 ± 0.148 | 0.491 ± 0.105 | 0.458 ± 0.046 | 0.481 ± 0.049 | 0.539 ± 0.041 | 0.621 ± 0.106 | 0.492 ± 0.061 | – |
| breast_cancer | 0.516 ± 0.100 | 0.500 ± 0.056 | 0.497 ± 0.059 | 0.497 ± 0.070 | 0.491 | 0.511 ± 0.140 | 0.521 ± 0.132 | 0.499 ± 0.101 | 0.496 ± 0.095 | 0.523 ± 0.106 | 0.749 ± 0.045 | 0.498 ± 0.064 | – |
| liver_disorders | 0.632 ± 0.045 | 0.511 ± 0.085 | 0.500 ± 0.052 | 0.498 ± 0.078 | 0.503 | 0.525 ± 0.087 | 0.494 ± 0.049 | 0.514 ± 0.055 | 0.500 ± 0.063 | 0.516 ± 0.046 | 0.546 ± 0.067 | 0.508 ± 0.130 | – |
| parkinsons | 0.516 ± 0.214 | 0.501 ± 0.058 | 0.483 ± 0.110 | 0.486 ± 0.071 | 0.510 | 0.499 ± 0.039 | 0.487 ± 0.073 | 0.536 ± 0.118 | 0.502 ± 0.076 | 0.527 ± 0.074 | 0.695 ± 0.006 | 0.536 ± 0.037 | – |
| obesity | 0.559 ± 0.026 | 0.494 ± 0.001 | 0.503 ± 0.007 | 0.488 ± 0.036 | 0.527 | 0.515 ± 0.067 | 0.492 ± 0.051 | 0.512 ± 0.024 | 0.498 ± 0.035 | 0.540 ± 0.015 | 0.621 ± 0.010 | 0.499 ± 0.019 | – |
| diabetes | 0.504 ± 0.016 | 0.508 ± 0.027 | 0.504 ± 0.014 | 0.497 ± 0.022 | – | 0.497 ± 0.010 | 0.498 ± 0.006 | 0.500 ± 0.012 | 0.490 ± 0.017 | 0.504 ± 0.033 | 0.504 ± 0.013 | 0.501 ± 0.022 | – |
| german_credit | 0.513 ± 0.056 | 0.520 ± 0.038 | 0.506 ± 0.061 | 0.522 ± 0.031 | – | 0.530 ± 0.026 | 0.497 ± 0.021 | 0.552 ± 0.028 | 0.511 ± 0.041 | 0.574 ± 0.026 | 0.776 ± 0.013 | 0.524 ± 0.044 | – |
| bank_marketing | 0.505 ± 0.021 | 0.503 ± 0.034 | 0.506 ± 0.022 | 0.504 ± 0.026 | – | 0.498 ± 0.006 | 0.502 ± 0.008 | 0.501 ± 0.006 | 0.503 ± 0.024 | 0.496 ± 0.004 | 0.504 ± 0.022 | 0.498 ± 0.017 | – |
| census_income | 0.503 ± 0.015 | 0.493 ± 0.015 | 0.501 ± 0.014 | 0.507 ± 0.020 | – | 0.507 ± 0.020 | 0.497 ± 0.013 | 0.507 ± 0.010 | 0.496 ± 0.028 | 0.508 ± 0.019 | 0.507 ± 0.027 | 0.500 ± 0.021 | – |
| surv_melanoma | 0.647 ± 0.090 | – | – | – | – | 0.538 ± 0.059 | 0.513 ± 0.057 | 0.563 ± 0.117 | 0.561 ± 0.084 | 0.585 ± 0.076 | 0.559 ± 0.059 | 0.486 ± 0.157 | – |
| surv_pbc | 0.571 ± 0.112 | – | – | – | – | 0.522 ± 0.028 | 0.511 ± 0.101 | 0.525 ± 0.083 | 0.518 ± 0.061 | 0.545 ± 0.038 | 0.581 ± 0.077 | 0.508 ± 0.061 | – |
| surv_pbc3 | 0.552 ± 0.054 | – | – | – | – | 0.555 ± 0.113 | 0.496 ± 0.065 | 0.533 ± 0.082 | 0.507 ± 0.094 | 0.547 ± 0.058 | 0.790 ± 0.034 | 0.501 ± 0.106 | – |
| surv_retinopathy | 0.601 ± 0.171 | – | – | – | – | 0.500 ± 0.056 | 0.491 ± 0.088 | 0.535 ± 0.078 | 0.496 ± 0.079 | 0.520 ± 0.054 | 0.532 ± 0.114 | 0.499 ± 0.022 | – |
| surv_whas500 | 0.516 ± 0.088 | – | – | – | – | 0.519 ± 0.034 | 0.507 ± 0.038 | 0.525 ± 0.030 | 0.499 ± 0.086 | 0.524 ± 0.056 | 0.742 ± 0.053 | 0.484 ± 0.057 | – |
| surv_gbsg2 | 0.527 ± 0.083 | – | – | – | – | 0.501 ± 0.067 | 0.497 ± 0.078 | 0.495 ± 0.094 | 0.481 ± 0.064 | 0.522 ± 0.027 | 0.503 ± 0.057 | 0.481 ± 0.053 | – |
| surv_colon | 0.520 ± 0.058 | – | – | – | – | 0.489 ± 0.075 | 0.482 ± 0.039 | 0.478 ± 0.028 | 0.468 ± 0.011 | 0.513 ± 0.058 | 0.489 ± 0.009 | 0.516 ± 0.076 | – |
| surv_grace | 0.571 ± 0.026 | – | – | – | – | 0.495 ± 0.064 | 0.514 ± 0.069 | 0.502 ± 0.057 | 0.489 ± 0.012 | 0.549 ± 0.037 | 0.530 ± 0.002 | 0.493 ± 0.006 | – |
| surv_trace | 0.578 ± 0.006 | – | – | – | – | 0.498 ± 0.059 | 0.497 ± 0.010 | 0.499 ± 0.006 | 0.508 ± 0.028 | 0.525 ± 0.025 | 0.571 ± 0.032 | 0.497 ± 0.042 | – |
| surv_framingham | 0.512 ± 0.025 | – | – | – | – | 0.509 ± 0.014 | 0.505 ± 0.030 | 0.508 ± 0.024 | 0.509 ± 0.051 | 0.527 ± 0.017 | 0.512 ± 0.048 | 0.507 ± 0.057 | – |
| tab_blood_transfusion | 0.533 ± 0.094 | – | – | – | – | 0.505 ± 0.006 | 0.517 ± 0.058 | 0.512 ± 0.104 | 0.526 ± 0.041 | 0.545 ± 0.045 | 0.524 ± 0.027 | 0.499 ± 0.102 | – |
| tab_credit_g | 0.516 ± 0.025 | – | – | – | – | 0.494 ± 0.050 | 0.488 ± 0.038 | 0.533 ± 0.059 | 0.500 ± 0.067 | 0.548 ± 0.063 | 0.774 ± 0.015 | 0.523 ± 0.061 | – |

| Dataset | DISCO-TAB | CTGAN | TVAE | GaussCop | TabDDPM | ADS-GAN | PATE-GAN | RTVAE | NFlow | ARF | BayesNet | Marginals | GReaT |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| tab_qsar_biodeg | $0.503 \pm 0.026$ | – | – | – | – | $0.511 \pm 0.015$ | $0.485 \pm 0.048$ | $0.507 \pm 0.021$ | $0.497 \pm 0.051$ | $0.527 \pm 0.047$ | $0.722 \pm 0.037$ | $0.484 \pm 0.025$ | – |
| tab_maternal_risk | $0.647 \pm 0.020$ | – | – | – | – | $0.501 \pm 0.142$ | $0.493 \pm 0.096$ | $0.502 \pm 0.088$ | $0.506 \pm 0.112$ | $0.525 \pm 0.180$ | $0.513 \pm 0.110$ | $0.500 \pm 0.046$ | – |
| tab_website_phishing | $0.530 \pm 0.028$ | – | – | – | – | $0.537 \pm 0.073$ | $0.498 \pm 0.112$ | $0.513 \pm 0.041$ | $0.520 \pm 0.030$ | $0.589 \pm 0.015$ | $0.511 \pm 0.031$ | $0.507 \pm 0.019$ | – |

Table 17: Duplicate rate (↓) — mean ± 95% CI over seeds

| Dataset | DISCO-TAB | CTGAN | TVAE | GaussCop | TabDDPM | ADS-GAN | PATE-GAN | RTVAE | NFlow | ARF | BayesNet | Marginals | GReaT |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| heart_failure | **$0.000 \pm 0.000$** | **$0.000 \pm 0.000$** | **$0.000 \pm 0.000$** | **$0.000 \pm 0.000$** | **0.000** | **$0.000 \pm 0.000$** | **$0.000 \pm 0.000$** | **$0.000 \pm 0.000$** | **$0.000 \pm 0.000$** | **$0.000 \pm 0.000$** | **$0.000 \pm 0.000$** | **$0.000 \pm 0.000$** | – |
| breast_cancer | **$0.000 \pm 0.000$** | **$0.000 \pm 0.000$** | **$0.000 \pm 0.000$** | **$0.000 \pm 0.000$** | **0.000** | **$0.000 \pm 0.000$** | **$0.000 \pm 0.000$** | **$0.000 \pm 0.000$** | **$0.000 \pm 0.000$** | **$0.000 \pm 0.000$** | **$0.000 \pm 0.000$** | **$0.000 \pm 0.000$** | – |
| liver_disorders | $0.307 \pm 0.675$ | **$0.000 \pm 0.000$** | **$0.000 \pm 0.000$** | **$0.000 \pm 0.000$** | **0.000** | **$0.000 \pm 0.000$** | **$0.000 \pm 0.000$** | **$0.000 \pm 0.000$** | **$0.000 \pm 0.000$** | **$0.000 \pm 0.000$** | $0.002 \pm 0.005$ | **$0.000 \pm 0.000$** | – |
| parkinsons | **$0.000 \pm 0.000$** | **$0.000 \pm 0.000$** | **$0.000 \pm 0.000$** | **$0.000 \pm 0.000$** | **0.000** | **$0.000 \pm 0.000$** | **$0.000 \pm 0.000$** | **$0.000 \pm 0.000$** | **$0.000 \pm 0.000$** | **$0.000 \pm 0.000$** | **$0.000 \pm 0.000$** | **$0.000 \pm 0.000$** | – |
| obesity | **$0.000 \pm 0.000$** | **$0.000 \pm 0.000$** | **$0.000 \pm 0.000$** | **$0.000 \pm 0.000$** | **0.000** | **$0.000 \pm 0.000$** | **$0.000 \pm 0.000$** | **$0.000 \pm 0.000$** | **$0.000 \pm 0.000$** | $0.000 \pm 0.002$ | **$0.000 \pm 0.000$** | **$0.000 \pm 0.000$** | – |
| diabetes | **$0.000 \pm 0.000$** | $0.000 \pm 0.000$ | $0.001 \pm 0.001$ | **$0.000 \pm 0.000$** | – | **$0.000 \pm 0.000$** | **$0.000 \pm 0.000$** | **$0.000 \pm 0.000$** | **$0.000 \pm 0.000$** | **$0.000 \pm 0.000$** | **$0.000 \pm 0.000$** | **$0.000 \pm 0.000$** | – |
| german_credit | **$0.000 \pm 0.000$** | **$0.000 \pm 0.000$** | **$0.000 \pm 0.000$** | **$0.000 \pm 0.000$** | – | **$0.000 \pm 0.000$** | **$0.000 \pm 0.000$** | **$0.000 \pm 0.000$** | **$0.000 \pm 0.000$** | **$0.000 \pm 0.000$** | **$0.000 \pm 0.000$** | **$0.000 \pm 0.000$** | – |
| bank_marketing | **$0.000 \pm 0.000$** | **$0.000 \pm 0.000$** | **$0.000 \pm 0.000$** | **$0.000 \pm 0.000$** | – | **$0.000 \pm 0.000$** | **$0.000 \pm 0.000$** | **$0.000 \pm 0.000$** | **$0.000 \pm 0.000$** | **$0.000 \pm 0.000$** | **$0.000 \pm 0.000$** | **$0.000 \pm 0.000$** | – |
| census_income | **$0.000 \pm 0.000$** | **$0.000 \pm 0.000$** | **$0.000 \pm 0.000$** | **$0.000 \pm 0.000$** | – | **$0.000 \pm 0.000$** | **$0.000 \pm 0.000$** | **$0.000 \pm 0.000$** | **$0.000 \pm 0.000$** | **$0.000 \pm 0.000$** | **$0.000 \pm 0.000$** | **$0.000 \pm 0.000$** | – |
| surv_melanoma | $0.006 \pm 0.026$ | – | – | – | – | **$0.000 \pm 0.000$** | **$0.000 \pm 0.000$** | **$0.000 \pm 0.000$** | **$0.000 \pm 0.000$** | **$0.000 \pm 0.000$** | **$0.000 \pm 0.000$** | **$0.000 \pm 0.000$** | – |
| surv_pbc | **$0.000 \pm 0.000$** | – | – | – | – | **$0.000 \pm 0.000$** | $0.003 \pm 0.006$ | $0.017 \pm 0.075$ | $0.004 \pm 0.010$ | **$0.000 \pm 0.000$** | **$0.000 \pm 0.000$** | **$0.000 \pm 0.000$** | – |
| surv_pbc3 | **$0.000 \pm 0.000$** | – | – | – | – | **$0.000 \pm 0.000$** | **$0.000 \pm 0.000$** | **$0.000 \pm 0.000$** | **$0.000 \pm 0.000$** | **$0.000 \pm 0.000$** | **$0.000 \pm 0.000$** | **$0.000 \pm 0.000$** | – |
| surv_retinopathy | $0.263 \pm 0.584$ | – | – | – | – | $0.119 \pm 0.110$ | $0.024 \pm 0.022$ | $0.101 \pm 0.047$ | $0.056 \pm 0.055$ | $0.127 \pm 0.043$ | $0.204 \pm 0.039$ | **$0.008 \pm 0.010$** | – |
| surv_whas500 | **$0.000 \pm 0.000$** | – | – | – | – | **$0.000 \pm 0.000$** | **$0.000 \pm 0.000$** | **$0.000 \pm 0.000$** | **$0.000 \pm 0.000$** | **$0.000 \pm 0.000$** | **$0.000 \pm 0.000$** | **$0.000 \pm 0.000$** | – |
| surv_gbsg2 | $0.001 \pm 0.003$ | – | – | – | – | **$0.000 \pm 0.000$** | **$0.000 \pm 0.000$** | **$0.000 \pm 0.000$** | **$0.000 \pm 0.000$** | **$0.000 \pm 0.000$** | **$0.000 \pm 0.000$** | **$0.000 \pm 0.000$** | – |
| surv_colon | **$0.000 \pm 0.000$** | – | – | – | – | **$0.000 \pm 0.000$** | **$0.000 \pm 0.000$** | **$0.000 \pm 0.000$** | **$0.000 \pm 0.000$** | $0.053 \pm 0.027$ | **$0.000 \pm 0.000$** | **$0.000 \pm 0.000$** | – |
| surv_grace | $0.052 \pm 0.115$ | – | – | – | – | $0.003 \pm 0.003$ | $0.001 \pm 0.003$ | $0.000 \pm 0.002$ | $0.003 \pm 0.005$ | $0.003 \pm 0.004$ | $0.010 \pm 0.018$ | **$0.000 \pm 0.000$** | – |
| surv_trace | **$0.000 \pm 0.000$** | – | – | – | – | **$0.000 \pm 0.000$** | **$0.000 \pm 0.000$** | **$0.000 \pm 0.000$** | **$0.000 \pm 0.000$** | **$0.000 \pm 0.000$** | **$0.000 \pm 0.000$** | **$0.000 \pm 0.000$** | – |
| surv_framingham | **$0.000 \pm 0.000$** | – | – | – | – | **$0.000 \pm 0.000$** | **$0.000 \pm 0.000$** | **$0.000 \pm 0.000$** | **$0.000 \pm 0.000$** | **$0.000 \pm 0.000$** | **$0.000 \pm 0.000$** | **$0.000 \pm 0.000$** | – |
| tab_blood_transfusion | **$0.000 \pm 0.000$** | – | – | – | – | **$0.000 \pm 0.000$** | **$0.000 \pm 0.000$** | **$0.000 \pm 0.000$** | **$0.000 \pm 0.000$** | $0.009 \pm 0.006$ | $0.005 \pm 0.003$ | **$0.000 \pm 0.000$** | – |
| tab_credit_g | **$0.000 \pm 0.000$** | – | – | – | – | **$0.000 \pm 0.000$** | **$0.000 \pm 0.000$** | **$0.000 \pm 0.000$** | **$0.000 \pm 0.000$** | **$0.000 \pm 0.000$** | **$0.000 \pm 0.000$** | **$0.000 \pm 0.000$** | – |
| tab_qsar_biodeg | **$0.000 \pm 0.000$** | – | – | – | – | **$0.000 \pm 0.000$** | **$0.000 \pm 0.000$** | **$0.000 \pm 0.000$** | **$0.000 \pm 0.000$** | **$0.000 \pm 0.000$** | **$0.000 \pm 0.000$** | **$0.000 \pm 0.000$** | – |
| tab_maternal_risk | **$0.000 \pm 0.000$** | – | – | – | – | **$0.000 \pm 0.000$** | **$0.000 \pm 0.000$** | **$0.000 \pm 0.000$** | **$0.000 \pm 0.000$** | **$0.000 \pm 0.000$** | **$0.000 \pm 0.000$** | **$0.000 \pm 0.000$** | – |
| tab_website_phishing | $0.402 \pm 0.866$ | – | – | – | – | $0.343 \pm 0.046$ | $0.157 \pm 0.079$ | $0.319 \pm 0.044$ | $0.298 \pm 0.129$ | $0.431 \pm 0.016$ | $0.318 \pm 0.024$ | **$0.055 \pm 0.005$** | – |

## Diversity / coverage (PRDC)

Table 18: Precision (↑) — mean ± 95% CI over seeds

| Dataset | DISCO-TAB | CTGAN | TVAE | GaussCop | TabDDPM | ADS-GAN | PATE-GAN | RTVAE | NFlow | ARF | BayesNet | Marginals | GReaT |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| heart_failure | $0.965 \pm 0.074$ | $0.812 \pm 0.120$ | **$1.000 \pm 0.000$** | $0.943 \pm 0.043$ | 0.477 | $0.967 \pm 0.021$ | $0.619 \pm 0.360$ | $0.996 \pm 0.018$ | $0.940 \pm 0.032$ | $0.969 \pm 0.012$ | $0.986 \pm 0.042$ | $0.457 \pm 0.146$ | – |
| breast_cancer | $0.575 \pm 0.156$ | $0.004 \pm 0.016$ | $0.221 \pm 0.149$ | $0.853 \pm 0.090$ | 0.433 | $0.350 \pm 0.035$ | $0.001 \pm 0.006$ | $0.541 \pm 0.033$ | $0.035 \pm 0.068$ | $0.453 \pm 0.024$ | **$0.969 \pm 0.036$** | $0.054 \pm 0.121$ | – |
| liver_disorders | $0.995 \pm 0.010$ | $0.571 \pm 0.315$ | **$0.999 \pm 0.005$** | $0.981 \pm 0.005$ | 0.978 | $0.982 \pm 0.024$ | $0.704 \pm 0.095$ | $0.998 \pm 0.005$ | $0.937 \pm 0.050$ | $0.982 \pm 0.024$ | $0.969 \pm 0.036$ | $0.594 \pm 0.172$ | – |
| parkinsons | $0.872 \pm 0.126$ | $0.004 \pm 0.018$ | $0.553 \pm 0.233$ | $0.848 \pm 0.112$ | 0.814 | $0.372 \pm 0.115$ | $0.013 \pm 0.016$ | $0.842 \pm 0.072$ | $0.158 \pm 0.113$ | $0.741 \pm 0.064$ | **$0.991 \pm 0.024$** | $0.107 \pm 0.096$ | – |
| obesity | **$0.937 \pm 0.007$** | $0.484 \pm 0.015$ | $0.874 \pm 0.047$ | $0.587 \pm 0.042$ | 0.688 | $0.684 \pm 0.331$ | $0.374 \pm 0.191$ | $0.702 \pm 0.062$ | $0.526 \pm 0.095$ | $0.813 \pm 0.027$ | $0.900 \pm 0.025$ | $0.098 \pm 0.120$ | – |
| diabetes | $0.982 \pm 0.011$ | $0.940 \pm 0.035$ | **$0.995 \pm 0.005$** | $0.874 \pm 0.099$ | – | $0.965 \pm 0.032$ | $0.757 \pm 0.675$ | $0.917 \pm 0.214$ | $0.916 \pm 0.041$ | $0.977 \pm 0.009$ | $0.960 \pm 0.009$ | $0.514 \pm 0.068$ | – |
| german_credit | $0.924 \pm 0.199$ | $0.734 \pm 0.078$ | **$1.000 \pm 0.002$** | $0.822 \pm 0.036$ | – | $0.888 \pm 0.016$ | $0.700 \pm 0.094$ | $0.926 \pm 0.022$ | $0.671 \pm 0.019$ | $0.917 \pm 0.016$ | $0.990 \pm 0.011$ | $0.150 \pm 0.053$ | – |
| bank_marketing | $0.971 \pm 0.023$ | $0.720 \pm 0.082$ | **$0.990 \pm 0.009$** | $0.664 \pm 0.165$ | – | $0.743 \pm 0.089$ | $0.392 \pm 0.130$ | $0.761 \pm 0.254$ | $0.636 \pm 0.049$ | $0.785 \pm 0.035$ | $0.776 \pm 0.022$ | $0.323 \pm 0.658$ | – |
| census_income | $0.967 \pm 0.025$ | $0.543 \pm 0.086$ | **$0.983 \pm 0.012$** | $0.405 \pm 0.417$ | – | $0.480 \pm 0.355$ | $0.186 \pm 0.298$ | $0.377 \pm 0.516$ | $0.321 \pm 0.141$ | $0.539 \pm 0.085$ | $0.552 \pm 0.019$ | $0.017 \pm 0.054$ | – |
| surv_melanoma | $0.990 \pm 0.009$ | – | – | – | – | **$0.994 \pm 0.026$** | $0.789 \pm 0.091$ | $0.986 \pm 0.038$ | $0.953 \pm 0.068$ | $0.986 \pm 0.009$ | $0.986 \pm 0.023$ | $0.831 \pm 0.057$ | – |
| surv_pbc | $0.771 \pm 0.755$ | – | – | – | – | $0.991 \pm 0.012$ | $0.914 \pm 0.168$ | $0.888 \pm 0.467$ | $0.993 \pm 0.021$ | $0.992 \pm 0.010$ | **$0.997 \pm 0.006$** | $0.841 \pm 0.066$ | – |
| surv_pbc3 | $0.982 \pm 0.036$ | – | – | – | – | $0.944 \pm 0.094$ | $0.456 \pm 0.253$ | **$0.995 \pm 0.014$** | $0.906 \pm 0.037$ | $0.953 \pm 0.050$ | $0.994 \pm 0.019$ | $0.225 \pm 0.181$ | – |
| surv_retinopathy | $0.704 \pm 0.774$ | – | – | – | – | $0.815 \pm 0.167$ | $0.515 \pm 0.182$ | $0.737 \pm 0.030$ | $0.639 \pm 0.156$ | $0.926 \pm 0.013$ | **$0.999 \pm 0.005$** | $0.442 \pm 0.086$ | – |
| surv_whas500 | $0.987 \pm 0.021$ | – | – | – | – | $0.960 \pm 0.031$ | $0.341 \pm 0.392$ | **$1.000 \pm 0.000$** | $0.938 \pm 0.054$ | $0.962 \pm 0.018$ | $0.991 \pm 0.014$ | $0.725 \pm 0.314$ | – |
| surv_gbsg2 | $0.892 \pm 0.368$ | – | – | – | – | $0.937 \pm 0.053$ | $0.648 \pm 0.300$ | $0.916 \pm 0.051$ | $0.871 \pm 0.072$ | **$0.951 \pm 0.020$** | $0.777 \pm 0.008$ | $0.540 \pm 0.203$ | – |
| surv_colon | **$0.994 \pm 0.000$** | – | – | – | – | $0.976 \pm 0.022$ | $0.712 \pm 0.280$ | $0.961 \pm 0.013$ | $0.928 \pm 0.083$ | $0.978 \pm 0.023$ | $0.986 \pm 0.009$ | $0.593 \pm 0.073$ | – |
| surv_grace | $0.994 \pm 0.016$ | – | – | – | – | $0.989 \pm 0.025$ | $0.560 \pm 1.175$ | **$1.000 \pm 0.002$** | $0.988 \pm 0.010$ | $0.991 \pm 0.007$ | $0.980 \pm 0.014$ | $0.665 \pm 0.016$ | – |
| surv_trace | **$0.999 \pm 0.002$** | – | – | – | – | $0.995 \pm 0.001$ | $0.755 \pm 0.208$ | $0.899 \pm 0.405$ | $0.982 \pm 0.004$ | $0.991 \pm 0.002$ | $0.990 \pm 0.005$ | $0.641 \pm 0.036$ | – |
| surv_framingham | **$0.980 \pm 0.013$** | – | – | – | – | $0.938 \pm 0.080$ | $0.301 \pm 0.513$ | $0.652 \pm 1.006$ | $0.904 \pm 0.071$ | $0.958 \pm 0.012$ | $0.936 \pm 0.024$ | $0.272 \pm 0.078$ | – |
| tab_blood_transfusion | $0.688 \pm 0.618$ | – | – | – | – | $0.829 \pm 0.126$ | $0.553 \pm 0.401$ | $0.581 \pm 0.036$ | $0.658 \pm 0.207$ | $0.879 \pm 0.078$ | **$0.972 \pm 0.015$** | $0.347 \pm 0.024$ | – |
| tab_credit_g | $0.791 \pm 0.094$ | – | – | – | – | $0.912 \pm 0.048$ | $0.673 \pm 0.327$ | $0.933 \pm 0.027$ | $0.734 \pm 0.032$ | $0.910 \pm 0.043$ | **$0.992 \pm 0.006$** | $0.122 \pm 0.031$ | – |
| tab_qsar_biodeg | $0.149 \pm 0.037$ | – | – | – | – | $0.251 \pm 0.076$ | $0.006 \pm 0.010$ | $0.348 \pm 0.041$ | $0.073 \pm 0.031$ | $0.536 \pm 0.025$ | **$0.961 \pm 0.014$** | $0.040 \pm 0.014$ | – |
| tab_maternal_risk | **$0.992 \pm 0.018$** | – | – | – | – | $0.757 \pm 0.314$ | $0.476 \pm 0.331$ | $0.838 \pm 0.111$ | $0.778 \pm 0.134$ | $0.905 \pm 0.077$ | $0.895 \pm 0.077$ | $0.778 \pm 0.042$ | – |
| tab_website_phishing | $0.990 \pm 0.042$ | – | – | – | – | $0.997 \pm 0.007$ | $0.969 \pm 0.064$ | $0.997 \pm 0.007$ | $0.992 \pm 0.013$ | **$0.999 \pm 0.005$** | $0.997 \pm 0.005$ | $0.882 \pm 0.071$ | – |

Table 19: Recall (↑) — mean ± 95% CI over seeds

| Dataset | DISCO-TAB | CTGAN | TVAE | GaussCop | TabDDPM | ADS-GAN | PATE-GAN | RTVAE | NFlow | ARF | BayesNet | Marginals | GReaT |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| heart_failure | $0.921 \pm 0.089$ | $0.886 \pm 0.037$ | $0.031 \pm 0.069$ | $0.918 \pm 0.042$ | **0.979** | $0.879 \pm 0.055$ | $0.907 \pm 0.094$ | $0.669 \pm 0.154$ | $0.923 \pm 0.049$ | $0.967 \pm 0.027$ | $0.971 \pm 0.018$ | $0.975 \pm 0.091$ | – |
| breast_cancer | $0.103 \pm 0.102$ | $0.651 \pm 0.274$ | $0.286 \pm 0.066$ | $0.895 \pm 0.068$ | 0.960 | $0.738 \pm 0.152$ | $0.588 \pm 0.549$ | $0.330 \pm 0.109$ | $0.913 \pm 0.023$ | $0.938 \pm 0.033$ | **$0.964 \pm 0.019$** | $0.525 \pm 0.210$ | – |
| liver_disorders | $0.918 \pm 0.026$ | $0.891 \pm 0.199$ | $0.563 \pm 0.307$ | $0.972 \pm 0.005$ | 0.971 | $0.762 \pm 0.185$ | $0.960 \pm 0.080$ | $0.199 \pm 0.293$ | $0.969 \pm 0.045$ | $0.978 \pm 0.031$ | $0.986 \pm 0.009$ | **$1.000 \pm 0.000$** | – |
| parkinsons | $0.372 \pm 0.073$ | $0.667 \pm 0.016$ | $0.434 \pm 0.199$ | $0.981 \pm 0.000$ | **0.987** | $0.684 \pm 0.189$ | $0.838 \pm 0.151$ | $0.363 \pm 0.161$ | $0.944 \pm 0.115$ | $0.979 \pm 0.018$ | $0.972 \pm 0.051$ | $0.447 \pm 0.160$ | – |
| obesity | $0.923 \pm 0.026$ | $0.915 \pm 0.020$ | $0.351 \pm 0.032$ | $0.854 \pm 0.197$ | 0.933 | $0.907 \pm 0.011$ | $0.810 \pm 0.145$ | $0.881 \pm 0.026$ | $0.907 \pm 0.059$ | $0.945 \pm 0.010$ | **$0.954 \pm 0.009$** | $0.949 \pm 0.054$ | – |
| diabetes | $0.945 \pm 0.063$ | $0.963 \pm 0.025$ | $0.704 \pm 0.143$ | $0.973 \pm 0.022$ | – | $0.940 \pm 0.070$ | $0.910 \pm 0.152$ | $0.946 \pm 0.056$ | $0.964 \pm 0.023$ | $0.978 \pm 0.006$ | $0.984 \pm 0.012$ | **$0.988 \pm 0.019$** | – |
| german_credit | $0.473 \pm 1.310$ | $0.907 \pm 0.079$ | $0.015 \pm 0.002$ | $0.932 \pm 0.022$ | – | $0.924 \pm 0.035$ | $0.917 \pm 0.097$ | $0.875 \pm 0.005$ | $0.940 \pm 0.078$ | $0.966 \pm 0.008$ | $0.958 \pm 0.014$ | **$0.988 \pm 0.025$** | – |
| bank_marketing | $0.790 \pm 0.146$ | $0.936 \pm 0.016$ | $0.298 \pm 0.184$ | $0.814 \pm 0.161$ | – | $0.840 \pm 0.125$ | $0.871 \pm 0.072$ | $0.775 \pm 0.157$ | $0.892 \pm 0.103$ | **$0.966 \pm 0.017$** | $0.949 \pm 0.009$ | $0.949 \pm 0.184$ | – |

| Dataset | DISCO-TAB | CTGAN | TVAE | GaussCop | TabDDPM | ADS-GAN | PATE-GAN | RTVAE | NFlow | ARF | BayesNet | Marginals | GReaT |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| census_income | $0.712 \pm 0.170$ | $0.911 \pm 0.096$ | $0.304 \pm 0.172$ | $0.883 \pm 0.062$ | – | $0.787 \pm 0.036$ | $0.631 \pm 1.049$ | $0.458 \pm 0.739$ | $0.840 \pm 0.097$ | $\mathbf{0.960 \pm 0.017}$ | $0.903 \pm 0.083$ | $0.917 \pm 0.265$ | – |
| surv_melanoma | $0.947 \pm 0.038$ | – | – | – | – | $0.872 \pm 0.215$ | $0.929 \pm 0.229$ | $0.522 \pm 0.273$ | $0.986 \pm 0.009$ | $0.988 \pm 0.015$ | $0.986 \pm 0.023$ | $\mathbf{1.000 \pm 0.000}$ | – |
| surv_pbc | $0.782 \pm 0.546$ | – | – | – | – | $0.968 \pm 0.043$ | $0.968 \pm 0.056$ | $0.647 \pm 0.709$ | $0.972 \pm 0.026$ | $0.995 \pm 0.006$ | $0.992 \pm 0.010$ | $\mathbf{1.000 \pm 0.000}$ | – |
| surv_pbc3 | $0.878 \pm 0.073$ | – | – | – | – | $0.842 \pm 0.098$ | $0.858 \pm 0.298$ | $0.455 \pm 0.099$ | $0.914 \pm 0.041$ | $0.962 \pm 0.021$ | $0.956 \pm 0.031$ | $\mathbf{0.986 \pm 0.032}$ | – |
| surv_retinopathy | $0.668 \pm 1.305$ | – | – | – | – | $0.929 \pm 0.022$ | $0.965 \pm 0.065$ | $0.982 \pm 0.042$ | $0.976 \pm 0.055$ | $0.995 \pm 0.013$ | $\mathbf{0.998 \pm 0.005}$ | $0.997 \pm 0.015$ | – |
| surv_whas500 | $0.684 \pm 0.124$ | – | – | – | – | $0.885 \pm 0.110$ | $0.906 \pm 0.135$ | $0.593 \pm 0.238$ | $0.925 \pm 0.045$ | $0.965 \pm 0.018$ | $0.957 \pm 0.041$ | $\mathbf{1.000 \pm 0.000}$ | – |
| surv_gbsg2 | $0.888 \pm 0.133$ | – | – | – | – | $0.853 \pm 0.142$ | $0.904 \pm 0.327$ | $0.763 \pm 0.053$ | $0.964 \pm 0.022$ | $0.974 \pm 0.003$ | $0.895 \pm 0.053$ | $\mathbf{0.985 \pm 0.032}$ | – |
| surv_colon | $0.948 \pm 0.063$ | – | – | – | – | $0.911 \pm 0.139$ | $0.788 \pm 0.419$ | $0.743 \pm 0.028$ | $0.978 \pm 0.019$ | $0.987 \pm 0.021$ | $0.980 \pm 0.013$ | $\mathbf{0.995 \pm 0.017}$ | – |
| surv_grace | $0.945 \pm 0.041$ | – | – | – | – | $0.811 \pm 0.330$ | $0.893 \pm 0.419$ | $0.299 \pm 0.225$ | $0.982 \pm 0.006$ | $0.994 \pm 0.002$ | $0.993 \pm 0.002$ | $\mathbf{0.998 \pm 0.002}$ | – |
| surv_trace | $0.950 \pm 0.064$ | – | – | – | – | $0.838 \pm 0.191$ | $0.977 \pm 0.041$ | $0.316 \pm 0.097$ | $0.993 \pm 0.005$ | $0.994 \pm 0.009$ | $0.996 \pm 0.004$ | $\mathbf{1.000 \pm 0.001}$ | – |
| surv_framingham | $0.905 \pm 0.016$ | – | – | – | – | $0.904 \pm 0.095$ | $0.869 \pm 0.253$ | $0.663 \pm 0.764$ | $0.951 \pm 0.026$ | $0.981 \pm 0.015$ | $0.973 \pm 0.009$ | $\mathbf{0.998 \pm 0.006}$ | – |
| tab_blood_transfusion | $0.613 \pm 0.912$ | – | – | – | – | $0.825 \pm 0.166$ | $0.928 \pm 0.171$ | $0.244 \pm 0.098$ | $0.980 \pm 0.045$ | $\mathbf{0.990 \pm 0.009}$ | $0.986 \pm 0.031$ | $0.978 \pm 0.055$ | – |
| tab_credit_g | $0.188 \pm 0.098$ | – | – | – | – | $0.870 \pm 0.093$ | $0.806 \pm 0.609$ | $0.876 \pm 0.072$ | $0.914 \pm 0.028$ | $0.967 \pm 0.012$ | $0.954 \pm 0.025$ | $\mathbf{0.976 \pm 0.017}$ | – |
| tab_qsar_biodeg | $0.278 \pm 0.104$ | – | – | – | – | $0.801 \pm 0.106$ | $0.119 \pm 0.281$ | $0.640 \pm 0.014$ | $0.772 \pm 0.120$ | $0.954 \pm 0.008$ | $\mathbf{0.973 \pm 0.025}$ | $0.472 \pm 1.128$ | – |
| tab_maternal_risk | $0.912 \pm 0.041$ | – | – | – | – | $0.809 \pm 0.260$ | $0.807 \pm 0.332$ | $0.700 \pm 0.072$ | $0.953 \pm 0.072$ | $\mathbf{0.990 \pm 0.020}$ | $0.970 \pm 0.021$ | $0.972 \pm 0.095$ | – |
| tab_website_phishing | $0.834 \pm 0.687$ | – | – | – | – | $\mathbf{1.000 \pm 0.000}$ | $0.957 \pm 0.168$ | $\mathbf{1.000 \pm 0.000}$ | $0.999 \pm 0.005$ | $\mathbf{1.000 \pm 0.000}$ | $\mathbf{1.000 \pm 0.000}$ | $\mathbf{1.000 \pm 0.000}$ | – |

Table 20: Density (↑) — mean ± 95% CI over seeds

| Dataset | DISCO-TAB | CTGAN | TVAE | GaussCop | TabDDPM | ADS-GAN | PATE-GAN | RTVAE | NFlow | ARF | BayesNet | Marginals | GReaT |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| heart_failure | $0.866 \pm 0.521$ | $0.489 \pm 0.259$ | $1.616 \pm 0.212$ | $0.904 \pm 0.147$ | 0.574 | $1.036 \pm 0.204$ | $0.250 \pm 0.195$ | $\mathbf{1.623 \pm 0.132}$ | $0.942 \pm 0.128$ | $0.954 \pm 0.055$ | $1.059 \pm 0.228$ | $0.139 \pm 0.040$ | – |
| breast_cancer | $0.311 \pm 0.185$ | $0.001 \pm 0.003$ | $0.078 \pm 0.068$ | $0.590 \pm 0.106$ | 0.507 | $0.135 \pm 0.012$ | $0.000 \pm 0.001$ | $0.556 \pm 0.164$ | $0.008 \pm 0.016$ | $0.192 \pm 0.003$ | $\mathbf{0.963 \pm 0.203}$ | $0.011 \pm 0.025$ | – |
| liver_disorders | $1.139 \pm 0.220$ | $0.268 \pm 0.231$ | $\mathbf{1.375 \pm 0.251}$ | $0.946 \pm 0.069$ | 1.162 | $1.090 \pm 0.271$ | $0.370 \pm 0.012$ | $1.100 \pm 0.161$ | $0.874 \pm 0.274$ | $1.019 \pm 0.030$ | $0.956 \pm 0.084$ | $0.258 \pm 0.097$ | – |
| parkinsons | $0.960 \pm 0.632$ | $0.001 \pm 0.006$ | $0.276 \pm 0.178$ | $0.575 \pm 0.217$ | 0.772 | $0.166 \pm 0.143$ | $0.003 \pm 0.005$ | $0.992 \pm 0.094$ | $0.054 \pm 0.058$ | $0.466 \pm 0.026$ | $\mathbf{1.035 \pm 0.045}$ | $0.027 \pm 0.032$ | – |
| obesity | $\mathbf{0.904 \pm 0.034}$ | $0.225 \pm 0.040$ | $0.808 \pm 0.126$ | $0.286 \pm 0.046$ | 0.565 | $0.430 \pm 0.399$ | $0.153 \pm 0.082$ | $0.458 \pm 0.060$ | $0.252 \pm 0.096$ | $0.626 \pm 0.053$ | $0.854 \pm 0.025$ | $0.023 \pm 0.027$ | – |
| diabetes | $1.021 \pm 0.093$ | $0.870 \pm 0.156$ | $\mathbf{1.140 \pm 0.043}$ | $0.641 \pm 0.120$ | – | $0.941 \pm 0.063$ | $0.619 \pm 0.783$ | $0.836 \pm 0.416$ | $0.815 \pm 0.133$ | $0.928 \pm 0.063$ | $0.928 \pm 0.038$ | $0.213 \pm 0.077$ | – |
| german_credit | $0.742 \pm 0.648$ | $0.439 \pm 0.132$ | $\mathbf{2.157 \pm 0.320}$ | $0.558 \pm 0.014$ | – | $0.784 \pm 0.114$ | $0.393 \pm 0.095$ | $0.911 \pm 0.210$ | $0.376 \pm 0.051$ | $0.838 \pm 0.070$ | $1.030 \pm 0.050$ | $0.039 \pm 0.016$ | – |
| bank_marketing | $1.104 \pm 0.343$ | $0.449 \pm 0.112$ | $\mathbf{1.565 \pm 0.044}$ | $0.347 \pm 0.157$ | – | $0.447 \pm 0.163$ | $0.147 \pm 0.076$ | $0.501 \pm 0.476$ | $0.319 \pm 0.041$ | $0.526 \pm 0.058$ | $0.551 \pm 0.026$ | $0.076 \pm 0.150$ | – |
| census_income | $1.189 \pm 0.481$ | $0.328 \pm 0.169$ | $\mathbf{1.667 \pm 0.184}$ | $0.156 \pm 0.170$ | – | $0.281 \pm 0.381$ | $0.055 \pm 0.102$ | $0.187 \pm 0.351$ | $0.132 \pm 0.077$ | $0.321 \pm 0.074$ | $0.434 \pm 0.191$ | $0.004 \pm 0.011$ | – |
| surv_melanoma | $1.107 \pm 0.292$ | – | – | – | – | $1.102 \pm 0.087$ | $0.512 \pm 0.121$ | $\mathbf{1.292 \pm 0.174}$ | $0.923 \pm 0.179$ | $1.027 \pm 0.055$ | $0.963 \pm 0.061$ | $0.504 \pm 0.131$ | – |
| surv_pbc | $0.777 \pm 0.939$ | – | – | – | – | $1.007 \pm 0.066$ | $0.779 \pm 0.305$ | $0.867 \pm 1.303$ | $1.016 \pm 0.063$ | $\mathbf{1.030 \pm 0.061}$ | $1.027 \pm 0.106$ | $0.592 \pm 0.136$ | – |
| surv_pbc3 | $1.111 \pm 0.614$ | – | – | – | – | $0.881 \pm 0.056$ | $0.187 \pm 0.100$ | $\mathbf{1.736 \pm 0.377}$ | $0.853 \pm 0.235$ | $0.907 \pm 0.108$ | $0.977 \pm 0.039$ | $0.066 \pm 0.052$ | – |
| surv_retinopathy | $0.706 \pm 1.299$ | – | – | – | – | $0.770 \pm 0.443$ | $0.341 \pm 0.138$ | $0.680 \pm 0.047$ | $0.508 \pm 0.255$ | $0.936 \pm 0.071$ | $\mathbf{1.053 \pm 0.071}$ | $0.281 \pm 0.122$ | – |
| surv_whas500 | $1.457 \pm 0.428$ | – | – | – | – | $1.035 \pm 0.207$ | $0.152 \pm 0.237$ | $\mathbf{1.693 \pm 0.512}$ | $0.889 \pm 0.054$ | $0.973 \pm 0.048$ | $1.039 \pm 0.066$ | $0.357 \pm 0.281$ | – |
| surv_gbsg2 | $0.870 \pm 0.692$ | – | – | – | – | $0.827 \pm 0.514$ | $0.304 \pm 0.173$ | $\mathbf{0.923 \pm 0.108}$ | $0.728 \pm 0.042$ | $0.905 \pm 0.096$ | $0.592 \pm 0.032$ | $0.168 \pm 0.090$ | – |
| surv_colon | $\mathbf{1.096 \pm 0.079}$ | – | – | – | – | $0.960 \pm 0.116$ | $0.372 \pm 0.570$ | $1.081 \pm 0.013$ | $0.893 \pm 0.279$ | $1.042 \pm 0.022$ | $0.982 \pm 0.011$ | $0.285 \pm 0.047$ | – |
| surv_grace | $1.056 \pm 0.122$ | – | – | – | – | $1.012 \pm 0.083$ | $0.376 \pm 0.806$ | $\mathbf{1.108 \pm 0.088}$ | $1.013 \pm 0.043$ | $1.021 \pm 0.032$ | $0.938 \pm 0.036$ | $0.360 \pm 0.071$ | – |
| surv_trace | $\mathbf{1.091 \pm 0.054}$ | – | – | – | – | $1.082 \pm 0.115$ | $0.525 \pm 0.228$ | $0.898 \pm 0.838$ | $1.023 \pm 0.058$ | $1.041 \pm 0.026$ | $1.059 \pm 0.027$ | $0.397 \pm 0.022$ | – |
| surv_framingham | $\mathbf{1.035 \pm 0.123}$ | – | – | – | – | $0.823 \pm 0.291$ | $0.141 \pm 0.272$ | $0.498 \pm 1.113$ | $0.782 \pm 0.154$ | $0.945 \pm 0.061$ | $0.868 \pm 0.012$ | $0.098 \pm 0.038$ | – |
| tab_blood_transfusion | $0.537 \pm 0.787$ | – | – | – | – | $0.758 \pm 0.137$ | $0.338 \pm 0.613$ | $0.556 \pm 0.052$ | $0.574 \pm 0.228$ | $0.738 \pm 0.035$ | $\mathbf{0.903 \pm 0.076}$ | $0.119 \pm 0.043$ | – |
| tab_credit_g | $0.436 \pm 0.148$ | – | – | – | – | $0.916 \pm 0.278$ | $0.390 \pm 0.481$ | $0.948 \pm 0.050$ | $0.457 \pm 0.079$ | $0.814 \pm 0.062$ | $\mathbf{1.093 \pm 0.042}$ | $0.029 \pm 0.006$ | – |
| tab_qsar_biodeg | $0.043 \pm 0.017$ | – | – | – | – | $0.091 \pm 0.030$ | $0.001 \pm 0.003$ | $0.134 \pm 0.031$ | $0.021 \pm 0.016$ | $0.290 \pm 0.027$ | $\mathbf{0.896 \pm 0.039}$ | $0.008 \pm 0.003$ | – |
| tab_maternal_risk | $\mathbf{1.084 \pm 0.118}$ | – | – | – | – | $0.603 \pm 0.727$ | $0.238 \pm 0.222$ | $0.994 \pm 0.262$ | $0.591 \pm 0.157$ | $0.842 \pm 0.105$ | $0.757 \pm 0.140$ | $0.218 \pm 0.034$ | – |
| tab_website_phishing | $2.197 \pm 2.674$ | – | – | – | – | $2.272 \pm 0.372$ | $1.588 \pm 0.531$ | $2.224 \pm 0.125$ | $2.064 \pm 0.492$ | $\mathbf{2.375 \pm 0.007}$ | $2.274 \pm 0.068$ | $0.979 \pm 0.018$ | – |

Table 21: Coverage (↑) — mean ± 95% CI over seeds

| Dataset | DISCO-TAB | CTGAN | TVAE | GaussCop | TabDDPM | ADS-GAN | PATE-GAN | RTVAE | NFlow | ARF | BayesNet | Marginals | GReaT |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| heart_failure | $0.841 \pm 0.320$ | $0.681 \pm 0.111$ | $0.202 \pm 0.097$ | $0.968 \pm 0.026$ | 0.933 | $0.925 \pm 0.027$ | $0.400 \pm 0.510$ | $0.940 \pm 0.026$ | $0.961 \pm 0.033$ | $\mathbf{0.979 \pm 0.021}$ | $0.947 \pm 0.088$ | $0.096 \pm 0.000$ | – |
| breast_cancer | $0.217 \pm 0.025$ | $0.001 \pm 0.003$ | $0.144 \pm 0.082$ | $0.789 \pm 0.055$ | 0.778 | $0.304 \pm 0.087$ | $0.001 \pm 0.003$ | $0.305 \pm 0.036$ | $0.017 \pm 0.025$ | $0.511 \pm 0.057$ | $\mathbf{0.978 \pm 0.062}$ | $0.008 \pm 0.003$ | – |
| liver_disorders | $0.899 \pm 0.039$ | $0.349 \pm 0.086$ | $0.686 \pm 0.218$ | $0.961 \pm 0.010$ | **0.989** | $0.798 \pm 0.090$ | $0.566 \pm 0.207$ | $0.460 \pm 0.289$ | $0.929 \pm 0.105$ | $0.984 \pm 0.029$ | $0.966 \pm 0.054$ | $0.165 \pm 0.034$ | – |
| parkinsons | $0.556 \pm 0.096$ | $0.004 \pm 0.018$ | $0.267 \pm 0.079$ | $0.812 \pm 0.144$ | 0.872 | $0.318 \pm 0.199$ | $0.015 \pm 0.024$ | $0.476 \pm 0.133$ | $0.171 \pm 0.187$ | $0.825 \pm 0.040$ | $\mathbf{0.981 \pm 0.042}$ | $0.021 \pm 0.009$ | – |
| obesity | $0.882 \pm 0.070$ | $0.304 \pm 0.070$ | $0.283 \pm 0.014$ | $0.295 \pm 0.169$ | 0.725 | $0.540 \pm 0.233$ | $0.221 \pm 0.115$ | $0.561 \pm 0.031$ | $0.354 \pm 0.048$ | $0.866 \pm 0.028$ | $\mathbf{0.941 \pm 0.008}$ | $0.028 \pm 0.004$ | – |
| diabetes | $0.907 \pm 0.047$ | $0.852 \pm 0.024$ | $0.763 \pm 0.123$ | $0.726 \pm 0.154$ | – | $0.902 \pm 0.061$ | $0.529 \pm 0.218$ | $0.861 \pm 0.189$ | $0.833 \pm 0.081$ | $0.952 \pm 0.028$ | $\mathbf{0.953 \pm 0.031}$ | $0.164 \pm 0.022$ | – |
| german_credit | $0.783 \pm 0.627$ | $0.674 \pm 0.127$ | $0.310 \pm 0.043$ | $0.826 \pm 0.006$ | – | $0.905 \pm 0.039$ | $0.522 \pm 0.128$ | $0.909 \pm 0.050$ | $0.670 \pm 0.065$ | $0.953 \pm 0.011$ | $\mathbf{0.984 \pm 0.011}$ | $0.076 \pm 0.018$ | – |
| bank_marketing | $\mathbf{0.898 \pm 0.058}$ | $0.693 \pm 0.099$ | $0.712 \pm 0.146$ | $0.581 \pm 0.015$ | – | $0.616 \pm 0.163$ | $0.237 \pm 0.197$ | $0.632 \pm 0.369$ | $0.502 \pm 0.087$ | $0.805 \pm 0.048$ | $0.788 \pm 0.035$ | $0.006 \pm 0.008$ | – |
| census_income | $\mathbf{0.841 \pm 0.081}$ | $0.527 \pm 0.060$ | $0.703 \pm 0.154$ | $0.288 \pm 0.317$ | – | $0.474 \pm 0.425$ | $0.100 \pm 0.186$ | $0.327 \pm 0.512$ | $0.306 \pm 0.164$ | $0.684 \pm 0.103$ | $0.767 \pm 0.350$ | $0.003 \pm 0.000$ | – |
| surv_melanoma | $0.911 \pm 0.197$ | – | – | – | – | $0.850 \pm 0.229$ | $0.516 \pm 0.441$ | $0.673 \pm 0.166$ | $0.931 \pm 0.168$ | $\mathbf{0.972 \pm 0.032}$ | $0.963 \pm 0.045$ | $0.557 \pm 0.155$ | – |
| surv_pbc | $0.477 \pm 0.977$ | – | – | – | – | $0.896 \pm 0.095$ | $0.778 \pm 0.075$ | $0.554 \pm 0.781$ | $0.940 \pm 0.020$ | $0.989 \pm 0.023$ | $\mathbf{0.993 \pm 0.015}$ | $0.708 \pm 0.120$ | – |
| surv_pbc3 | $0.919 \pm 0.105$ | – | – | – | – | $0.872 \pm 0.029$ | $0.331 \pm 0.294$ | $0.888 \pm 0.005$ | $0.889 \pm 0.079$ | $0.974 \pm 0.031$ | $\mathbf{0.994 \pm 0.005}$ | $0.082 \pm 0.031$ | – |
| surv_retinopathy | $0.702 \pm 1.147$ | – | – | – | – | $0.867 \pm 0.250$ | $0.591 \pm 0.117$ | $0.917 \pm 0.020$ | $0.773 \pm 0.148$ | $\mathbf{0.984 \pm 0.048}$ | $0.980 \pm 0.051$ | $0.612 \pm 0.228$ | – |
| surv_whas500 | $0.915 \pm 0.052$ | – | – | – | – | $0.936 \pm 0.016$ | $0.281 \pm 0.290$ | $0.914 \pm 0.068$ | $0.894 \pm 0.007$ | $0.976 \pm 0.017$ | $\mathbf{0.983 \pm 0.010}$ | $0.113 \pm 0.044$ | – |
| surv_gbsg2 | $0.814 \pm 0.291$ | – | – | – | – | $0.691 \pm 0.396$ | $0.411 \pm 0.316$ | $0.804 \pm 0.046$ | $0.881 \pm 0.121$ | $\mathbf{0.952 \pm 0.026}$ | $0.651 \pm 0.011$ | $0.069 \pm 0.016$ | – |
| surv_colon | $0.958 \pm 0.170$ | – | – | – | – | $0.794 \pm 0.301$ | $0.399 \pm 0.498$ | $0.817 \pm 0.036$ | $0.935 \pm 0.025$ | $\mathbf{0.983 \pm 0.019}$ | $0.938 \pm 0.052$ | $0.255 \pm 0.073$ | – |
| surv_grace | $0.940 \pm 0.040$ | – | – | – | – | $0.739 \pm 0.269$ | $0.464 \pm 0.996$ | $0.481 \pm 0.041$ | $0.942 \pm 0.035$ | $\mathbf{0.981 \pm 0.006}$ | $0.952 \pm 0.034$ | $0.447 \pm 0.081$ | – |
| surv_trace | $0.933 \pm 0.081$ | – | – | – | – | $0.673 \pm 0.156$ | $0.519 \pm 0.165$ | $0.297 \pm 0.427$ | $0.890 \pm 0.048$ | $\mathbf{0.979 \pm 0.025}$ | $0.971 \pm 0.012$ | $0.463 \pm 0.053$ | – |
| surv_framingham | $0.926 \pm 0.034$ | – | – | – | – | $0.844 \pm 0.185$ | $0.202 \pm 0.072$ | $0.405 \pm 0.946$ | $0.828 \pm 0.036$ | $\mathbf{0.962 \pm 0.014}$ | $0.933 \pm 0.023$ | $0.138 \pm 0.027$ | – |
| tab_blood_transfusion | $0.376 \pm 0.375$ | – | – | – | – | $0.592 \pm 0.222$ | $0.282 \pm 0.426$ | $0.251 \pm 0.157$ | $0.785 \pm 0.141$ | $\mathbf{0.933 \pm 0.053}$ | $0.915 \pm 0.029$ | $0.104 \pm 0.022$ | – |
| tab_credit_g | $0.623 \pm 0.075$ | – | – | – | – | $0.909 \pm 0.044$ | $0.422 \pm 0.297$ | $0.917 \pm 0.032$ | $0.735 \pm 0.117$ | $0.947 \pm 0.027$ | $\mathbf{0.988 \pm 0.009}$ | $0.062 \pm 0.010$ | – |
| tab_qsar_biodeg | $0.085 \pm 0.044$ | – | – | – | – | $0.212 \pm 0.065$ | $0.003 \pm 0.005$ | $0.248 \pm 0.042$ | $0.048 \pm 0.032$ | $0.651 \pm 0.046$ | $\mathbf{0.973 \pm 0.014}$ | $0.001 \pm 0.000$ | – |
| tab_maternal_risk | $0.899 \pm 0.021$ | – | – | – | – | $0.597 \pm 0.484$ | $0.270 \pm 0.409$ | $0.661 \pm 0.068$ | $0.715 \pm 0.113$ | $\mathbf{0.918 \pm 0.067}$ | $0.770 \pm 0.068$ | $0.173 \pm 0.046$ | – |
| tab_website_phishing | $0.883 \pm 0.440$ | – | – | – | – | $0.988 \pm 0.005$ | $0.752 \pm 0.176$ | $0.982 \pm 0.020$ | $0.983 \pm 0.013$ | $\mathbf{0.999 \pm 0.002}$ | $0.990 \pm 0.011$ | $0.838 \pm 0.031$ | – |

## Detection / distribution

Table 22: C2ST AUROC (↓, 0.5 best) — mean ± 95% CI over seeds

| Dataset | DISCO-TAB | CTGAN | TVAE | GaussCop | TabDDPM | ADS-GAN | PATE-GAN | RTVAE | NFlow | ARF | BayesNet | Marginals | GReaT |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| heart_failure | 0.969 ± 0.039 | 0.956 ± 0.039 | 0.999 ± 0.003 | **0.902 ± 0.033** | 0.958 | 0.956 ± 0.014 | 0.976 ± 0.012 | 0.999 ± 0.003 | 0.939 ± 0.014 | 0.921 ± 0.048 | 0.932 ± 0.013 | 0.987 ± 0.003 | – |
| breast_cancer | 1.000 ± 0.000 | 1.000 ± 0.000 | 0.993 ± 0.005 | 0.817 ± 0.265 | 0.828 | 0.989 ± 0.014 | 0.998 ± 0.001 | 1.000 ± 0.000 | 0.988 ± 0.008 | 0.860 ± 0.004 | **0.462 ± 0.065** | 0.998 ± 0.002 | – |
| liver_disorders | 0.662 ± 0.068 | 0.960 ± 0.054 | 0.822 ± 0.062 | 0.593 ± 0.034 | **0.461** | 0.879 ± 0.055 | 0.914 ± 0.088 | 1.000 ± 0.000 | 0.648 ± 0.074 | 0.477 ± 0.005 | 0.560 ± 0.024 | 0.972 ± 0.002 | – |
| parkinsons | 0.999 ± 0.005 | 0.995 ± 0.016 | 0.984 ± 0.012 | 0.732 ± 0.150 | 0.728 | 0.976 ± 0.021 | 0.993 ± 0.006 | 0.993 ± 0.016 | 0.951 ± 0.043 | 0.793 ± 0.066 | **0.483 ± 0.053** | 0.983 ± 0.010 | – |
| obesity | 0.998 ± 0.002 | 0.993 ± 0.002 | 0.999 ± 0.001 | 0.991 ± 0.008 | **0.839** | 0.994 ± 0.003 | 0.997 ± 0.008 | 1.000 ± 0.000 | 0.993 ± 0.002 | 0.889 ± 0.002 | 0.977 ± 0.002 | 1.000 ± 0.000 | – |
| diabetes | 1.000 ± 0.001 | 0.951 ± 0.022 | **0.927 ± 0.011** | 0.984 ± 0.002 | – | 0.999 ± 0.001 | 0.997 ± 0.002 | 1.000 ± 0.000 | 0.997 ± 0.002 | 0.989 ± 0.004 | 0.996 ± 0.004 | 0.997 ± 0.002 | – |
| german_credit | 1.000 ± 0.000 | 0.937 ± 0.035 | 0.999 ± 0.000 | 0.877 ± 0.013 | – | 0.892 ± 0.078 | 0.977 ± 0.013 | 1.000 ± 0.001 | 0.924 ± 0.011 | 0.843 ± 0.020 | **0.775 ± 0.028** | 0.996 ± 0.002 | – |
| bank_marketing | 0.888 ± 0.048 | 0.910 ± 0.025 | 0.964 ± 0.023 | 0.985 ± 0.026 | – | 0.969 ± 0.062 | 0.998 ± 0.005 | 1.000 ± 0.000 | 0.968 ± 0.020 | **0.848 ± 0.021** | 0.966 ± 0.006 | 1.000 ± 0.000 | – |
| census_income | 1.000 ± 0.000 | 0.965 ± 0.020 | 0.980 ± 0.014 | 0.990 ± 0.034 | – | 0.995 ± 0.004 | 0.998 ± 0.002 | 1.000 ± 0.001 | 0.993 ± 0.001 | **0.942 ± 0.009** | 0.995 ± 0.008 | 1.000 ± 0.000 | – |
| surv_melanoma | 0.853 ± 0.047 | – | – | – | – | 0.841 ± 0.123 | 0.922 ± 0.147 | 0.999 ± 0.002 | 0.717 ± 0.075 | **0.682 ± 0.048** | 0.714 ± 0.107 | 0.890 ± 0.077 | – |
| surv_pbc | 0.729 ± 0.575 | – | – | – | – | 0.633 ± 0.068 | 0.790 ± 0.093 | 0.875 ± 0.458 | 0.536 ± 0.115 | **0.476 ± 0.077** | 0.485 ± 0.067 | 0.832 ± 0.045 | – |
| surv_pbc3 | 0.913 ± 0.005 | – | – | – | – | 0.912 ± 0.017 | 0.970 ± 0.064 | 1.000 ± 0.000 | 0.849 ± 0.014 | 0.761 ± 0.057 | **0.758 ± 0.013** | 0.986 ± 0.001 | – |
| surv_retinopathy | 0.744 ± 0.580 | – | – | – | – | 0.795 ± 0.202 | 0.948 ± 0.023 | 0.981 ± 0.014 | 0.881 ± 0.065 | 0.653 ± 0.066 | **0.469 ± 0.122** | 0.944 ± 0.012 | – |
| surv_whas500 | 1.000 ± 0.000 | – | – | – | – | 0.785 ± 0.049 | 0.971 ± 0.066 | 1.000 ± 0.000 | 0.675 ± 0.032 | 0.568 ± 0.023 | **0.503 ± 0.058** | 0.985 ± 0.004 | – |
| surv_gbsg2 | 0.857 ± 0.256 | – | – | – | – | 0.914 ± 0.151 | 0.959 ± 0.066 | 1.000 ± 0.000 | 0.767 ± 0.045 | **0.636 ± 0.042** | 0.878 ± 0.016 | 0.994 ± 0.002 | – |
| surv_colon | **0.600 ± 0.062** | – | – | – | – | 0.994 ± 0.003 | 0.997 ± 0.004 | 1.000 ± 0.000 | 0.995 ± 0.004 | 0.910 ± 0.040 | 0.989 ± 0.003 | 0.989 ± 0.005 | – |
| surv_grace | 0.671 ± 0.038 | – | – | – | – | 0.867 ± 0.080 | 0.927 ± 0.167 | 1.000 ± 0.000 | 0.734 ± 0.040 | **0.654 ± 0.030** | 0.699 ± 0.033 | 0.942 ± 0.018 | – |
| surv_trace | 0.999 ± 0.002 | – | – | – | – | 0.995 ± 0.003 | 0.992 ± 0.002 | 0.999 ± 0.003 | 0.993 ± 0.003 | **0.902 ± 0.024** | 0.994 ± 0.004 | 0.985 ± 0.004 | – |
| surv_framingham | 0.804 ± 0.036 | – | – | – | – | 0.910 ± 0.081 | 0.988 ± 0.003 | 0.996 ± 0.012 | 0.864 ± 0.031 | **0.785 ± 0.017** | 0.792 ± 0.007 | 0.988 ± 0.004 | – |
| tab_blood_transfusion | 0.992 ± 0.027 | – | – | – | – | 0.986 ± 0.010 | 0.989 ± 0.014 | 1.000 ± 0.000 | 0.979 ± 0.016 | **0.898 ± 0.017** | 0.971 ± 0.009 | 0.992 ± 0.005 | – |
| tab_credit_g | 1.000 ± 0.000 | – | – | – | – | 0.894 ± 0.044 | 0.977 ± 0.057 | 1.000 ± 0.001 | 0.920 ± 0.014 | 0.847 ± 0.019 | **0.784 ± 0.007** | 0.997 ± 0.001 | – |
| tab_qsar_biodeg | 1.000 ± 0.002 | – | – | – | – | 0.999 ± 0.001 | 0.999 ± 0.003 | 1.000 ± 0.000 | 0.999 ± 0.001 | **0.967 ± 0.011** | 0.987 ± 0.002 | 1.000 ± 0.000 | – |
| tab_maternal_risk | 0.999 ± 0.002 | – | – | – | – | 0.993 ± 0.004 | 0.995 ± 0.008 | 0.999 ± 0.004 | 0.983 ± 0.002 | 0.974 ± 0.015 | **0.973 ± 0.023** | 0.995 ± 0.003 | – |
| tab_website_phishing | 0.771 ± 0.501 | – | – | – | – | 0.635 ± 0.052 | 0.927 ± 0.034 | 0.664 ± 0.047 | 0.707 ± 0.081 | **0.495 ± 0.038** | 0.584 ± 0.034 | 0.904 ± 0.029 | – |

Table 23: $MMD^2$ (↓) — mean ± 95% CI over seeds

| Dataset | DISCO-TAB | CTGAN | TVAE | GaussCop | TabDDPM | ADS-GAN | PATE-GAN | RTVAE | NFlow | ARF | BayesNet | Marginals | GReaT |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| heart_failure | 0.050 ± 0.074 | 0.099 ± 0.049 | 0.392 ± 0.088 | **0.000 ± 0.000** | 0.188 | 0.033 ± 0.010 | 0.242 ± 0.210 | 0.027 ± 0.007 | 0.007 ± 0.011 | 0.002 ± 0.004 | 0.009 ± 0.023 | 0.399 ± 0.025 | – |
| breast_cancer | 0.104 ± 0.055 | 0.218 ± 0.041 | 0.110 ± 0.068 | 0.009 ± 0.003 | 0.227 | 0.058 ± 0.013 | 0.806 ± 0.460 | 0.101 ± 0.028 | 0.039 ± 0.057 | 0.004 ± 0.005 | **0.001 ± 0.002** | 0.573 ± 0.050 | – |
| liver_disorders | 0.051 ± 0.056 | 0.276 ± 0.126 | 0.118 ± 0.148 | 0.001 ± 0.002 | **0.000** | 0.066 ± 0.048 | 0.132 ± 0.156 | 0.066 ± 0.030 | 0.021 ± 0.034 | 0.001 ± 0.002 | 0.003 ± 0.002 | 0.480 ± 0.041 | – |
| parkinsons | 0.083 ± 0.117 | 0.202 ± 0.086 | 0.140 ± 0.050 | **0.000 ± 0.000** | 0.121 | 0.096 ± 0.059 | 0.145 ± 0.265 | 0.119 ± 0.032 | 0.022 ± 0.016 | 0.004 ± 0.010 | **0.000 ± 0.000** | 0.488 ± 0.007 | – |
| obesity | 0.005 ± 0.006 | 0.037 ± 0.027 | 0.081 ± 0.022 | 0.031 ± 0.042 | 0.014 | 0.008 ± 0.020 | 0.109 ± 0.200 | 0.006 ± 0.002 | 0.019 ± 0.010 | 0.004 ± 0.003 | **0.000 ± 0.000** | 0.202 ± 0.030 | – |
| diabetes | 0.024 ± 0.072 | 0.019 ± 0.007 | 0.017 ± 0.019 | 0.019 ± 0.030 | – | 0.007 ± 0.007 | 0.157 ± 0.159 | 0.025 ± 0.086 | 0.019 ± 0.019 | **0.000 ± 0.000** | 0.001 ± 0.001 | 0.381 ± 0.085 | – |
| german_credit | 0.046 ± 0.027 | 0.024 ± 0.030 | 0.299 ± 0.013 | 0.001 ± 0.003 | – | 0.008 ± 0.009 | 0.049 ± 0.039 | 0.007 ± 0.004 | 0.015 ± 0.010 | 0.000 ± 0.000 | **0.000 ± 0.000** | 0.238 ± 0.022 | – |
| bank_marketing | 0.012 ± 0.012 | 0.010 ± 0.006 | 0.033 ± 0.014 | 0.012 ± 0.009 | – | 0.021 ± 0.015 | 0.090 ± 0.130 | 0.019 ± 0.017 | 0.027 ± 0.018 | **0.005 ± 0.003** | 0.007 ± 0.001 | 0.873 ± 0.104 | – |
| census_income | 0.039 ± 0.047 | 0.012 ± 0.010 | 0.039 ± 0.040 | 0.123 ± 0.294 | – | 0.048 ± 0.057 | 0.224 ± 0.647 | 0.065 ± 0.111 | 0.044 ± 0.032 | **0.009 ± 0.004** | 0.026 ± 0.006 | 0.676 ± 0.035 | – |
| surv_melanoma | 0.063 ± 0.192 | – | – | – | – | 0.036 ± 0.064 | 0.223 ± 0.252 | 0.031 ± 0.043 | 0.008 ± 0.013 | **0.000 ± 0.000** | 0.006 ± 0.010 | 0.179 ± 0.051 | – |
| surv_pbc | 0.038 ± 0.054 | – | – | – | – | 0.029 ± 0.032 | 0.106 ± 0.069 | 0.109 ± 0.386 | 0.024 ± 0.023 | 0.000 ± 0.001 | **0.000 ± 0.000** | 0.153 ± 0.025 | – |
| surv_pbc3 | 0.026 ± 0.021 | – | – | – | – | 0.040 ± 0.015 | 0.225 ± 0.202 | 0.049 ± 0.023 | 0.016 ± 0.009 | 0.001 ± 0.002 | **0.000 ± 0.000** | 0.340 ± 0.021 | – |
| surv_retinopathy | 0.042 ± 0.129 | – | – | – | – | 0.018 ± 0.035 | 0.052 ± 0.049 | 0.007 ± 0.006 | 0.028 ± 0.019 | **0.000 ± 0.000** | **0.000 ± 0.000** | 0.045 ± 0.019 | – |
| surv_whas500 | 0.027 ± 0.036 | – | – | – | – | 0.017 ± 0.021 | 0.259 ± 0.275 | 0.034 ± 0.043 | 0.007 ± 0.003 | 0.000 ± 0.000 | **0.000 ± 0.000** | 0.394 ± 0.016 | – |
| surv_gbsg2 | 0.038 ± 0.070 | – | – | – | – | 0.087 ± 0.190 | 0.200 ± 0.235 | 0.020 ± 0.008 | 0.006 ± 0.008 | **0.004 ± 0.001** | 0.039 ± 0.004 | 0.661 ± 0.073 | – |
| surv_colon | 0.007 ± 0.012 | – | – | – | – | 0.052 ± 0.097 | 0.187 ± 0.228 | 0.009 ± 0.010 | 0.007 ± 0.005 | **0.000 ± 0.000** | 0.003 ± 0.002 | 0.386 ± 0.055 | – |
| surv_grace | 0.013 ± 0.029 | – | – | – | – | 0.059 ± 0.072 | 0.422 ± 1.087 | 0.011 ± 0.001 | 0.011 ± 0.020 | **0.000 ± 0.000** | 0.002 ± 0.002 | 0.271 ± 0.073 | – |
| surv_trace | 0.008 ± 0.028 | – | – | – | – | 0.017 ± 0.020 | 0.213 ± 0.094 | 0.140 ± 0.430 | 0.013 ± 0.010 | 0.000 ± 0.001 | **0.000 ± 0.000** | 0.232 ± 0.022 | – |
| surv_framingham | 0.007 ± 0.002 | – | – | – | – | 0.027 ± 0.036 | 0.380 ± 0.213 | 0.188 ± 0.399 | 0.019 ± 0.018 | **0.001 ± 0.002** | 0.003 ± 0.003 | 0.365 ± 0.077 | – |
| tab_blood_transfusion | 0.108 ± 0.355 | – | – | – | – | 0.089 ± 0.049 | 0.292 ± 0.265 | 0.026 ± 0.017 | 0.014 ± 0.017 | **0.000 ± 0.000** | 0.012 ± 0.008 | 0.632 ± 0.039 | – |
| tab_credit_g | 0.085 ± 0.016 | – | – | – | – | 0.010 ± 0.017 | 0.079 ± 0.127 | 0.006 ± 0.003 | 0.014 ± 0.007 | 0.001 ± 0.002 | **0.000 ± 0.000** | 0.244 ± 0.031 | – |
| tab_qsar_biodeg | 0.067 ± 0.073 | – | – | – | – | 0.034 ± 0.027 | 0.568 ± 0.387 | 0.025 ± 0.016 | 0.036 ± 0.056 | 0.004 ± 0.003 | **0.002 ± 0.001** | 0.680 ± 0.030 | – |
| tab_maternal_risk | 0.007 ± 0.010 | – | – | – | – | 0.066 ± 0.165 | 0.316 ± 0.525 | 0.019 ± 0.023 | 0.017 ± 0.019 | **0.000 ± 0.001** | 0.026 ± 0.010 | 0.360 ± 0.064 | – |
| tab_website_phishing | 0.043 ± 0.155 | – | – | – | – | 0.005 ± 0.009 | 0.083 ± 0.069 | 0.003 ± 0.005 | 0.009 ± 0.008 | **0.000 ± 0.000** | 0.000 ± 0.000 | 0.034 ± 0.006 | – |

Table 24: Fréchet-PCA (↓) — mean ± 95% CI over seeds

| Dataset | DISCO-TAB | CTGAN | TVAE | GaussCop | TabDDPM | ADS-GAN | PATE-GAN | RTVAE | NFlow | ARF | BayesNet | Marginals | GReaT |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| heart_failure | 2.197 ± 1.768 | 4.871 ± 3.551 | 10.777 ± 1.890 | **0.309 ± 0.172** | 36.258 | 1.474 ± 0.435 | 11.078 ± 7.624 | 1.620 ± 0.978 | 0.651 ± 0.427 | 0.331 ± 0.176 | 0.664 ± 0.694 | 36.526 ± 6.670 | – |
| breast_cancer | 24.230 ± 35.796 | 29.409 ± 5.766 | 13.849 ± 4.611 | 0.790 ± 0.154 | 262.006 | 5.478 ± 2.510 | 107.847 ± 59.046 | 12.971 ± 1.348 | 11.090 ± 12.826 | 1.360 ± 0.375 | **0.471 ± 0.211** | 216.031 ± 50.704 | – |
| liver_disorders | 0.799 ± 0.631 | 5.592 ± 4.179 | 2.911 ± 0.827 | 0.102 ± 0.040 | 0.163 | 1.003 ± 1.006 | 3.041 ± 2.091 | 1.369 ± 0.855 | 0.689 ± 0.564 | **0.071 ± 0.009** | 0.171 ± 0.140 | 26.046 ± 2.381 | – |
| parkinsons | 6.899 ± 3.221 | 20.827 ± 6.418 | 12.444 ± 2.154 | 0.514 ± 0.286 | 77.241 | 7.191 ± 3.318 | 19.322 ± 15.333 | 12.159 ± 1.849 | 5.990 ± 2.586 | 1.317 ± 0.600 | **0.304 ± 0.113** | 83.740 ± 5.762 | – |
| obesity | 0.287 ± 0.337 | 1.640 ± 0.992 | 2.601 ± 0.085 | 1.213 ± 1.935 | 2.692 | 0.349 ± 0.768 | 4.702 ± 5.667 | 0.368 ± 0.063 | 0.948 ± 0.500 | 0.193 ± 0.057 | **0.036 ± 0.023** | 10.496 ± 2.717 | – |
| diabetes | 1.776 ± 5.212 | 1.105 ± 0.353 | 0.953 ± 0.723 | 0.542 ± 0.953 | – | 0.263 ± 0.077 | 3.167 ± 1.559 | 0.796 ± 1.901 | 0.996 ± 1.917 | **0.058 ± 0.023** | 0.062 ± 0.015 | 27.853 ± 17.394 | – |
| german_credit | 1.858 ± 5.398 | 1.687 ± 1.583 | 8.982 ± 0.810 | 0.254 ± 0.129 | – | 0.435 ± 0.334 | 2.723 ± 1.713 | 0.581 ± 0.146 | 1.105 ± 0.711 | 0.103 ± 0.027 | **0.071 ± 0.031** | 16.535 ± 1.822 | – |
| bank_marketing | 0.856 ± 0.799 | 0.585 ± 0.580 | 2.557 ± 1.284 | 3.818 ± 12.599 | – | 1.044 ± 0.846 | 12.075 ± 30.720 | 0.902 ± 0.377 | 8.494 ± 29.701 | **0.127 ± 0.061** | 0.237 ± 0.236 | 1835.097 ± 5924.108 | – |
| census_income | 1.344 ± 1.265 | 0.799 ± 0.778 | 3.091 ± 1.218 | 21.677 ± 39.564 | – | 1.784 ± 1.003 | 7.884 ± 15.023 | 2.709 ± 1.327 | 1.540 ± 1.072 | **0.188 ± 0.072** | 0.891 ± 0.055 | 113.086 ± 33.070 | – |
| surv_melanoma | 0.975 ± 2.300 | – | – | – | – | 0.801 ± 1.452 | 4.625 ± 4.366 | 0.859 ± 0.984 | 0.386 ± 0.476 | **0.073 ± 0.027** | 0.271 ± 0.304 | 4.857 ± 2.184 | – |
| surv_pbc | 1.520 ± 2.591 | – | – | – | – | 0.475 ± 0.287 | 1.865 ± 0.658 | 2.770 ± 9.431 | 0.439 ± 0.332 | 0.057 ± 0.052 | **0.057 ± 0.030** | 4.204 ± 0.987 | – |
| surv_pbc3 | 1.247 ± 0.927 | – | – | – | – | 1.710 ± 1.121 | 11.014 ± 5.263 | 2.785 ± 0.549 | 1.009 ± 0.265 | 0.287 ± 0.064 | **0.222 ± 0.061** | 28.665 ± 2.528 | – |
| surv_retinopathy | 0.982 ± 2.703 | – | – | – | – | 0.824 ± 0.867 | 2.196 ± 1.095 | 0.966 ± 0.197 | 1.526 ± 0.668 | 0.320 ± 0.102 | **0.035 ± 0.050** | 2.611 ± 0.508 | – |
| surv_whas500 | 1.544 ± 1.357 | – | – | – | – | 0.955 ± 0.983 | 16.775 ± 10.154 | 1.925 ± 1.097 | 0.928 ± 1.290 | 0.244 ± 0.080 | **0.205 ± 0.061** | 67.247 ± 23.059 | – |
| surv_gbsg2 | 11.501 ± 46.501 | – | – | – | – | 2.080 ± 3.255 | 8.209 ± 14.802 | 1.058 ± 0.462 | 0.470 ± 0.379 | **0.134 ± 0.020** | 0.835 ± 0.070 | 62.397 ± 16.244 | – |
| surv_colon | 0.417 ± 1.688 | – | – | – | – | 1.398 ± 2.134 | 5.128 ± 3.767 | 0.730 ± 0.497 | 0.493 ± 0.805 | **0.054 ± 0.009** | 0.180 ± 0.032 | 26.894 ± 8.398 | – |
| surv_grace | 0.219 ± 0.264 | – | – | – | – | 0.998 ± 0.921 | 9.721 ± 23.582 | 0.345 ± 0.082 | 0.194 ± 0.296 | **0.014 ± 0.016** | 0.086 ± 0.064 | 11.076 ± 5.627 | – |
| surv_trace | 0.308 ± 1.020 | – | – | – | – | 0.659 ± 0.398 | 3.817 ± 2.281 | 3.200 ± 8.118 | 0.410 ± 0.376 | 0.022 ± 0.017 | **0.016 ± 0.008** | 8.624 ± 0.478 | – |
| surv_framingham | 0.209 ± 0.105 | – | – | – | – | 0.623 ± 0.708 | 10.909 ± 10.687 | 4.702 ± 9.962 | 0.590 ± 0.403 | **0.048 ± 0.019** | 0.168 ± 0.038 | 29.902 ± 8.845 | – |
| tab_blood_transfusion | 1.560 ± 3.196 | – | – | – | – | 1.406 ± 0.086 | 3.423 ± 2.467 | 1.012 ± 0.429 | 0.921 ± 0.542 | 0.212 ± 0.073 | **0.146 ± 0.084** | 38.142 ± 4.669 | – |
| tab_credit_g | 2.865 ± 0.397 | – | – | – | – | 0.532 ± 0.671 | 3.872 ± 4.041 | 0.574 ± 0.243 | 0.919 ± 0.376 | 0.153 ± 0.105 | **0.064 ± 0.013** | 17.004 ± 2.819 | – |

| Dataset | DISCO-TAB | CTGAN | TVAE | GaussCop | TabDDPM | ADS-GAN | PATE-GAN | RTVAE | NFlow | ARF | BayesNet | Marginals | GReaT |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| tab_qsar_biodeg | 14.141 ± 1.739 | – | – | – | – | 4.580 ± 1.868 | 59.061 ± 54.876 | 9.061 ± 5.276 | 10.922 ± 9.358 | 2.540 ± 1.324 | **0.348 ± 0.255** | 754.443 ± 102.561 | – |
| tab_maternal_risk | 0.275 ± 0.121 | – | – | – | – | 1.322 ± 3.054 | 5.393 ± 6.872 | 0.583 ± 0.543 | 0.632 ± 0.218 | **0.082 ± 0.052** | 0.710 ± 0.568 | 18.273 ± 4.293 | – |
| tab_website_phishing | 0.562 ± 1.887 | – | – | – | – | 0.119 ± 0.090 | 1.272 ± 0.917 | 0.099 ± 0.087 | 0.198 ± 0.118 | **0.031 ± 0.011** | 0.052 ± 0.014 | 0.612 ± 0.098 | – |

Table 25: Wasserstein (↓) — mean ± 95% CI over seeds

| Dataset | DISCO-TAB | CTGAN | TVAE | GaussCop | TabDDPM | ADS-GAN | PATE-GAN | RTVAE | NFlow | ARF | BayesNet | Marginals | GReaT |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| heart_failure | 0.319 ± 0.238 | 0.391 ± 0.066 | 0.597 ± 0.056 | **0.111 ± 0.027** | 0.958 | 0.247 ± 0.038 | 0.685 ± 0.228 | 0.308 ± 0.058 | 0.142 ± 0.063 | 0.125 ± 0.023 | 0.124 ± 0.096 | 1.085 ± 0.106 | – |
| breast_cancer | 0.309 ± 0.054 | 0.692 ± 0.122 | 0.393 ± 0.103 | 0.150 ± 0.026 | 1.851 | 0.309 ± 0.043 | 1.567 ± 0.561 | 0.510 ± 0.065 | 0.167 ± 0.172 | 0.107 ± 0.063 | **0.090 ± 0.021** | 2.098 ± 0.235 | – |
| liver_disorders | 0.378 ± 0.237 | 0.863 ± 0.148 | 0.423 ± 0.139 | **0.109 ± 0.024** | 0.113 | 0.356 ± 0.066 | 0.486 ± 0.236 | 0.425 ± 0.089 | 0.236 ± 0.149 | 0.112 ± 0.019 | 0.121 ± 0.019 | 1.872 ± 0.095 | – |
| parkinsons | 0.264 ± 0.139 | 0.622 ± 0.122 | 0.406 ± 0.061 | 0.120 ± 0.014 | 1.227 | 0.384 ± 0.126 | 0.471 ± 0.482 | 0.540 ± 0.058 | 0.164 ± 0.045 | 0.138 ± 0.076 | **0.081 ± 0.029** | 1.477 ± 0.048 | – |
| obesity | 0.119 ± 0.042 | 0.308 ± 0.151 | 0.358 ± 0.054 | 0.296 ± 0.123 | 0.300 | 0.172 ± 0.133 | 0.539 ± 0.618 | 0.175 ± 0.016 | 0.187 ± 0.091 | 0.146 ± 0.030 | **0.037 ± 0.002** | 0.727 ± 0.069 | – |
| diabetes | 0.235 ± 0.361 | 0.245 ± 0.016 | 0.164 ± 0.063 | 0.184 ± 0.096 | – | 0.144 ± 0.036 | 0.471 ± 0.165 | 0.185 ± 0.196 | 0.207 ± 0.098 | 0.067 ± 0.008 | **0.026 ± 0.000** | 1.483 ± 0.399 | – |
| german_credit | 0.225 ± 0.341 | 0.252 ± 0.099 | 0.663 ± 0.091 | 0.077 ± 0.028 | – | 0.157 ± 0.070 | 0.406 ± 0.157 | 0.185 ± 0.024 | 0.190 ± 0.115 | 0.069 ± 0.015 | **0.056 ± 0.006** | 1.113 ± 0.097 | – |
| bank_marketing | 0.162 ± 0.050 | 0.133 ± 0.066 | 0.245 ± 0.048 | 0.272 ± 0.184 | – | 0.184 ± 0.067 | 0.648 ± 0.882 | 0.201 ± 0.019 | 0.221 ± 0.179 | 0.087 ± 0.012 | **0.072 ± 0.022** | 7.511 ± 8.825 | – |
| census_income | 0.279 ± 0.245 | 0.141 ± 0.084 | 0.264 ± 0.113 | 0.762 ± 1.192 | – | 0.248 ± 0.047 | 0.623 ± 0.925 | 0.202 ± 0.050 | 0.194 ± 0.096 | 0.113 ± 0.007 | **0.059 ± 0.001** | 3.209 ± 0.490 | – |
| surv_melanoma | 0.327 ± 0.570 | – | – | – | – | 0.273 ± 0.278 | 0.726 ± 0.538 | 0.333 ± 0.108 | 0.183 ± 0.068 | **0.108 ± 0.016** | 0.138 ± 0.082 | 0.686 ± 0.139 | – |
| surv_pbc | 0.262 ± 0.135 | – | – | – | – | 0.226 ± 0.064 | 0.417 ± 0.154 | 0.354 ± 0.853 | 0.215 ± 0.095 | 0.058 ± 0.076 | **0.054 ± 0.044** | 0.510 ± 0.041 | – |
| surv_pbc3 | 0.227 ± 0.144 | – | – | – | – | 0.290 ± 0.122 | 0.718 ± 0.237 | 0.385 ± 0.055 | 0.168 ± 0.022 | 0.113 ± 0.028 | **0.086 ± 0.026** | 1.194 ± 0.035 | – |
| surv_retinopathy | 0.164 ± 0.153 | – | – | – | – | 0.204 ± 0.179 | 0.305 ± 0.222 | 0.193 ± 0.075 | 0.215 ± 0.153 | 0.066 ± 0.017 | **0.049 ± 0.017** | 0.323 ± 0.024 | – |
| surv_whas500 | 0.223 ± 0.155 | – | – | – | – | 0.176 ± 0.083 | 0.615 ± 0.359 | 0.298 ± 0.126 | 0.127 ± 0.031 | 0.070 ± 0.015 | **0.063 ± 0.018** | 1.128 ± 0.056 | – |
| surv_gbsg2 | 0.270 ± 0.367 | – | – | – | – | 0.405 ± 0.369 | 0.801 ± 0.599 | 0.321 ± 0.052 | 0.124 ± 0.032 | 0.132 ± 0.017 | **0.069 ± 0.034** | 2.844 ± 0.265 | – |
| surv_colon | 0.137 ± 0.011 | – | – | – | – | 0.268 ± 0.234 | 0.501 ± 0.337 | 0.166 ± 0.051 | 0.117 ± 0.081 | **0.043 ± 0.008** | 0.071 ± 0.007 | 1.267 ± 0.162 | – |
| surv_grace | 0.159 ± 0.104 | – | – | – | – | 0.336 ± 0.199 | 0.967 ± 1.322 | 0.247 ± 0.036 | 0.153 ± 0.104 | **0.048 ± 0.018** | 0.080 ± 0.017 | 0.941 ± 0.126 | – |
| surv_trace | 0.099 ± 0.143 | – | – | – | – | 0.209 ± 0.034 | 0.547 ± 0.186 | 0.432 ± 0.568 | 0.175 ± 0.088 | 0.040 ± 0.016 | **0.037 ± 0.010** | 0.885 ± 0.017 | – |
| surv_framingham | 0.135 ± 0.036 | – | – | – | – | 0.232 ± 0.127 | 0.985 ± 0.375 | 0.581 ± 0.872 | 0.214 ± 0.130 | 0.053 ± 0.010 | **0.052 ± 0.005** | 1.341 ± 0.154 | – |
| tab_blood_transfusion | 0.397 ± 0.744 | – | – | – | – | 0.459 ± 0.094 | 0.658 ± 0.258 | 0.323 ± 0.090 | 0.167 ± 0.116 | **0.084 ± 0.036** | 0.126 ± 0.008 | 2.406 ± 0.154 | – |
| tab_credit_g | 0.178 ± 0.076 | – | – | – | – | 0.177 ± 0.186 | 0.464 ± 0.445 | 0.194 ± 0.033 | 0.173 ± 0.042 | 0.086 ± 0.035 | **0.060 ± 0.019** | 1.133 ± 0.086 | – |
| tab_qsar_biodeg | 0.281 ± 0.076 | – | – | – | – | 0.192 ± 0.058 | 0.787 ± 0.514 | 0.244 ± 0.025 | 0.184 ± 0.084 | 0.104 ± 0.020 | **0.068 ± 0.013** | 3.328 ± 0.200 | – |
| tab_maternal_risk | 0.146 ± 0.084 | – | – | – | – | 0.323 ± 0.394 | 0.694 ± 0.565 | 0.304 ± 0.062 | 0.174 ± 0.060 | **0.124 ± 0.016** | 0.255 ± 0.059 | 1.182 ± 0.096 | – |
| tab_website_phishing | – | – | – | – | – | – | – | – | – | – | – | – | – |

Table 26: Association error (↓) — mean ± 95% CI over seeds

| Dataset | DISCO-TAB | CTGAN | TVAE | GaussCop | TabDDPM | ADS-GAN | PATE-GAN | RTVAE | NFlow | ARF | BayesNet | Marginals | GReaT |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| heart_failure | 0.057 ± 0.005 | 0.059 ± 0.005 | 0.070 ± 0.006 | 0.055 ± 0.007 | 0.150 | 0.069 ± 0.012 | 0.094 ± 0.022 | 0.060 ± 0.012 | 0.058 ± 0.008 | **0.050 ± 0.020** | 0.071 ± 0.013 | 0.063 ± 0.009 | – |
| breast_cancer | 0.278 ± 0.019 | 0.347 ± 0.010 | 0.195 ± 0.007 | **0.051 ± 0.017** | 0.188 | 0.107 ± 0.028 | 0.290 ± 0.050 | 0.240 ± 0.022 | 0.298 ± 0.045 | 0.099 ± 0.008 | 0.053 ± 0.028 | 0.357 ± 0.012 | – |
| liver_disorders | 0.077 ± 0.095 | 0.222 ± 0.028 | 0.156 ± 0.099 | 0.052 ± 0.005 | 0.080 | 0.118 ± 0.122 | 0.216 ± 0.077 | 0.207 ± 0.025 | 0.171 ± 0.019 | **0.051 ± 0.010** | 0.095 ± 0.055 | 0.232 ± 0.048 | – |
| parkinsons | 0.242 ± 0.046 | 0.444 ± 0.026 | 0.335 ± 0.008 | 0.056 ± 0.018 | 0.224 | 0.175 ± 0.029 | 0.366 ± 0.025 | 0.369 ± 0.044 | 0.378 ± 0.054 | 0.133 ± 0.021 | **0.047 ± 0.007** | 0.450 ± 0.016 | – |
| obesity | 0.047 ± 0.002 | 0.103 ± 0.004 | 0.086 ± 0.008 | 0.069 ± 0.006 | 0.115 | 0.039 ± 0.058 | 0.091 ± 0.017 | 0.074 ± 0.008 | 0.087 ± 0.013 | 0.037 ± 0.010 | **0.031 ± 0.001** | 0.105 ± 0.002 | – |
| diabetes | 0.055 ± 0.056 | 0.090 ± 0.048 | 0.056 ± 0.028 | 0.071 ± 0.014 | – | 0.027 ± 0.003 | 0.079 ± 0.027 | 0.075 ± 0.018 | 0.065 ± 0.010 | **0.016 ± 0.004** | 0.037 ± 0.008 | 0.098 ± 0.006 | – |
| german_credit | 0.108 ± 0.509 | 0.085 ± 0.004 | 0.111 ± 0.014 | 0.072 ± 0.006 | – | 0.048 ± 0.010 | 0.082 ± 0.003 | 0.063 ± 0.013 | 0.080 ± 0.004 | 0.047 ± 0.009 | **0.033 ± 0.005** | 0.086 ± 0.003 | – |
| bank_marketing | 0.038 ± 0.004 | 0.075 ± 0.021 | 0.079 ± 0.023 | 0.067 ± 0.004 | – | **0.034 ± 0.014** | 0.084 ± 0.009 | 0.043 ± 0.023 | 0.072 ± 0.005 | 0.038 ± 0.005 | 0.048 ± 0.009 | 0.089 ± 0.003 | – |
| census_income | 0.059 ± 0.024 | 0.081 ± 0.007 | 0.075 ± 0.013 | 0.108 ± 0.017 | – | 0.044 ± 0.011 | 0.116 ± 0.015 | 0.070 ± 0.036 | 0.104 ± 0.007 | 0.056 ± 0.004 | **0.025 ± 0.007** | 0.127 ± 0.002 | – |
| surv_melanoma | 0.074 ± 0.023 | – | – | – | – | 0.086 ± 0.068 | 0.119 ± 0.076 | 0.113 ± 0.023 | 0.109 ± 0.086 | 0.059 ± 0.026 | **0.055 ± 0.005** | 0.130 ± 0.008 | – |
| surv_pbc | 0.109 ± 0.112 | – | – | – | – | 0.060 ± 0.045 | 0.087 ± 0.016 | 0.082 ± 0.046 | 0.073 ± 0.026 | **0.051 ± 0.037** | 0.056 ± 0.025 | 0.090 ± 0.037 | – |
| surv_pbc3 | 0.081 ± 0.019 | – | – | – | – | 0.068 ± 0.034 | 0.099 ± 0.013 | 0.086 ± 0.009 | 0.096 ± 0.002 | 0.056 ± 0.019 | **0.051 ± 0.007** | 0.097 ± 0.010 | – |
| surv_retinopathy | 0.082 ± 0.086 | – | – | – | – | 0.088 ± 0.029 | 0.120 ± 0.010 | 0.093 ± 0.022 | 0.097 ± 0.038 | 0.056 ± 0.020 | **0.040 ± 0.048** | 0.120 ± 0.007 | – |
| surv_whas500 | 0.069 ± 0.020 | – | – | – | – | 0.055 ± 0.015 | 0.084 ± 0.029 | 0.064 ± 0.005 | 0.066 ± 0.009 | 0.048 ± 0.003 | **0.046 ± 0.008** | 0.071 ± 0.004 | – |
| surv_gbsg2 | 0.062 ± 0.042 | – | – | – | – | 0.098 ± 0.088 | 0.126 ± 0.008 | 0.090 ± 0.043 | 0.092 ± 0.036 | **0.044 ± 0.021** | 0.090 ± 0.002 | 0.121 ± 0.030 | – |
| surv_colon | 0.038 ± 0.017 | – | – | – | – | 0.082 ± 0.055 | 0.061 ± 0.013 | 0.043 ± 0.013 | 0.049 ± 0.015 | **0.034 ± 0.013** | 0.048 ± 0.007 | 0.057 ± 0.003 | – |
| surv_grace | 0.039 ± 0.020 | – | – | – | – | 0.044 ± 0.018 | 0.088 ± 0.060 | 0.060 ± 0.023 | 0.040 ± 0.039 | **0.021 ± 0.012** | 0.061 ± 0.033 | 0.055 ± 0.010 | – |
| surv_trace | 0.034 ± 0.019 | – | – | – | – | 0.062 ± 0.053 | 0.095 ± 0.030 | 0.104 ± 0.045 | 0.066 ± 0.040 | 0.021 ± 0.016 | **0.019 ± 0.007** | 0.109 ± 0.007 | – |
| surv_framingham | 0.023 ± 0.014 | – | – | – | – | 0.032 ± 0.005 | 0.144 ± 0.045 | 0.107 ± 0.003 | 0.113 ± 0.020 | **0.018 ± 0.004** | 0.073 ± 0.011 | 0.152 ± 0.008 | – |
| tab_blood_transfusion | 0.233 ± 0.628 | – | – | – | – | 0.120 ± 0.110 | 0.369 ± 0.086 | 0.315 ± 0.176 | 0.266 ± 0.145 | 0.066 ± 0.039 | **0.064 ± 0.033** | 0.402 ± 0.057 | – |
| tab_credit_g | 0.064 ± 0.003 | – | – | – | – | 0.047 ± 0.008 | 0.081 ± 0.012 | 0.061 ± 0.007 | 0.077 ± 0.006 | 0.047 ± 0.005 | **0.034 ± 0.005** | 0.086 ± 0.003 | – |
| tab_qsar_biodeg | 0.154 ± 0.022 | – | – | – | – | 0.083 ± 0.009 | 0.147 ± 0.019 | 0.117 ± 0.001 | 0.142 ± 0.008 | 0.063 ± 0.008 | **0.032 ± 0.005** | 0.147 ± 0.003 | – |
| tab_maternal_risk | 0.079 ± 0.068 | – | – | – | – | 0.081 ± 0.037 | 0.166 ± 0.044 | 0.114 ± 0.044 | 0.160 ± 0.034 | **0.041 ± 0.010** | 0.084 ± 0.041 | 0.205 ± 0.010 | – |
| tab_website_phishing | 0.081 ± 0.082 | – | – | – | – | 0.058 ± 0.013 | 0.105 ± 0.005 | 0.076 ± 0.012 | 0.067 ± 0.014 | **0.047 ± 0.014** | 0.051 ± 0.010 | 0.106 ± 0.008 | – |

## SDMetrics quality report

Table 27: Overall quality (↑) — mean ± 95% CI over seeds

| Dataset | DISCO-TAB | CTGAN | TVAE | GaussCop | TabDDPM | ADS-GAN | PATE-GAN | RTVAE | NFlow | ARF | BayesNet | Marginals | GReaT |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| heart_failure | 0.374 ± 0.062 | 0.369 ± 0.021 | 0.307 ± 0.016 | **0.423 ± 0.010** | 0.341 | 0.387 ± 0.010 | 0.297 ± 0.060 | 0.348 ± 0.017 | 0.415 ± 0.021 | 0.419 ± 0.004 | 0.420 ± 0.028 | 0.317 ± 0.015 | – |
| breast_cancer | 0.707 ± 0.025 | 0.600 ± 0.005 | 0.719 ± 0.033 | 0.849 ± 0.004 | 0.704 | 0.783 ± 0.017 | 0.404 ± 0.149 | 0.614 ± 0.031 | 0.747 ± 0.003 | 0.840 ± 0.022 | **0.876 ± 0.008** | 0.522 ± 0.009 | – |
| liver_disorders | 0.805 ± 0.080 | 0.597 ± 0.056 | 0.705 ± 0.040 | 0.870 ± 0.003 | 0.866 | 0.787 ± 0.024 | 0.635 ± 0.104 | 0.646 ± 0.040 | 0.760 ± 0.052 | **0.871 ± 0.014** | 0.851 ± 0.024 | 0.564 ± 0.021 | – |
| parkinsons | 0.766 ± 0.041 | 0.556 ± 0.036 | 0.669 ± 0.012 | 0.847 ± 0.012 | 0.726 | 0.731 ± 0.059 | 0.603 ± 0.128 | 0.564 ± 0.020 | 0.724 ± 0.019 | 0.813 ± 0.030 | **0.861 ± 0.006** | 0.520 ± 0.018 | – |
| obesity | 0.843 ± 0.054 | 0.775 ± 0.035 | 0.747 ± 0.050 | 0.816 ± 0.008 | 0.850 | 0.836 ± 0.053 | 0.718 ± 0.200 | 0.695 ± 0.033 | 0.792 ± 0.019 | 0.894 ± 0.032 | **0.914 ± 0.004** | 0.578 ± 0.006 | – |
| diabetes | 0.385 ± 0.023 | 0.391 ± 0.005 | 0.399 ± 0.009 | 0.404 ± 0.014 | – | 0.396 ± 0.016 | 0.336 ± 0.060 | 0.388 ± 0.022 | 0.388 ± 0.025 | 0.421 ± 0.003 | **0.428 ± 0.000** | 0.272 ± 0.011 | – |
| german_credit | 0.715 ± 0.129 | 0.761 ± 0.019 | 0.557 ± 0.073 | 0.821 ± 0.007 | – | 0.816 ± 0.015 | 0.732 ± 0.022 | 0.785 ± 0.023 | 0.750 ± 0.014 | 0.840 ± 0.017 | **0.863 ± 0.013** | 0.542 ± 0.010 | – |
| bank_marketing | 0.704 ± 0.005 | 0.709 ± 0.017 | 0.690 ± 0.024 | 0.702 ± 0.014 | – | 0.648 ± 0.100 | 0.511 ± 0.111 | 0.586 ± 0.116 | 0.614 ± 0.053 | **0.718 ± 0.013** | 0.692 ± 0.007 | 0.404 ± 0.004 | – |
| census_income | 0.518 ± 0.085 | 0.590 ± 0.026 | 0.607 ± 0.044 | 0.597 ± 0.060 | – | 0.541 ± 0.089 | 0.402 ± 0.289 | 0.531 ± 0.106 | 0.502 ± 0.049 | **0.611 ± 0.010** | 0.585 ± 0.033 | 0.305 ± 0.002 | – |

| Dataset | DISCO-TAB | CTGAN | TVAE | GaussCop | TabDDPM | ADS-GAN | PATE-GAN | RTVAE | NFlow | ARF | BayesNet | Marginals | GReaT |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| surv_melanoma | $0.341 \pm 0.124$ | – | – | – | – | $0.352 \pm 0.049$ | $0.285 \pm 0.119$ | $0.317 \pm 0.033$ | $0.376 \pm 0.007$ | $\mathbf{0.388 \pm 0.006}$ | $0.387 \pm 0.014$ | $0.318 \pm 0.006$ | – |
| surv_pbc | $0.447 \pm 0.127$ | – | – | – | – | $0.472 \pm 0.224$ | $0.445 \pm 0.228$ | $0.390 \pm 0.086$ | $0.500 \pm 0.280$ | $0.520 \pm 0.263$ | $\mathbf{0.528 \pm 0.287}$ | $0.404 \pm 0.150$ | – |
| surv_pbc3 | $0.431 \pm 0.184$ | – | – | – | – | $0.434 \pm 0.227$ | $0.354 \pm 0.198$ | $0.378 \pm 0.205$ | $0.452 \pm 0.198$ | $0.475 \pm 0.216$ | $\mathbf{0.488 \pm 0.220}$ | $0.344 \pm 0.158$ | – |
| surv_retinopathy | $0.768 \pm 0.324$ | – | – | – | – | $0.763 \pm 0.106$ | $0.606 \pm 0.073$ | $0.618 \pm 0.016$ | $0.720 \pm 0.021$ | $0.846 \pm 0.013$ | $\mathbf{0.899 \pm 0.016}$ | $0.623 \pm 0.030$ | – |
| surv_whas500 | $0.516 \pm 0.068$ | – | – | – | – | $0.534 \pm 0.018$ | $0.424 \pm 0.078$ | $0.467 \pm 0.061$ | $0.530 \pm 0.044$ | $0.546 \pm 0.028$ | $\mathbf{0.560 \pm 0.033}$ | $0.419 \pm 0.030$ | – |
| surv_gbsg2 | $0.671 \pm 0.222$ | – | – | – | – | $0.605 \pm 0.208$ | $0.562 \pm 0.115$ | $0.457 \pm 0.120$ | $0.720 \pm 0.110$ | $\mathbf{0.742 \pm 0.128}$ | $0.605 \pm 0.083$ | $0.437 \pm 0.135$ | – |
| surv_colon | $0.876 \pm 0.360$ | – | – | – | – | $0.864 \pm 0.065$ | $0.669 \pm 0.094$ | $0.842 \pm 0.007$ | $0.827 \pm 0.105$ | $0.926 \pm 0.022$ | $\mathbf{0.929 \pm 0.003}$ | $0.634 \pm 0.037$ | – |
| surv_grace | $0.381 \pm 0.017$ | – | – | – | – | $0.340 \pm 0.048$ | $0.265 \pm 0.200$ | $0.350 \pm 0.012$ | $0.379 \pm 0.026$ | $\mathbf{0.404 \pm 0.005}$ | $0.396 \pm 0.004$ | $0.316 \pm 0.010$ | – |
| surv_trace | $0.403 \pm 0.028$ | – | – | – | – | $0.381 \pm 0.018$ | $0.330 \pm 0.025$ | $0.347 \pm 0.099$ | $0.392 \pm 0.016$ | $0.418 \pm 0.003$ | $\mathbf{0.418 \pm 0.001}$ | $0.307 \pm 0.006$ | – |
| surv_framingham | $\mathbf{0.898 \pm 0.008}$ | – | – | – | – | $0.867 \pm 0.060$ | $0.547 \pm 0.021$ | $0.692 \pm 0.174$ | $0.751 \pm 0.053$ | $0.898 \pm 0.001$ | $0.877 \pm 0.031$ | $0.600 \pm 0.025$ | – |
| tab_blood_transfusion | $0.610 \pm 0.344$ | – | – | – | – | $0.608 \pm 0.043$ | $0.474 \pm 0.129$ | $0.561 \pm 0.056$ | $0.643 \pm 0.065$ | $0.732 \pm 0.024$ | $\mathbf{0.733 \pm 0.005}$ | $0.427 \pm 0.026$ | – |
| tab_credit_g | $0.667 \pm 0.044$ | – | – | – | – | $0.899 \pm 0.057$ | $0.748 \pm 0.202$ | $0.871 \pm 0.056$ | $0.856 \pm 0.019$ | $0.928 \pm 0.014$ | $\mathbf{0.964 \pm 0.017}$ | $0.612 \pm 0.003$ | – |
| tab_qsar_biodeg | $0.707 \pm 0.050$ | – | – | – | – | $0.768 \pm 0.010$ | $0.550 \pm 0.116$ | $0.694 \pm 0.024$ | $0.712 \pm 0.028$ | $0.813 \pm 0.028$ | $\mathbf{0.851 \pm 0.023}$ | $0.475 \pm 0.013$ | – |
| tab_maternal_risk | $0.746 \pm 0.100$ | – | – | – | – | $0.689 \pm 0.112$ | $0.539 \pm 0.194$ | $0.573 \pm 0.005$ | $0.782 \pm 0.021$ | $\mathbf{0.850 \pm 0.017}$ | $0.690 \pm 0.036$ | $0.537 \pm 0.045$ | – |
| tab_website_phishing | $0.852 \pm 0.243$ | – | – | – | – | $0.910 \pm 0.025$ | $0.692 \pm 0.099$ | $0.902 \pm 0.031$ | $0.886 \pm 0.005$ | $0.956 \pm 0.011$ | $\mathbf{0.959 \pm 0.011}$ | $0.750 \pm 0.009$ | – |

Table 28: Column shapes (↑) — mean ± 95% CI over seeds

| Dataset | DISCO-TAB | CTGAN | TVAE | GaussCop | TabDDPM | ADS-GAN | PATE-GAN | RTVAE | NFlow | ARF | BayesNet | Marginals | GReaT |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| heart_failure | $0.748 \pm 0.124$ | $0.738 \pm 0.042$ | $0.614 \pm 0.032$ | $\mathbf{0.846 \pm 0.020}$ | 0.681 | $0.773 \pm 0.019$ | $0.595 \pm 0.120$ | $0.696 \pm 0.033$ | $0.830 \pm 0.041$ | $0.838 \pm 0.007$ | $0.841 \pm 0.056$ | $0.634 \pm 0.029$ | – |
| breast_cancer | $0.756 \pm 0.052$ | $0.649 \pm 0.016$ | $0.724 \pm 0.082$ | $0.868 \pm 0.015$ | 0.631 | $0.782 \pm 0.028$ | $0.255 \pm 0.287$ | $0.550 \pm 0.060$ | $0.882 \pm 0.039$ | $0.897 \pm 0.034$ | $\mathbf{0.911 \pm 0.016}$ | $0.495 \pm 0.027$ | – |
| liver_disorders | $0.643 \pm 0.117$ | $0.487 \pm 0.106$ | $0.612 \pm 0.083$ | $0.771 \pm 0.011$ | 0.772 | $0.632 \pm 0.018$ | $0.601 \pm 0.179$ | $0.535 \pm 0.082$ | $0.731 \pm 0.068$ | $\mathbf{0.774 \pm 0.012}$ | $0.767 \pm 0.016$ | $0.434 \pm 0.052$ | – |
| parkinsons | $0.773 \pm 0.044$ | $0.591 \pm 0.048$ | $0.700 \pm 0.027$ | $0.863 \pm 0.015$ | 0.676 | $0.714 \pm 0.066$ | $0.669 \pm 0.236$ | $0.522 \pm 0.048$ | $0.849 \pm 0.016$ | $0.856 \pm 0.057$ | $\mathbf{0.886 \pm 0.013}$ | $0.522 \pm 0.035$ | – |
| obesity | $0.851 \pm 0.016$ | $0.856 \pm 0.045$ | $0.821 \pm 0.027$ | $0.893 \pm 0.011$ | 0.889 | $0.885 \pm 0.003$ | $0.794 \pm 0.185$ | $0.848 \pm 0.024$ | $0.879 \pm 0.007$ | $0.931 \pm 0.016$ | $\mathbf{0.937 \pm 0.003}$ | $0.664 \pm 0.013$ | – |
| diabetes | $0.770 \pm 0.046$ | $0.782 \pm 0.009$ | $0.798 \pm 0.017$ | $0.808 \pm 0.028$ | – | $0.791 \pm 0.032$ | $0.672 \pm 0.119$ | $0.776 \pm 0.044$ | $0.776 \pm 0.050$ | $0.843 \pm 0.007$ | $\mathbf{0.856 \pm 0.001}$ | $0.543 \pm 0.022$ | – |
| german_credit | $0.799 \pm 0.052$ | $0.872 \pm 0.017$ | $0.639 \pm 0.011$ | $0.925 \pm 0.007$ | – | $0.885 \pm 0.019$ | $0.817 \pm 0.061$ | $0.870 \pm 0.016$ | $0.854 \pm 0.020$ | $\mathbf{0.927 \pm 0.002}$ | $0.925 \pm 0.002$ | $0.622 \pm 0.009$ | – |
| bank_marketing | $0.856 \pm 0.020$ | $0.852 \pm 0.019$ | $0.812 \pm 0.023$ | $0.852 \pm 0.021$ | – | $0.809 \pm 0.085$ | $0.667 \pm 0.139$ | $0.767 \pm 0.041$ | $0.783 \pm 0.052$ | $\mathbf{0.881 \pm 0.005}$ | $0.849 \pm 0.015$ | $0.515 \pm 0.007$ | – |
| census_income | $0.663 \pm 0.088$ | $0.796 \pm 0.057$ | $0.782 \pm 0.031$ | $0.818 \pm 0.109$ | – | $0.726 \pm 0.076$ | $0.572 \pm 0.403$ | $0.730 \pm 0.102$ | $0.710 \pm 0.102$ | $\mathbf{0.822 \pm 0.010}$ | $0.771 \pm 0.021$ | $0.426 \pm 0.008$ | – |
| surv_melanoma | $0.682 \pm 0.247$ | – | – | – | – | $0.703 \pm 0.097$ | $0.569 \pm 0.237$ | $0.634 \pm 0.066$ | $0.752 \pm 0.013$ | $\mathbf{0.777 \pm 0.013}$ | $0.774 \pm 0.028$ | $0.637 \pm 0.013$ | – |
| surv_pbc | $0.767 \pm 0.042$ | – | – | – | – | $0.767 \pm 0.038$ | $0.712 \pm 0.053$ | $0.744 \pm 0.213$ | $0.777 \pm 0.025$ | $\mathbf{0.832 \pm 0.024}$ | $\mathbf{0.832 \pm 0.016}$ | $0.679 \pm 0.016$ | – |
| surv_pbc3 | $0.781 \pm 0.060$ | – | – | – | – | $0.766 \pm 0.031$ | $0.619 \pm 0.190$ | $0.661 \pm 0.043$ | $0.819 \pm 0.028$ | $0.850 \pm 0.012$ | $\mathbf{0.871 \pm 0.016}$ | $0.613 \pm 0.008$ | – |
| surv_retinopathy | $0.786 \pm 0.090$ | – | – | – | – | $0.794 \pm 0.063$ | $0.747 \pm 0.078$ | $0.795 \pm 0.017$ | $0.794 \pm 0.016$ | $0.846 \pm 0.007$ | $\mathbf{0.851 \pm 0.005}$ | $0.768 \pm 0.001$ | – |
| surv_whas500 | $0.826 \pm 0.083$ | – | – | – | – | $0.855 \pm 0.039$ | $0.725 \pm 0.145$ | $0.767 \pm 0.057$ | $0.880 \pm 0.016$ | $0.903 \pm 0.011$ | $\mathbf{0.906 \pm 0.004}$ | $0.689 \pm 0.016$ | – |
| surv_gbsg2 | $0.771 \pm 0.154$ | – | – | – | – | $0.687 \pm 0.205$ | $0.635 \pm 0.242$ | $0.701 \pm 0.024$ | $0.824 \pm 0.030$ | $\mathbf{0.830 \pm 0.016}$ | $0.801 \pm 0.007$ | $0.512 \pm 0.023$ | – |
| surv_colon | $0.799 \pm 0.582$ | – | – | – | – | $0.754 \pm 0.118$ | $0.699 \pm 0.154$ | $0.815 \pm 0.022$ | $0.838 \pm 0.021$ | $\mathbf{0.874 \pm 0.005}$ | $0.860 \pm 0.005$ | $0.630 \pm 0.017$ | – |
| surv_grace | $0.761 \pm 0.033$ | – | – | – | – | $0.679 \pm 0.097$ | $0.531 \pm 0.400$ | $0.700 \pm 0.024$ | $0.757 \pm 0.053$ | $\mathbf{0.808 \pm 0.011}$ | $0.791 \pm 0.007$ | $0.632 \pm 0.020$ | – |
| surv_trace | $0.805 \pm 0.056$ | – | – | – | – | $0.763 \pm 0.035$ | $0.660 \pm 0.049$ | $0.695 \pm 0.198$ | $0.783 \pm 0.031$ | $0.836 \pm 0.005$ | $\mathbf{0.836 \pm 0.003}$ | $0.613 \pm 0.011$ | – |
| surv_framingham | $0.806 \pm 0.034$ | – | – | – | – | $0.759 \pm 0.058$ | $0.494 \pm 0.018$ | $0.615 \pm 0.313$ | $0.768 \pm 0.049$ | $0.839 \pm 0.008$ | $\mathbf{0.847 \pm 0.001}$ | $0.599 \pm 0.016$ | – |
| tab_blood_transfusion | $0.579 \pm 0.307$ | – | – | – | – | $0.531 \pm 0.006$ | $0.434 \pm 0.207$ | $0.576 \pm 0.070$ | $0.694 \pm 0.052$ | $\mathbf{0.746 \pm 0.030}$ | $0.721 \pm 0.007$ | $0.370 \pm 0.023$ | – |
| tab_credit_g | $0.792 \pm 0.018$ | – | – | – | – | $0.918 \pm 0.056$ | $0.823 \pm 0.180$ | $0.919 \pm 0.006$ | $0.911 \pm 0.022$ | $0.967 \pm 0.005$ | $\mathbf{0.971 \pm 0.013}$ | $0.654 \pm 0.008$ | – |
| tab_qsar_biodeg | $0.789 \pm 0.064$ | – | – | – | – | $0.849 \pm 0.029$ | $0.574 \pm 0.167$ | $0.775 \pm 0.019$ | $0.873 \pm 0.005$ | $0.910 \pm 0.012$ | $\mathbf{0.916 \pm 0.003}$ | $0.417 \pm 0.015$ | – |
| tab_maternal_risk | $0.779 \pm 0.023$ | – | – | – | – | $0.732 \pm 0.233$ | $0.588 \pm 0.268$ | $0.724 \pm 0.031$ | $0.867 \pm 0.044$ | $\mathbf{0.899 \pm 0.021}$ | $0.788 \pm 0.014$ | $0.585 \pm 0.011$ | – |
| tab_website_phishing | $0.908 \pm 0.097$ | – | – | – | – | $0.936 \pm 0.032$ | $0.769 \pm 0.094$ | $0.953 \pm 0.033$ | $0.920 \pm 0.012$ | $\mathbf{0.982 \pm 0.007}$ | $0.972 \pm 0.011$ | $0.841 \pm 0.014$ | – |

Table 29: Column-pair trends (↑) — mean ± 95% CI over seeds

| Dataset | DISCO-TAB | CTGAN | TVAE | GaussCop | TabDDPM | ADS-GAN | PATE-GAN | RTVAE | NFlow | ARF | BayesNet | Marginals | GReaT |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| heart_failure | $\mathbf{0.000 \pm 0.000}$ | $\mathbf{0.000 \pm 0.000}$ | $\mathbf{0.000 \pm 0.000}$ | $\mathbf{-0.000 \pm 0.000}$ | **0.000** | $\mathbf{0.000 \pm 0.000}$ | $\mathbf{0.000 \pm 0.000}$ | $\mathbf{0.000 \pm 0.000}$ | $\mathbf{0.000 \pm 0.000}$ | $\mathbf{0.000 \pm 0.000}$ | $\mathbf{0.000 \pm 0.000}$ | $\mathbf{-0.000 \pm 0.000}$ | – |
| breast_cancer | $0.658 \pm 0.015$ | $0.550 \pm 0.009$ | $0.714 \pm 0.017$ | $0.830 \pm 0.015$ | 0.777 | $0.784 \pm 0.021$ | $0.553 \pm 0.016$ | $0.678 \pm 0.008$ | $0.611 \pm 0.039$ | $0.782 \pm 0.010$ | $\mathbf{0.841 \pm 0.001}$ | $0.549 \pm 0.010$ | – |
| liver_disorders | $0.967 \pm 0.044$ | $0.707 \pm 0.008$ | $0.798 \pm 0.098$ | $\mathbf{0.969 \pm 0.006}$ | 0.959 | $0.942 \pm 0.060$ | $0.668 \pm 0.035$ | $0.757 \pm 0.004$ | $0.790 \pm 0.036$ | $0.968 \pm 0.017$ | $0.934 \pm 0.062$ | $0.694 \pm 0.018$ | – |
| parkinsons | $0.759 \pm 0.038$ | $0.520 \pm 0.025$ | $0.637 \pm 0.017$ | $0.831 \pm 0.009$ | 0.777 | $0.749 \pm 0.052$ | $0.537 \pm 0.028$ | $0.606 \pm 0.030$ | $0.599 \pm 0.042$ | $0.771 \pm 0.008$ | $\mathbf{0.836 \pm 0.006}$ | $0.518 \pm 0.008$ | – |
| obesity | $0.834 \pm 0.099$ | $0.695 \pm 0.026$ | $0.673 \pm 0.074$ | $0.738 \pm 0.008$ | 0.812 | $0.788 \pm 0.104$ | $0.642 \pm 0.217$ | $0.541 \pm 0.070$ | $0.705 \pm 0.037$ | $0.856 \pm 0.048$ | $\mathbf{0.890 \pm 0.005}$ | $0.492 \pm 0.010$ | – |
| diabetes | $\mathbf{0.000 \pm 0.000}$ | $\mathbf{0.000 \pm 0.000}$ | $\mathbf{0.000 \pm 0.000}$ | $\mathbf{0.000 \pm 0.000}$ | – | $\mathbf{-0.000 \pm 0.000}$ | $\mathbf{-0.000 \pm 0.000}$ | $\mathbf{-0.000 \pm 0.000}$ | $\mathbf{0.000 \pm 0.000}$ | $\mathbf{-0.000 \pm 0.000}$ | $\mathbf{0.000 \pm 0.000}$ | $\mathbf{0.000 \pm 0.000}$ | – |
| german_credit | $0.630 \pm 0.310$ | $0.650 \pm 0.033$ | $0.475 \pm 0.151$ | $0.716 \pm 0.020$ | – | $0.746 \pm 0.011$ | $0.647 \pm 0.045$ | $0.701 \pm 0.059$ | $0.646 \pm 0.039$ | $0.753 \pm 0.033$ | $\mathbf{0.802 \pm 0.024}$ | $0.462 \pm 0.015$ | – |
| bank_marketing | $0.552 \pm 0.017$ | $0.566 \pm 0.016$ | $\mathbf{0.568 \pm 0.027}$ | $0.553 \pm 0.014$ | – | $0.488 \pm 0.116$ | $0.356 \pm 0.117$ | $0.406 \pm 0.192$ | $0.445 \pm 0.054$ | $0.556 \pm 0.021$ | $0.535 \pm 0.009$ | $0.293 \pm 0.007$ | – |
| census_income | $0.373 \pm 0.083$ | $0.383 \pm 0.025$ | $\mathbf{0.431 \pm 0.062}$ | $0.375 \pm 0.027$ | – | $0.355 \pm 0.103$ | $0.233 \pm 0.177$ | $0.333 \pm 0.124$ | $0.293 \pm 0.011$ | $0.400 \pm 0.009$ | $0.399 \pm 0.044$ | $0.185 \pm 0.003$ | – |
| surv_melanoma | $\mathbf{0.000 \pm 0.000}$ | – | – | – | – | $\mathbf{0.000 \pm 0.000}$ | $\mathbf{0.000 \pm 0.000}$ | $\mathbf{-0.000 \pm 0.000}$ | $\mathbf{0.000 \pm 0.000}$ | $\mathbf{0.000 \pm 0.000}$ | $\mathbf{0.000 \pm 0.000}$ | $\mathbf{-0.000 \pm 0.000}$ | – |
| surv_pbc | $0.127 \pm 0.289$ | – | – | – | – | $0.178 \pm 0.449$ | $0.178 \pm 0.505$ | $0.035 \pm 0.076$ | $0.222 \pm 0.559$ | $0.207 \pm 0.516$ | $\mathbf{0.223 \pm 0.559}$ | $0.129 \pm 0.306$ | – |
| surv_pbc3 | $0.082 \pm 0.353$ | – | – | – | – | $0.102 \pm 0.439$ | $0.090 \pm 0.386$ | $0.096 \pm 0.411$ | $0.086 \pm 0.368$ | $0.099 \pm 0.427$ | $\mathbf{0.106 \pm 0.454}$ | $0.075 \pm 0.324$ | – |
| surv_retinopathy | $0.751 \pm 0.558$ | – | – | – | – | $0.733 \pm 0.153$ | $0.465 \pm 0.103$ | $0.440 \pm 0.024$ | $0.646 \pm 0.036$ | $0.846 \pm 0.026$ | $\mathbf{0.947 \pm 0.028}$ | $0.479 \pm 0.062$ | – |
| surv_whas500 | $0.206 \pm 0.067$ | – | – | – | – | $\mathbf{0.214 \pm 0.069}$ | $0.123 \pm 0.023$ | $0.166 \pm 0.067$ | $0.181 \pm 0.074$ | $0.189 \pm 0.067$ | $0.213 \pm 0.070$ | $0.148 \pm 0.072$ | – |
| surv_gbsg2 | $0.572 \pm 0.326$ | – | – | – | – | $0.522 \pm 0.234$ | $0.489 \pm 0.109$ | $0.213 \pm 0.264$ | $0.616 \pm 0.192$ | $\mathbf{0.655 \pm 0.261}$ | $0.408 \pm 0.169$ | $0.362 \pm 0.256$ | – |
| surv_colon | $0.954 \pm 0.138$ | – | – | – | – | $0.974 \pm 0.064$ | $0.639 \pm 0.044$ | $0.869 \pm 0.017$ | $0.816 \pm 0.201$ | $0.978 \pm 0.049$ | $\mathbf{0.998 \pm 0.001}$ | $0.638 \pm 0.083$ | – |
| surv_grace | $\mathbf{0.000 \pm 0.000}$ | – | – | – | – | $\mathbf{0.000 \pm 0.000}$ | $\mathbf{0.000 \pm 0.000}$ | $\mathbf{0.000 \pm 0.000}$ | $\mathbf{0.000 \pm 0.000}$ | $\mathbf{0.000 \pm 0.000}$ | $\mathbf{0.000 \pm 0.000}$ | $\mathbf{0.000 \pm 0.000}$ | – |
| surv_trace | $\mathbf{-0.000 \pm 0.000}$ | – | – | – | – | $\mathbf{0.000 \pm 0.000}$ | $\mathbf{-0.000 \pm 0.000}$ | $\mathbf{0.000 \pm 0.000}$ | $\mathbf{-0.000 \pm 0.000}$ | $\mathbf{-0.000 \pm 0.000}$ | $\mathbf{-0.000 \pm 0.000}$ | $\mathbf{0.000 \pm 0.000}$ | – |
| surv_framingham | $\mathbf{0.991 \pm 0.018}$ | – | – | – | – | $0.975 \pm 0.071$ | $0.600 \pm 0.061$ | $0.769 \pm 0.040$ | $0.734 \pm 0.093$ | $0.957 \pm 0.007$ | $0.908 \pm 0.061$ | $0.602 \pm 0.040$ | – |
| tab_blood_transfusion | $0.642 \pm 0.408$ | – | – | – | – | $0.685 \pm 0.085$ | $0.514 \pm 0.076$ | $0.547 \pm 0.101$ | $0.593 \pm 0.129$ | $0.718 \pm 0.019$ | $\mathbf{0.746 \pm 0.007}$ | $0.484 \pm 0.040$ | – |
| tab_credit_g | $0.541 \pm 0.069$ | – | – | – | – | $0.879 \pm 0.061$ | $0.673 \pm 0.228$ | $0.823 \pm 0.119$ | $0.801 \pm 0.022$ | $0.890 \pm 0.028$ | $\mathbf{0.957 \pm 0.022}$ | $0.569 \pm 0.006$ | – |
| tab_qsar_biodeg | $0.625 \pm 0.036$ | – | – | – | – | $0.688 \pm 0.042$ | $0.526 \pm 0.088$ | $0.613 \pm 0.039$ | $0.551 \pm 0.051$ | $0.716 \pm 0.066$ | $\mathbf{0.787 \pm 0.049}$ | $0.532 \pm 0.035$ | – |
| tab_maternal_risk | $0.712 \pm 0.178$ | – | – | – | – | $0.645 \pm 0.014$ | $0.491 \pm 0.137$ | $0.422 \pm 0.023$ | $0.698 \pm 0.068$ | $\mathbf{0.801 \pm 0.044}$ | $0.593 \pm 0.082$ | $0.490 \pm 0.079$ | – |
| tab_website_phishing | $0.796 \pm 0.390$ | – | – | – | – | $0.883 \pm 0.021$ | $0.616 \pm 0.106$ | $0.851 \pm 0.031$ | $0.851 \pm 0.005$ | $0.929 \pm 0.018$ | $\mathbf{0.947 \pm 0.014}$ | $0.660 \pm 0.021$ | – |

## FAITH framework

Table 30: FAITH overall (↑) — mean ± 95% CI over seeds

| Dataset | DISCO-TAB | CTGAN | TVAE | GaussCop | TabDDPM | ADS-GAN | PATE-GAN | RTVAE | NFlow | ARF | BayesNet | Marginals | GReaT |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| heart_failure | 0.813 ± 0.090 | 0.884 ± 0.084 | 0.739 ± 0.014 | 0.827 ± 0.035 | 0.870 | 0.808 ± 0.057 | 0.916 ± 0.054 | 0.758 ± 0.044 | 0.804 ± 0.075 | 0.834 ± 0.048 | 0.817 ± 0.059 | **0.967 ± 0.002** | – |
| breast_cancer | 0.388 ± 0.057 | 0.883 ± 0.001 | 0.804 ± 0.131 | 0.897 ± 0.084 | 0.856 | **0.925 ± 0.036** | 0.870 ± 0.019 | 0.770 ± 0.070 | 0.900 ± 0.032 | 0.899 ± 0.081 | 0.854 ± 0.046 | 0.883 ± 0.002 | – |
| liver_disorders | 0.870 ± 0.089 | 0.903 ± 0.010 | 0.710 ± 0.013 | 0.859 ± 0.035 | 0.835 | 0.800 ± 0.039 | 0.845 ± 0.046 | 0.804 ± 0.038 | 0.819 ± 0.071 | 0.852 ± 0.032 | 0.837 ± 0.062 | **0.923 ± 0.008** | – |
| parkinsons | 0.298 ± 0.133 | 0.852 ± 0.013 | 0.763 ± 0.091 | 0.868 ± 0.003 | 0.825 | **0.877 ± 0.086** | 0.847 ± 0.018 | 0.715 ± 0.063 | 0.860 ± 0.057 | 0.862 ± 0.056 | 0.818 ± 0.030 | 0.852 ± 0.008 | – |
| obesity | 0.561 ± 0.212 | 0.944 ± 0.014 | 0.790 ± 0.042 | 0.957 ± 0.012 | 0.854 | 0.902 ± 0.133 | 0.941 ± 0.047 | 0.907 ± 0.034 | 0.947 ± 0.004 | 0.877 ± 0.011 | 0.855 ± 0.015 | **0.961 ± 0.003** | – |
| diabetes | 0.745 ± 0.090 | 0.862 ± 0.037 | 0.769 ± 0.034 | 0.924 ± 0.015 | 0.841 ± 0.002 | 0.853 ± 0.015 | 0.877 ± 0.168 | 0.841 ± 0.056 | 0.865 ± 0.024 | 0.886 ± 0.002 | 0.842 ± 0.007 | **0.964 ± 0.003** | – |
| german_credit | 0.374 ± 0.180 | 0.881 ± 0.025 | 0.724 ± 0.004 | 0.859 ± 0.014 | 0.809 ± 0.019 | 0.847 ± 0.019 | 0.853 ± 0.022 | 0.814 ± 0.009 | 0.878 ± 0.027 | 0.852 ± 0.011 | 0.829 ± 0.016 | **0.960 ± 0.003** | – |
| bank_marketing | 0.810 ± 0.054 | 0.922 ± 0.021 | 0.749 ± 0.024 | 0.936 ± 0.026 | 0.822 ± 0.038 | 0.932 ± 0.050 | 0.960 ± 0.018 | 0.903 ± 0.066 | 0.941 ± 0.028 | 0.927 ± 0.018 | 0.919 ± 0.016 | **0.974 ± 0.002** | – |
| census_income | 0.350 ± 0.158 | 0.949 ± 0.021 | 0.769 ± 0.010 | 0.982 ± 0.021 | 0.852 ± 0.016 | 0.957 ± 0.072 | 0.974 ± 0.009 | 0.967 ± 0.059 | 0.976 ± 0.013 | 0.959 ± 0.012 | 0.935 ± 0.036 | **0.984 ± 0.001** | – |
| surv_melanoma | 0.791 ± 0.042 | 0.862 ± 0.073 | 0.771 ± 0.002 | 0.804 ± 0.056 | 0.800 ± 0.035 | 0.772 ± 0.065 | 0.880 ± 0.067 | 0.763 ± 0.012 | 0.804 ± 0.050 | 0.806 ± 0.014 | 0.808 ± 0.011 | **0.902 ± 0.025** | – |
| surv_pbc | 0.655 ± 0.432 | 0.842 ± 0.051 | 0.772 ± 0.114 | 0.826 ± 0.023 | 0.810 ± 0.049 | 0.797 ± 0.082 | 0.828 ± 0.084 | 0.836 ± 0.209 | 0.797 ± 0.074 | 0.812 ± 0.068 | 0.812 ± 0.072 | **0.913 ± 0.041** | – |
| surv_pbc3 | 0.795 ± 0.064 | 0.885 ± 0.034 | 0.728 ± 0.018 | 0.831 ± 0.016 | 0.868 ± 0.018 | 0.827 ± 0.039 | 0.907 ± 0.025 | 0.742 ± 0.026 | 0.808 ± 0.024 | 0.839 ± 0.038 | 0.833 ± 0.024 | **0.959 ± 0.007** | 0.755 |
| surv_retinopathy | 0.657 ± 0.382 | 0.847 ± 0.025 | 0.736 ± 0.053 | 0.827 ± 0.026 | 0.811 ± 0.026 | 0.807 ± 0.031 | 0.830 ± 0.068 | 0.818 ± 0.033 | 0.817 ± 0.052 | 0.809 ± 0.014 | 0.804 ± 0.028 | **0.859 ± 0.016** | – |
| surv_whas500 | 0.295 ± 0.046 | 0.927 ± 0.021 | 0.738 ± 0.006 | 0.846 ± 0.021 | 0.841 ± 0.021 | 0.822 ± 0.053 | 0.942 ± 0.032 | 0.761 ± 0.044 | 0.830 ± 0.040 | 0.838 ± 0.024 | 0.839 ± 0.028 | **0.965 ± 0.003** | – |
| surv_gbsg2 | 0.664 ± 0.492 | 0.884 ± 0.068 | 0.729 ± 0.043 | 0.851 ± 0.053 | 0.790 ± 0.029 | 0.840 ± 0.046 | 0.887 ± 0.076 | 0.802 ± 0.076 | 0.843 ± 0.056 | 0.853 ± 0.044 | 0.869 ± 0.060 | **0.954 ± 0.014** | 0.782 |
| surv_colon | 0.824 ± 0.074 | 0.852 ± 0.091 | 0.728 ± 0.017 | 0.840 ± 0.062 | 0.830 ± 0.069 | 0.845 ± 0.093 | 0.887 ± 0.038 | 0.805 ± 0.056 | 0.822 ± 0.064 | 0.841 ± 0.060 | 0.830 ± 0.052 | **0.956 ± 0.004** | – |
| surv_grace | 0.803 ± 0.016 | 0.865 ± 0.111 | 0.736 ± 0.016 | 0.818 ± 0.013 | 0.806 ± 0.012 | 0.792 ± 0.059 | 0.915 ± 0.123 | 0.767 ± 0.031 | 0.795 ± 0.018 | 0.817 ± 0.005 | 0.805 ± 0.007 | **0.953 ± 0.017** | – |
| surv_trace | 0.804 ± 0.052 | 0.884 ± 0.027 | 0.793 ± 0.116 | 0.827 ± 0.001 | 0.825 ± 0.008 | 0.801 ± 0.026 | 0.878 ± 0.063 | 0.867 ± 0.137 | 0.820 ± 0.046 | 0.827 ± 0.003 | 0.826 ± 0.012 | **0.944 ± 0.001** | – |
| surv_framingham | 0.816 ± 0.014 | 0.880 ± 0.015 | 0.730 ± 0.021 | 0.841 ± 0.009 | 0.821 ± 0.010 | 0.839 ± 0.050 | 0.916 ± 0.026 | 0.873 ± 0.223 | 0.823 ± 0.022 | 0.839 ± 0.005 | 0.826 ± 0.005 | **0.939 ± 0.003** | – |
| tab_blood_transfusion | 0.474 ± 0.440 | 0.862 ± 0.013 | 0.723 ± 0.021 | 0.836 ± 0.030 | 0.811 ± 0.027 | 0.798 ± 0.084 | 0.836 ± 0.111 | 0.811 ± 0.025 | 0.811 ± 0.007 | 0.834 ± 0.029 | 0.787 ± 0.036 | **0.881 ± 0.020** | – |
| tab_credit_g | 0.379 ± 0.070 | 0.881 ± 0.043 | 0.730 ± 0.014 | 0.871 ± 0.031 | 0.812 ± 0.010 | 0.840 ± 0.079 | 0.888 ± 0.144 | 0.826 ± 0.023 | 0.872 ± 0.016 | 0.866 ± 0.020 | 0.841 ± 0.030 | **0.958 ± 0.001** | – |
| tab_qsar_biodeg | 0.363 ± 0.113 | 0.939 ± 0.001 | 0.811 ± 0.066 | **0.967 ± 0.006** | 0.940 ± 0.005 | 0.921 ± 0.042 | 0.938 ± 0.014 | 0.865 ± 0.033 | 0.938 ± 0.008 | 0.883 ± 0.046 | 0.847 ± 0.029 | 0.938 ± 0.001 | – |
| tab_maternal_risk | 0.527 ± 0.550 | 0.881 ± 0.056 | 0.741 ± 0.068 | 0.892 ± 0.037 | 0.872 ± 0.024 | 0.834 ± 0.007 | 0.872 ± 0.072 | 0.789 ± 0.053 | 0.847 ± 0.038 | 0.853 ± 0.031 | 0.844 ± 0.042 | **0.925 ± 0.007** | – |
| tab_website_phishing | 0.351 ± 0.645 | 0.063 ± 0.022 | 0.012 ± 0.020 | 0.053 ± 0.011 | 0.503 ± 0.001 | 0.509 ± 0.009 | 0.530 ± 0.024 | 0.509 ± 0.005 | 0.514 ± 0.012 | 0.507 ± 0.004 | 0.509 ± 0.003 | **0.581 ± 0.014** | – |

Table 31: Factuality (↑) — mean ± 95% CI over seeds

| Dataset | DISCO-TAB | CTGAN | TVAE | GaussCop | TabDDPM | ADS-GAN | PATE-GAN | RTVAE | NFlow | ARF | BayesNet | Marginals | GReaT |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| heart_failure | 0.978 ± 0.043 | **1.000 ± 0.000** | **1.000 ± 0.000** | **1.000 ± 0.000** | 1.000 | **1.000 ± 0.000** | **1.000 ± 0.000** | **1.000 ± 0.000** | **1.000 ± 0.000** | **1.000 ± 0.000** | **1.000 ± 0.000** | **1.000 ± 0.000** | – |
| breast_cancer | 0.000 ± 0.000 | **1.000 ± 0.000** | **1.000 ± 0.000** | **1.000 ± 0.000** | 1.000 | **1.000 ± 0.000** | **1.000 ± 0.000** | **1.000 ± 0.000** | **1.000 ± 0.000** | **1.000 ± 0.000** | **1.000 ± 0.000** | **1.000 ± 0.000** | – |
| liver_disorders | 0.999 ± 0.005 | **1.000 ± 0.000** | **1.000 ± 0.000** | **1.000 ± 0.000** | 1.000 | **1.000 ± 0.000** | **1.000 ± 0.000** | **1.000 ± 0.000** | **1.000 ± 0.000** | **1.000 ± 0.000** | **1.000 ± 0.000** | **1.000 ± 0.000** | – |
| parkinsons | 0.000 ± 0.000 | **1.000 ± 0.000** | **1.000 ± 0.000** | **1.000 ± 0.000** | 1.000 | **1.000 ± 0.000** | **1.000 ± 0.000** | **1.000 ± 0.000** | **1.000 ± 0.000** | **1.000 ± 0.000** | **1.000 ± 0.000** | **1.000 ± 0.000** | – |
| obesity | 0.416 ± 0.407 | **1.000 ± 0.000** | **1.000 ± 0.000** | **1.000 ± 0.000** | 1.000 | **1.000 ± 0.000** | **1.000 ± 0.000** | **1.000 ± 0.000** | **1.000 ± 0.000** | **1.000 ± 0.000** | **1.000 ± 0.000** | **1.000 ± 0.000** | – |
| diabetes | 0.859 ± 0.302 | **1.000 ± 0.000** | **1.000 ± 0.000** | **1.000 ± 0.000** | **1.000 ± 0.000** | **1.000 ± 0.000** | **1.000 ± 0.000** | **1.000 ± 0.000** | **1.000 ± 0.000** | **1.000 ± 0.000** | **1.000 ± 0.000** | **1.000 ± 0.000** | – |
| german_credit | 0.000 ± 0.000 | **1.000 ± 0.000** | **1.000 ± 0.000** | **1.000 ± 0.000** | **1.000 ± 0.000** | **1.000 ± 0.000** | **1.000 ± 0.000** | **1.000 ± 0.000** | **1.000 ± 0.000** | **1.000 ± 0.000** | **1.000 ± 0.000** | **1.000 ± 0.000** | – |
| bank_marketing | **1.000 ± 0.000** | **1.000 ± 0.000** | **1.000 ± 0.000** | **1.000 ± 0.000** | **1.000 ± 0.000** | **1.000 ± 0.000** | **1.000 ± 0.000** | **1.000 ± 0.000** | **1.000 ± 0.000** | **1.000 ± 0.000** | **1.000 ± 0.000** | **1.000 ± 0.000** | – |
| census_income | 0.091 ± 0.248 | **1.000 ± 0.000** | **1.000 ± 0.000** | **1.000 ± 0.000** | **1.000 ± 0.000** | **1.000 ± 0.000** | **1.000 ± 0.000** | **1.000 ± 0.000** | **1.000 ± 0.000** | **1.000 ± 0.000** | **1.000 ± 0.000** | **1.000 ± 0.000** | – |
| surv_melanoma | 0.998 ± 0.009 | **1.000 ± 0.000** | **1.000 ± 0.000** | **1.000 ± 0.000** | **1.000 ± 0.000** | **1.000 ± 0.000** | **1.000 ± 0.000** | **1.000 ± 0.000** | **1.000 ± 0.000** | **1.000 ± 0.000** | **1.000 ± 0.000** | **1.000 ± 0.000** | – |
| surv_pbc | 0.621 ± 1.316 | **1.000 ± 0.000** | **1.000 ± 0.000** | **1.000 ± 0.000** | **1.000 ± 0.000** | **1.000 ± 0.000** | **1.000 ± 0.000** | **1.000 ± 0.000** | **1.000 ± 0.000** | **1.000 ± 0.000** | **1.000 ± 0.000** | **1.000 ± 0.000** | – |
| surv_pbc3 | 0.986 ± 0.024 | **1.000 ± 0.000** | **1.000 ± 0.000** | **1.000 ± 0.000** | **1.000 ± 0.000** | **1.000 ± 0.000** | **1.000 ± 0.000** | **1.000 ± 0.000** | **1.000 ± 0.000** | **1.000 ± 0.000** | **1.000 ± 0.000** | **1.000 ± 0.000** | 0.978 |
| surv_retinopathy | 0.544 ± 1.251 | **1.000 ± 0.000** | **1.000 ± 0.000** | **1.000 ± 0.000** | **1.000 ± 0.000** | **1.000 ± 0.000** | **1.000 ± 0.000** | **1.000 ± 0.000** | **1.000 ± 0.000** | **1.000 ± 0.000** | **1.000 ± 0.000** | **1.000 ± 0.000** | – |
| surv_whas500 | 0.000 ± 0.000 | **1.000 ± 0.000** | **1.000 ± 0.000** | **1.000 ± 0.000** | **1.000 ± 0.000** | **1.000 ± 0.000** | **1.000 ± 0.000** | **1.000 ± 0.000** | **1.000 ± 0.000** | **1.000 ± 0.000** | **1.000 ± 0.000** | **1.000 ± 0.000** | – |
| surv_gbsg2 | 0.631 ± 1.122 | **1.000 ± 0.000** | **1.000 ± 0.000** | **1.000 ± 0.000** | **1.000 ± 0.000** | **1.000 ± 0.000** | **1.000 ± 0.000** | **1.000 ± 0.000** | **1.000 ± 0.000** | **1.000 ± 0.000** | **1.000 ± 0.000** | **1.000 ± 0.000** | 0.996 |
| surv_colon | 0.998 ± 0.010 | **1.000 ± 0.000** | **1.000 ± 0.000** | **1.000 ± 0.000** | **1.000 ± 0.000** | **1.000 ± 0.000** | **1.000 ± 0.000** | **1.000 ± 0.000** | **1.000 ± 0.000** | **1.000 ± 0.000** | **1.000 ± 0.000** | **1.000 ± 0.000** | – |
| surv_grace | **1.000 ± 0.000** | **1.000 ± 0.000** | **1.000 ± 0.000** | **1.000 ± 0.000** | **1.000 ± 0.000** | **1.000 ± 0.000** | **1.000 ± 0.000** | **1.000 ± 0.000** | **1.000 ± 0.000** | **1.000 ± 0.000** | **1.000 ± 0.000** | **1.000 ± 0.000** | – |
| surv_trace | 1.000 ± 0.001 | **1.000 ± 0.000** | **1.000 ± 0.000** | **1.000 ± 0.000** | **1.000 ± 0.000** | **1.000 ± 0.000** | **1.000 ± 0.000** | **1.000 ± 0.000** | **1.000 ± 0.000** | **1.000 ± 0.000** | **1.000 ± 0.000** | **1.000 ± 0.000** | – |
| surv_framingham | 1.000 ± 0.000 | **1.000 ± 0.000** | **1.000 ± 0.000** | **1.000 ± 0.000** | **1.000 ± 0.000** | **1.000 ± 0.000** | **1.000 ± 0.000** | **1.000 ± 0.000** | **1.000 ± 0.000** | **1.000 ± 0.000** | **1.000 ± 0.000** | **1.000 ± 0.000** | – |
| tab_blood_transfusion | 0.313 ± 1.016 | **1.000 ± 0.000** | **1.000 ± 0.000** | **1.000 ± 0.000** | **1.000 ± 0.000** | **1.000 ± 0.000** | **1.000 ± 0.000** | **1.000 ± 0.000** | **1.000 ± 0.000** | **1.000 ± 0.000** | **1.000 ± 0.000** | **1.000 ± 0.000** | – |
| tab_credit_g | 0.000 ± 0.000 | **1.000 ± 0.000** | **1.000 ± 0.000** | **1.000 ± 0.000** | **1.000 ± 0.000** | **1.000 ± 0.000** | **1.000 ± 0.000** | **1.000 ± 0.000** | **1.000 ± 0.000** | **1.000 ± 0.000** | **1.000 ± 0.000** | **1.000 ± 0.000** | – |
| tab_qsar_biodeg | 0.029 ± 0.120 | **1.000 ± 0.000** | **1.000 ± 0.000** | **1.000 ± 0.000** | **1.000 ± 0.000** | **1.000 ± 0.000** | **1.000 ± 0.000** | **1.000 ± 0.000** | **1.000 ± 0.000** | **1.000 ± 0.000** | **1.000 ± 0.000** | **1.000 ± 0.000** | – |
| tab_maternal_risk | 0.453 ± 1.138 | **1.000 ± 0.000** | **1.000 ± 0.000** | **1.000 ± 0.000** | **1.000 ± 0.000** | **1.000 ± 0.000** | **1.000 ± 0.000** | **1.000 ± 0.000** | **1.000 ± 0.000** | **1.000 ± 0.000** | **1.000 ± 0.000** | **1.000 ± 0.000** | – |
| tab_website_phishing | 0.667 ± 1.434 | 0.000 ± 0.000 | 0.000 ± 0.000 | 0.000 ± 0.000 | **1.000 ± 0.000** | **1.000 ± 0.000** | **1.000 ± 0.000** | **1.000 ± 0.000** | **1.000 ± 0.000** | **1.000 ± 0.000** | **1.000 ± 0.000** | **1.000 ± 0.000** | – |

Table 32: Alignment (↑) — mean ± 95% CI over seeds

| Dataset | DISCO-TAB | CTGAN | TVAE | GaussCop | TabDDPM | ADS-GAN | PATE-GAN | RTVAE | NFlow | ARF | BayesNet | Marginals | GReaT |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| heart_failure | 0.910 ± 0.018 | 0.877 ± 0.029 | **0.944 ± 0.019** | 0.916 ± 0.011 | 0.790 | 0.893 ± 0.025 | 0.808 ± 0.034 | 0.904 ± 0.014 | 0.887 ± 0.008 | 0.927 ± 0.024 | 0.887 ± 0.047 | 0.879 ± 0.010 | – |
| breast_cancer | 0.882 ± 0.021 | 0.532 ± 0.002 | 0.747 ± 0.010 | 0.933 ± 0.022 | 0.722 | 0.861 ± 0.037 | 0.479 ± 0.077 | 0.713 ± 0.032 | 0.627 ± 0.055 | 0.873 ± 0.005 | **0.934 ± 0.029** | 0.532 ± 0.007 | – |
| liver_disorders | 0.917 ± 0.089 | 0.707 ± 0.020 | 0.789 ± 0.149 | **0.943 ± 0.020** | 0.901 | 0.858 ± 0.101 | 0.600 ± 0.138 | 0.725 ± 0.035 | 0.782 ± 0.052 | 0.941 ± 0.004 | 0.896 ± 0.071 | 0.693 ± 0.028 | – |
| parkinsons | 0.832 ± 0.080 | 0.411 ± 0.052 | 0.598 ± 0.001 | 0.919 ± 0.009 | 0.678 | 0.767 ± 0.060 | 0.393 ± 0.072 | 0.545 ± 0.051 | 0.547 ± 0.077 | 0.844 ± 0.022 | **0.933 ± 0.007** | 0.409 ± 0.033 | – |
| obesity | 0.936 ± 0.030 | 0.845 ± 0.012 | 0.855 ± 0.011 | 0.929 ± 0.020 | 0.824 | 0.946 ± 0.098 | 0.819 ± 0.098 | 0.903 ± 0.007 | 0.866 ± 0.012 | 0.958 ± 0.016 | **0.971 ± 0.013** | 0.845 ± 0.011 | – |
| diabetes | 0.912 ± 0.116 | 0.888 ± 0.031 | 0.915 ± 0.046 | 0.910 ± 0.008 | **0.983 ± 0.005** | 0.965 ± 0.008 | 0.868 ± 0.057 | 0.903 ± 0.016 | 0.906 ± 0.031 | 0.979 ± 0.003 | 0.946 ± 0.004 | 0.859 ± 0.008 | – |
| german_credit | 0.920 ± 0.094 | 0.841 ± 0.016 | 0.892 ± 0.010 | 0.914 ± 0.027 | 0.903 ± 0.055 | 0.919 ± 0.014 | 0.858 ± 0.013 | 0.901 ± 0.021 | 0.863 ± 0.023 | 0.943 ± 0.012 | **0.956 ± 0.021** | 0.843 ± 0.010 | – |
| bank_marketing | 0.932 ± 0.017 | 0.936 ± 0.027 | 0.877 ± 0.074 | 0.919 ± 0.069 | 0.782 ± 0.171 | 0.960 ± 0.035 | 0.891 ± 0.003 | 0.959 ± 0.062 | 0.912 ± 0.012 | **0.970 ± 0.027** | 0.959 ± 0.020 | 0.897 ± 0.007 | – |
| census_income | 0.938 ± 0.035 | 0.941 ± 0.027 | 0.929 ± 0.012 | 0.974 ± 0.005 | 0.847 ± 0.053 | 0.964 ± 0.015 | 0.905 ± 0.020 | 0.956 ± 0.014 | 0.939 ± 0.014 | 0.978 ± 0.010 | **0.979 ± 0.004** | 0.935 ± 0.005 | – |
| surv_melanoma | 0.903 ± 0.042 | 0.821 ± 0.034 | 0.894 ± 0.075 | 0.867 ± 0.039 | 0.876 ± 0.011 | 0.841 ± 0.132 | 0.772 ± 0.032 | 0.831 ± 0.024 | 0.853 ± 0.100 | **0.931 ± 0.010** | 0.911 ± 0.035 | 0.811 ± 0.084 | – |
| surv_pbc | 0.879 ± 0.124 | 0.848 ± 0.018 | 0.842 ± 0.113 | 0.890 ± 0.043 | 0.936 ± 0.020 | 0.913 ± 0.031 | 0.853 ± 0.067 | 0.877 ± 0.008 | 0.891 ± 0.050 | 0.935 ± 0.049 | **0.940 ± 0.044** | 0.856 ± 0.044 | – |
| surv_pbc3 | 0.892 ± 0.042 | 0.837 ± 0.009 | 0.895 ± 0.052 | 0.903 ± 0.023 | 0.827 ± 0.017 | 0.898 ± 0.028 | 0.816 ± 0.042 | 0.863 ± 0.011 | 0.859 ± 0.020 | 0.925 ± 0.012 | **0.929 ± 0.013** | 0.836 ± 0.032 | 0.835 |
| surv_retinopathy | 0.953 ± 0.037 | 0.653 ± 0.057 | 0.832 ± 0.256 | 0.739 ± 0.134 | **0.959 ± 0.059** | 0.830 ± 0.057 | 0.642 ± 0.040 | 0.775 ± 0.068 | 0.693 ± 0.038 | 0.890 ± 0.028 | 0.957 ± 0.105 | 0.630 ± 0.052 | – |
| surv_whas500 | 0.897 ± 0.050 | 0.862 ± 0.019 | **0.935 ± 0.033** | 0.915 ± 0.014 | 0.869 ± 0.037 | 0.917 ± 0.006 | 0.824 ± 0.038 | 0.900 ± 0.011 | 0.892 ± 0.014 | 0.928 ± 0.006 | 0.933 ± 0.010 | 0.865 ± 0.012 | – |
| surv_gbsg2 | 0.898 ± 0.088 | 0.821 ± 0.028 | 0.865 ± 0.044 | 0.903 ± 0.067 | 0.805 ± 0.118 | 0.865 ± 0.118 | 0.777 ± 0.101 | 0.862 ± 0.089 | 0.865 ± 0.050 | **0.944 ± 0.063** | 0.903 ± 0.063 | 0.818 ± 0.054 | 0.868 |
| surv_colon | 0.949 ± 0.030 | 0.842 ± 0.011 | 0.882 ± 0.040 | 0.920 ± 0.031 | 0.954 ± 0.018 | 0.901 ± 0.085 | 0.840 ± 0.012 | 0.927 ± 0.005 | 0.900 ± 0.045 | **0.960 ± 0.027** | 0.932 ± 0.007 | 0.842 ± 0.027 | – |
| surv_grace | 0.946 ± 0.037 | 0.901 ± 0.023 | 0.869 ± 0.089 | 0.948 ± 0.003 | 0.942 ± 0.064 | 0.937 ± 0.032 | 0.855 ± 0.044 | 0.914 ± 0.028 | 0.947 ± 0.045 | **0.975 ± 0.015** | 0.913 ± 0.043 | 0.903 ± 0.056 | – |
| surv_trace | 0.958 ± 0.030 | 0.838 ± 0.003 | 0.942 ± 0.033 | 0.921 ± 0.004 | **0.978 ± 0.007** | 0.922 ± 0.069 | 0.824 ± 0.009 | 0.859 ± 0.042 | 0.913 ± 0.062 | 0.974 ± 0.017 | 0.975 ± 0.012 | 0.852 ± 0.004 | – |
| surv_framingham | 0.964 ± 0.014 | 0.786 ± 0.036 | 0.721 ± 0.060 | **0.980 ± 0.005** | 0.962 ± 0.038 | 0.957 ± 0.001 | 0.724 ± 0.011 | 0.849 ± 0.021 | 0.842 ± 0.042 | 0.974 ± 0.005 | 0.898 ± 0.014 | 0.766 ± 0.009 | – |
| tab_blood_transfusion | 0.718 ± 0.340 | 0.527 ± 0.019 | 0.716 ± 0.041 | 0.927 ± 0.049 | **0.968 ± 0.056** | 0.867 ± 0.136 | 0.564 ± 0.110 | 0.649 ± 0.173 | 0.718 ± 0.196 | 0.926 ± 0.010 | 0.831 ± 0.094 | 0.530 ± 0.077 | – |
| tab_credit_g | 0.953 ± 0.031 | 0.841 ± 0.022 | 0.910 ± 0.049 | 0.911 ± 0.022 | 0.898 ± 0.067 | 0.920 ± 0.037 | 0.853 ± 0.078 | 0.903 ± 0.001 | 0.865 ± 0.020 | 0.944 ± 0.025 | **0.963 ± 0.003** | 0.831 ± 0.003 | – |

| Dataset | DISCO-TAB | CTGAN | TVAE | GaussCop | TabDDPM | ADS-GAN | PATE-GAN | RTVAE | NFlow | ARF | BayesNet | Marginals | GReaT |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| tab_qsar_biodeg | 0.732 ± 0.077 | 0.755 ± 0.005 | 0.883 ± 0.009 | 0.870 ± 0.019 | 0.761 ± 0.018 | 0.864 ± 0.011 | 0.751 ± 0.057 | 0.820 ± 0.003 | 0.768 ± 0.018 | 0.903 ± 0.020 | **0.952 ± 0.003** | 0.754 ± 0.005 | – |
| tab_maternal_risk | 0.908 ± 0.071 | 0.708 ± 0.012 | 0.819 ± 0.043 | 0.923 ± 0.017 | 0.770 ± 0.153 | 0.896 ± 0.076 | 0.699 ± 0.032 | 0.853 ± 0.063 | 0.789 ± 0.009 | **0.948 ± 0.023** | 0.897 ± 0.054 | 0.713 ± 0.041 | – |
| tab_website_phishing | – | – | – | – | – | – | – | – | – | – | – | – | – |

Table 33: Integrity (↑) — mean ± 95% CI over seeds

| Dataset | DISCO-TAB | CTGAN | TVAE | GaussCop | TabDDPM | ADS-GAN | PATE-GAN | RTVAE | NFlow | ARF | BayesNet | Marginals | GReaT |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| heart_failure | 0.386 ± 0.310 | 0.660 ± 0.307 | 0.013 ± 0.045 | 0.393 ± 0.144 | 0.690 | 0.338 ± 0.251 | 0.856 ± 0.248 | 0.126 ± 0.181 | 0.329 ± 0.304 | 0.407 ± 0.169 | 0.381 ± 0.210 | **0.990 ± 0.016** | – |
| breast_cancer | 0.670 ± 0.217 | **1.000 ± 0.000** | 0.470 ± 0.530 | 0.656 ± 0.360 | 0.703 | 0.838 ± 0.176 | **1.000 ± 0.000** | 0.367 ± 0.291 | 0.971 ± 0.079 | 0.724 ± 0.328 | 0.482 ± 0.178 | **1.000 ± 0.000** | – |
| liver_disorders | 0.565 ± 0.398 | 0.906 ± 0.059 | 0.051 ± 0.101 | 0.494 ± 0.153 | 0.438 | 0.342 ± 0.245 | 0.781 ± 0.058 | 0.493 ± 0.136 | 0.494 ± 0.293 | 0.465 ± 0.131 | 0.450 ± 0.320 | **0.998 ± 0.005** | – |
| parkinsons | 0.361 ± 0.468 | 0.998 ± 0.009 | 0.453 ± 0.365 | 0.553 ± 0.009 | 0.622 | 0.739 ± 0.330 | 0.994 ± 0.016 | 0.314 ± 0.207 | 0.895 ± 0.189 | 0.605 ± 0.223 | 0.338 ± 0.113 | **1.000 ± 0.000** | – |
| obesity | 0.478 ± 0.071 | 0.931 ± 0.065 | 0.304 ± 0.179 | 0.900 ± 0.048 | 0.590 | 0.663 ± 0.433 | 0.943 ± 0.092 | 0.725 ± 0.128 | 0.921 ± 0.004 | 0.549 ± 0.060 | 0.451 ± 0.071 | **1.000 ± 0.000** | – |
| diabetes | 0.350 ± 0.368 | 0.558 ± 0.125 | 0.160 ± 0.091 | 0.784 ± 0.061 | 0.381 ± 0.011 | 0.446 ± 0.053 | 0.638 ± 0.725 | 0.459 ± 0.239 | 0.555 ± 0.122 | 0.566 ± 0.007 | 0.420 ± 0.025 | **0.996 ± 0.004** | – |
| german_credit | 0.574 ± 0.816 | 0.685 ± 0.097 | 0.006 ± 0.007 | 0.521 ± 0.045 | 0.332 ± 0.024 | 0.470 ± 0.064 | 0.553 ± 0.083 | 0.356 ± 0.049 | 0.647 ± 0.109 | 0.464 ± 0.042 | 0.361 ± 0.070 | **0.998 ± 0.004** | – |
| bank_marketing | 0.309 ± 0.231 | 0.754 ± 0.079 | 0.118 ± 0.025 | 0.823 ± 0.040 | 0.508 ± 0.035 | 0.769 ± 0.171 | 0.951 ± 0.069 | 0.654 ± 0.306 | 0.853 ± 0.107 | 0.737 ± 0.050 | 0.716 ± 0.066 | **1.000 ± 0.000** | – |
| census_income | 0.280 ± 0.256 | 0.855 ± 0.110 | 0.147 ± 0.030 | 0.955 ± 0.080 | 0.561 ± 0.011 | 0.863 ± 0.273 | 0.991 ± 0.031 | 0.913 ± 0.235 | 0.964 ± 0.042 | 0.858 ± 0.039 | 0.762 ± 0.145 | **1.000 ± 0.000** | – |
| surv_melanoma | 0.264 ± 0.114 | 0.626 ± 0.258 | 0.191 ± 0.075 | 0.350 ± 0.192 | 0.325 ± 0.149 | 0.246 ± 0.189 | 0.750 ± 0.244 | 0.220 ± 0.055 | 0.364 ± 0.101 | 0.295 ± 0.061 | 0.321 ± 0.078 | **0.799 ± 0.026** | – |
| surv_pbc | 0.498 ± 1.079 | 0.518 ± 0.185 | 0.248 ± 0.388 | 0.416 ± 0.057 | 0.304 ± 0.185 | 0.276 ± 0.325 | 0.458 ± 0.404 | 0.469 ± 0.834 | 0.299 ± 0.311 | 0.312 ± 0.241 | 0.309 ± 0.285 | **0.795 ± 0.174** | – |
| surv_pbc3 | 0.315 ± 0.309 | 0.701 ± 0.136 | 0.017 ± 0.031 | 0.419 ± 0.073 | 0.643 ± 0.063 | 0.410 ± 0.182 | 0.811 ± 0.095 | 0.103 ± 0.095 | 0.372 ± 0.107 | 0.429 ± 0.145 | 0.401 ± 0.103 | **0.999 ± 0.005** | 0.229 |
| surv_retinopathy | 0.585 ± 0.981 | 0.734 ± 0.080 | 0.112 ± 0.184 | 0.568 ± 0.202 | 0.283 ± 0.085 | 0.398 ± 0.169 | 0.677 ± 0.260 | 0.495 ± 0.077 | 0.576 ± 0.228 | 0.347 ± 0.039 | 0.261 ± 0.102 | **0.805 ± 0.074** | – |
| surv_whas500 | 0.283 ± 0.182 | 0.844 ± 0.101 | 0.017 ± 0.027 | 0.469 ± 0.092 | 0.494 ± 0.074 | 0.371 ± 0.209 | 0.942 ± 0.100 | 0.145 ± 0.164 | 0.426 ± 0.152 | 0.426 ± 0.095 | 0.424 ± 0.113 | **0.995 ± 0.007** | – |
| surv_gbsg2 | 0.495 ± 0.350 | 0.714 ± 0.278 | 0.050 ± 0.130 | 0.499 ± 0.236 | 0.355 ± 0.149 | 0.495 ± 0.274 | 0.769 ± 0.312 | 0.345 ± 0.286 | 0.505 ± 0.263 | 0.467 ± 0.231 | 0.574 ± 0.211 | **0.998 ± 0.005** | 0.266 |
| surv_colon | 0.351 ± 0.325 | 0.568 ± 0.355 | 0.030 ± 0.031 | 0.440 ± 0.268 | 0.368 ± 0.265 | 0.477 ± 0.457 | 0.706 ± 0.160 | 0.293 ± 0.226 | 0.387 ± 0.231 | 0.405 ± 0.244 | 0.388 ± 0.208 | **0.983 ± 0.012** | – |
| surv_grace | 0.265 ± 0.096 | 0.560 ± 0.429 | 0.075 ± 0.062 | 0.323 ± 0.053 | 0.282 ± 0.094 | 0.233 ± 0.261 | 0.805 ± 0.451 | 0.154 ± 0.116 | 0.233 ± 0.069 | 0.294 ± 0.031 | 0.308 ± 0.035 | **0.909 ± 0.022** | – |
| surv_trace | 0.259 ± 0.214 | 0.700 ± 0.107 | 0.229 ± 0.432 | 0.388 ± 0.004 | 0.320 ± 0.030 | 0.280 ± 0.165 | 0.687 ± 0.241 | 0.610 ± 0.505 | 0.366 ± 0.124 | 0.336 ± 0.007 | 0.329 ± 0.045 | **0.924 ± 0.006** | – |
| surv_framingham | 0.301 ± 0.069 | 0.736 ± 0.034 | 0.198 ± 0.050 | 0.383 ± 0.032 | 0.324 ± 0.027 | 0.399 ± 0.201 | 0.941 ± 0.103 | 0.643 ± 0.905 | 0.450 ± 0.130 | 0.383 ± 0.022 | 0.405 ± 0.024 | **0.991 ± 0.004** | – |
| tab_blood_transfusion | 0.551 ± 0.523 | 0.923 ± 0.057 | 0.178 ± 0.074 | 0.418 ± 0.157 | 0.277 ± 0.056 | 0.326 ± 0.411 | 0.782 ± 0.554 | 0.596 ± 0.129 | 0.528 ± 0.170 | 0.410 ± 0.104 | 0.316 ± 0.066 | **0.995 ± 0.003** | – |
| tab_credit_g | 0.561 ± 0.279 | 0.683 ± 0.174 | 0.010 ± 0.031 | 0.572 ± 0.124 | 0.348 ± 0.098 | 0.438 ± 0.336 | 0.700 ± 0.573 | 0.399 ± 0.093 | 0.624 ± 0.081 | 0.520 ± 0.095 | 0.400 ± 0.117 | **0.999 ± 0.002** | – |
| tab_qsar_biodeg | 0.663 ± 0.446 | **1.000 ± 0.000** | 0.361 ± 0.260 | 0.999 ± 0.003 | **1.000 ± 0.000** | 0.821 ± 0.181 | **1.000 ± 0.000** | 0.638 ± 0.133 | 0.984 ± 0.030 | 0.629 ± 0.182 | 0.437 ± 0.115 | **1.000 ± 0.000** | – |
| tab_maternal_risk | 0.295 ± 0.262 | 0.814 ± 0.234 | 0.147 ± 0.274 | 0.644 ± 0.139 | 0.717 ± 0.133 | 0.442 ± 0.090 | 0.789 ± 0.298 | 0.302 ± 0.151 | 0.599 ± 0.161 | 0.463 ± 0.111 | 0.481 ± 0.220 | **0.989 ± 0.012** | – |
| tab_website_phishing | 0.069 ± 0.290 | 0.252 ± 0.086 | 0.047 ± 0.081 | 0.214 ± 0.043 | 0.010 ± 0.004 | 0.035 ± 0.034 | 0.121 ± 0.094 | 0.038 ± 0.020 | 0.058 ± 0.049 | 0.029 ± 0.015 | 0.035 ± 0.011 | **0.325 ± 0.058** | – |

Table 34: Tracking (↑) — mean ± 95% CI over seeds

| Dataset | DISCO-TAB | CTGAN | TVAE | GaussCop | TabDDPM | ADS-GAN | PATE-GAN | RTVAE | NFlow | ARF | BayesNet | Marginals | GReaT |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| heart_failure | 0.978 ± 0.043 | **1.000 ± 0.000** | **1.000 ± 0.000** | **1.000 ± 0.000** | 1.000 | **1.000 ± 0.000** | **1.000 ± 0.000** | **1.000 ± 0.000** | **1.000 ± 0.000** | **1.000 ± 0.000** | **1.000 ± 0.000** | **1.000 ± 0.000** | – |
| breast_cancer | 0.000 ± 0.000 | **1.000 ± 0.000** | **1.000 ± 0.000** | **1.000 ± 0.000** | 1.000 | **1.000 ± 0.000** | **1.000 ± 0.000** | **1.000 ± 0.000** | **1.000 ± 0.000** | **1.000 ± 0.000** | **1.000 ± 0.000** | **1.000 ± 0.000** | – |
| liver_disorders | 0.999 ± 0.005 | **1.000 ± 0.000** | **1.000 ± 0.000** | **1.000 ± 0.000** | 1.000 | **1.000 ± 0.000** | **1.000 ± 0.000** | **1.000 ± 0.000** | **1.000 ± 0.000** | **1.000 ± 0.000** | **1.000 ± 0.000** | **1.000 ± 0.000** | – |
| parkinsons | 0.000 ± 0.000 | **1.000 ± 0.000** | **1.000 ± 0.000** | **1.000 ± 0.000** | 1.000 | **1.000 ± 0.000** | **1.000 ± 0.000** | **1.000 ± 0.000** | **1.000 ± 0.000** | **1.000 ± 0.000** | **1.000 ± 0.000** | **1.000 ± 0.000** | – |
| obesity | 0.416 ± 0.407 | **1.000 ± 0.000** | **1.000 ± 0.000** | **1.000 ± 0.000** | 1.000 | **1.000 ± 0.000** | **1.000 ± 0.000** | **1.000 ± 0.000** | **1.000 ± 0.000** | **1.000 ± 0.000** | **1.000 ± 0.000** | **1.000 ± 0.000** | – |
| diabetes | 0.859 ± 0.302 | **1.000 ± 0.000** | **1.000 ± 0.000** | **1.000 ± 0.000** | **1.000 ± 0.000** | **1.000 ± 0.000** | **1.000 ± 0.000** | **1.000 ± 0.000** | **1.000 ± 0.000** | **1.000 ± 0.000** | **1.000 ± 0.000** | **1.000 ± 0.000** | – |
| german_credit | 0.000 ± 0.000 | **1.000 ± 0.000** | **1.000 ± 0.000** | **1.000 ± 0.000** | **1.000 ± 0.000** | **1.000 ± 0.000** | **1.000 ± 0.000** | **1.000 ± 0.000** | **1.000 ± 0.000** | **1.000 ± 0.000** | **1.000 ± 0.000** | **1.000 ± 0.000** | – |
| bank_marketing | **1.000 ± 0.000** | **1.000 ± 0.000** | **1.000 ± 0.000** | **1.000 ± 0.000** | **1.000 ± 0.000** | **1.000 ± 0.000** | **1.000 ± 0.000** | **1.000 ± 0.000** | **1.000 ± 0.000** | **1.000 ± 0.000** | **1.000 ± 0.000** | **1.000 ± 0.000** | – |
| census_income | 0.091 ± 0.248 | **1.000 ± 0.000** | **1.000 ± 0.000** | **1.000 ± 0.000** | **1.000 ± 0.000** | **1.000 ± 0.000** | **1.000 ± 0.000** | **1.000 ± 0.000** | **1.000 ± 0.000** | **1.000 ± 0.000** | **1.000 ± 0.000** | **1.000 ± 0.000** | – |
| surv_melanoma | 0.998 ± 0.009 | **1.000 ± 0.000** | **1.000 ± 0.000** | **1.000 ± 0.000** | **1.000 ± 0.000** | **1.000 ± 0.000** | **1.000 ± 0.000** | **1.000 ± 0.000** | **1.000 ± 0.000** | **1.000 ± 0.000** | **1.000 ± 0.000** | **1.000 ± 0.000** | – |
| surv_pbc | 0.621 ± 1.316 | **1.000 ± 0.000** | **1.000 ± 0.000** | **1.000 ± 0.000** | **1.000 ± 0.000** | **1.000 ± 0.000** | **1.000 ± 0.000** | **1.000 ± 0.000** | **1.000 ± 0.000** | **1.000 ± 0.000** | **1.000 ± 0.000** | **1.000 ± 0.000** | – |
| surv_pbc3 | 0.986 ± 0.024 | **1.000 ± 0.000** | **1.000 ± 0.000** | **1.000 ± 0.000** | **1.000 ± 0.000** | **1.000 ± 0.000** | **1.000 ± 0.000** | **1.000 ± 0.000** | **1.000 ± 0.000** | **1.000 ± 0.000** | **1.000 ± 0.000** | **1.000 ± 0.000** | 0.978 |
| surv_retinopathy | 0.544 ± 1.251 | **1.000 ± 0.000** | **1.000 ± 0.000** | **1.000 ± 0.000** | **1.000 ± 0.000** | **1.000 ± 0.000** | **1.000 ± 0.000** | **1.000 ± 0.000** | **1.000 ± 0.000** | **1.000 ± 0.000** | **1.000 ± 0.000** | **1.000 ± 0.000** | – |
| surv_whas500 | 0.000 ± 0.000 | **1.000 ± 0.000** | **1.000 ± 0.000** | **1.000 ± 0.000** | **1.000 ± 0.000** | **1.000 ± 0.000** | **1.000 ± 0.000** | **1.000 ± 0.000** | **1.000 ± 0.000** | **1.000 ± 0.000** | **1.000 ± 0.000** | **1.000 ± 0.000** | – |
| surv_gbsg2 | 0.631 ± 1.122 | **1.000 ± 0.000** | **1.000 ± 0.000** | **1.000 ± 0.000** | **1.000 ± 0.000** | **1.000 ± 0.000** | **1.000 ± 0.000** | **1.000 ± 0.000** | **1.000 ± 0.000** | **1.000 ± 0.000** | **1.000 ± 0.000** | **1.000 ± 0.000** | 0.996 |
| surv_colon | 0.998 ± 0.010 | **1.000 ± 0.000** | **1.000 ± 0.000** | **1.000 ± 0.000** | **1.000 ± 0.000** | **1.000 ± 0.000** | **1.000 ± 0.000** | **1.000 ± 0.000** | **1.000 ± 0.000** | **1.000 ± 0.000** | **1.000 ± 0.000** | **1.000 ± 0.000** | – |
| surv_grace | **1.000 ± 0.000** | **1.000 ± 0.000** | **1.000 ± 0.000** | **1.000 ± 0.000** | **1.000 ± 0.000** | **1.000 ± 0.000** | **1.000 ± 0.000** | **1.000 ± 0.000** | **1.000 ± 0.000** | **1.000 ± 0.000** | **1.000 ± 0.000** | **1.000 ± 0.000** | – |
| surv_trace | 1.000 ± 0.001 | **1.000 ± 0.000** | **1.000 ± 0.000** | **1.000 ± 0.000** | **1.000 ± 0.000** | **1.000 ± 0.000** | **1.000 ± 0.000** | **1.000 ± 0.000** | **1.000 ± 0.000** | **1.000 ± 0.000** | **1.000 ± 0.000** | **1.000 ± 0.000** | – |
| surv_framingham | 1.000 ± 0.000 | **1.000 ± 0.000** | **1.000 ± 0.000** | **1.000 ± 0.000** | **1.000 ± 0.000** | **1.000 ± 0.000** | **1.000 ± 0.000** | **1.000 ± 0.000** | **1.000 ± 0.000** | **1.000 ± 0.000** | **1.000 ± 0.000** | **1.000 ± 0.000** | – |
| tab_blood_transfusion | 0.313 ± 1.016 | **1.000 ± 0.000** | **1.000 ± 0.000** | **1.000 ± 0.000** | **1.000 ± 0.000** | **1.000 ± 0.000** | **1.000 ± 0.000** | **1.000 ± 0.000** | **1.000 ± 0.000** | **1.000 ± 0.000** | **1.000 ± 0.000** | **1.000 ± 0.000** | – |
| tab_credit_g | 0.000 ± 0.000 | **1.000 ± 0.000** | **1.000 ± 0.000** | **1.000 ± 0.000** | **1.000 ± 0.000** | **1.000 ± 0.000** | **1.000 ± 0.000** | **1.000 ± 0.000** | **1.000 ± 0.000** | **1.000 ± 0.000** | **1.000 ± 0.000** | **1.000 ± 0.000** | – |
| tab_qsar_biodeg | 0.029 ± 0.120 | **1.000 ± 0.000** | **1.000 ± 0.000** | **1.000 ± 0.000** | **1.000 ± 0.000** | **1.000 ± 0.000** | **1.000 ± 0.000** | **1.000 ± 0.000** | **1.000 ± 0.000** | **1.000 ± 0.000** | **1.000 ± 0.000** | **1.000 ± 0.000** | – |
| tab_maternal_risk | 0.453 ± 1.138 | **1.000 ± 0.000** | **1.000 ± 0.000** | **1.000 ± 0.000** | **1.000 ± 0.000** | **1.000 ± 0.000** | **1.000 ± 0.000** | **1.000 ± 0.000** | **1.000 ± 0.000** | **1.000 ± 0.000** | **1.000 ± 0.000** | **1.000 ± 0.000** | – |
| tab_website_phishing | 0.667 ± 1.434 | 0.000 ± 0.000 | 0.000 ± 0.000 | 0.000 ± 0.000 | **1.000 ± 0.000** | **1.000 ± 0.000** | **1.000 ± 0.000** | **1.000 ± 0.000** | **1.000 ± 0.000** | **1.000 ± 0.000** | **1.000 ± 0.000** | **1.000 ± 0.000** | – |

# Other

Table 35: Parse valid-rate (↑) — mean ± 95% CI over seeds

| Dataset | DISCO-TAB | CTGAN | TVAE | GaussCop | TabDDPM | ADS-GAN | PATE-GAN | RTVAE | NFlow | ARF | BayesNet | Marginals | GReaT |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| heart_failure | **0.675 ± 0.000** | – | – | – | – | – | – | – | – | – | – | – | – |
| breast_cancer | **0.693 ± 0.000** | – | – | – | – | – | – | – | – | – | – | – | – |
| liver_disorders | **0.680 ± 0.000** | – | – | – | – | – | – | – | – | – | – | – | – |
| parkinsons | **0.655 ± 0.000** | – | – | – | – | – | – | – | – | – | – | – | – |
| obesity | **0.708 ± 0.000** | – | – | – | – | – | – | – | – | – | – | – | – |
| diabetes | **0.711 ± 0.000** | – | – | – | – | – | – | – | – | – | – | – | – |
| german_credit | **0.537 ± 0.708** | – | – | – | – | – | – | – | – | – | – | – | – |
| bank_marketing | **0.711 ± 0.000** | – | – | – | – | – | – | – | – | – | – | – | – |
| census_income | **0.711 ± 0.000** | – | – | – | – | – | – | – | – | – | – | – | – |

| Dataset | DISCO-TAB | CTGAN | TVAE | GaussCop | TabDDPM | ADS-GAN | PATE-GAN | RTVAE | NFlow | ARF | BayesNet | Marginals | GReaT |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| surv_melanoma | **0.659 ± 0.000** | – | – | – | – | – | – | – | – | – | – | – | – |
| surv_pbc | **0.474 ± 0.873** | – | – | – | – | – | – | – | – | – | – | – | – |
| surv_pbc3 | **0.680 ± 0.000** | – | – | – | – | – | – | – | – | – | – | – | – |
| surv_retinopathy | **0.682 ± 0.000** | – | – | – | – | – | – | – | – | – | – | – | – |
| surv_whas500 | **0.688 ± 0.000** | – | – | – | – | – | – | – | – | – | – | – | – |
| surv_gbsg2 | **0.696 ± 0.000** | – | – | – | – | – | – | – | – | – | – | – | – |
| surv_colon | **0.700 ± 0.000** | – | – | – | – | – | – | – | – | – | – | – | – |
| surv_grace | **0.702 ± 0.000** | – | – | – | – | – | – | – | – | – | – | – | – |
| surv_trace | **0.708 ± 0.000** | – | – | – | – | – | – | – | – | – | – | – | – |
| surv_framingham | **0.711 ± 0.000** | – | – | – | – | – | – | – | – | – | – | – | – |
| tab_blood_transfusion | **0.529 ± 0.701** | – | – | – | – | – | – | – | – | – | – | – | – |
| tab_credit_g | **0.702 ± 0.000** | – | – | – | – | – | – | – | – | – | – | – | – |
| tab_qsar_biodeg | **0.703 ± 0.000** | – | – | – | – | – | – | – | – | – | – | – | – |
| tab_maternal_risk | **0.688 ± 0.000** | – | – | – | – | – | – | – | – | – | – | – | – |
| tab_website_phishing | **0.698 ± 0.000** | – | – | – | – | – | – | – | – | – | – | – | – |